



# DUXBURY BAY MANAGEMENT PLAN

A comprehensive management plan to enhance and preserve the ecological health, pristine waters and natural beauty of Duxbury Bay while sustaining harmony among all its uses



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## Overall Recommendations

|                                | Priority | Description                                                                                                                                                                                           |
|--------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>I. Water Quality</b>        | 1        | Create and maintain a central source of available scientific data                                                                                                                                     |
|                                | 1        | Establish a program for monitoring and addressing water quality issues. Designate one responsible party for overseeing such programming. Establish a standard procedures to identify problem sources. |
|                                | 2        | Establish a program to inform the officials of results of triennial survey conducted by the Division of Marine Fisheries.                                                                             |
|                                | 2        | Create a standing program to specifically identify potential pollution sources and recommended mitigation measures.                                                                                   |
|                                | 3        | Support the Open Space Plans to reduce or eliminate the application of pesticides and herbicides on Town owned properties                                                                             |
|                                | 3        | Support the Stormwater Management Program. Establish a program to prioritize storm drains for Best Management Practices ("BMP")                                                                       |
|                                | 3        | Support the Harbormaster's Office in the establishment and maintenance of a No Discharge Zone for marine sanitary devices in Duxbury waters.                                                          |
|                                | 3        | Apply for Funding for an Initial Survey to Measure and Catalog Pollution Parameters.                                                                                                                  |
| <b>II Shellfishing</b>         |          |                                                                                                                                                                                                       |
|                                | 2        | Identify and prioritize activities that threaten shellfish populations and human consumption that should be managed more closely                                                                      |
|                                | 2        | Collaborate with the Duxbury Shellfish Advisory Committee (DSAC) on the potential need for shellfish disease monitoring                                                                               |
|                                | 2        | Collaborate with DSAC to consider and initiate carrying capacity study                                                                                                                                |
|                                | 1        | Consider studies including boat ramp use, channel traffic, parking capacities and mooring capacities to enable future recommendations regarding aquaculture and coastal use management.               |
|                                | 2        | Explore feasibility of municipal shellfish propagation program                                                                                                                                        |
| <b>III. Boating and Safety</b> |          |                                                                                                                                                                                                       |
|                                | 2        | Encourage all boating Bay users to take personnel responsibility and to seek education to improve boating safety, operation, navigation and on the water etiquette.                                   |
|                                | 1        | Establish temporary alternative storage areas and locations for storm moorings and monitor and maintain Storm Management Plan relating capacity and facilities for safe vessel hauling                |
|                                | 2        | While improvements have been made to utilize aquaculture buoys to identify cage structures and mark grant boundaries, more can be done to ensure consistency in the markings as well                  |

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as educating the boating community on the buoys designation.

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| 2 | Explore the need to develop recommendations and define a methodology to address water craft conflicts of Bay use. |
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| 1 | The charting of sensitive areas should be considered and displayed for public education access points to the bay. |
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| 1 | Facilitate adoption by the Harbormaster of regulations formalizing existing limits on the number of moorings in designated anchorages and adopting limits for any designated anchorages that lack mooring limits |
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| 1 | Implement best practices recommendations for amphibious vessels (e.g. SeaLegs) by informing owners and monitoring vessel impact |
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#### IV. Moorings

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| 2 | Develop and publish a waterways guide for all bay users to include: <ul style="list-style-type: none"> <li>a. Location of public access areas (landings, ramps, parking);</li> <li>b. Information on tides in the bay (spring vs. neap; mean depths and currents);</li> <li>c. Description of aquaculture lease areas, practices, and laws;</li> <li>d. Delineation of "caution areas".</li> <li>e. Safety information, including locations and phone numbers of emergency personnel and resources.</li> </ul> |
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| 1 | Prior to the creation of any additional designated anchorages or any increase in the number of authorized moorings for existing designated anchorages, the Harbormaster should consult with the DBMC, which should provide notice and an opportunity for public comment on the proposal. |
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| 2 | The Harbormaster and the DBMC should evaluate all mooring field changes in the last 10 years and any future proposed changes |
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| 1 | Encourage the Harbormaster to adopt regulations requiring that the review of all mooring applications consider the impact of proposed moorings on existing and historical eelgrass beds, aquaculture licenses, and natural shellfish populations. |
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| 2 | Encourage the Harbormaster to evaluate the future use of helical moorings and other advances in mooring technology with Authorized Mooring Services. |
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## V. Bay Access

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| 1 | Encourage the Harbormaster to evaluate a reconfiguration of the Basin mooring grid to minimize shoaling problems from gradual silting between dredging.                                                          |
| 2 | Charge the Duxbury Bay Management Commission with exploring its taking coordination responsibility for all public access to the bay                                                                              |
| 2 | Identify and reduce any obstructions to public passage along the shoreline and in the intertidal area through regulation, enforcement, and voluntary incentives to property owners.                              |
| 2 | Conduct a shoreline access survey to inspect structures that have been identified as causing obstructions to public passage and take appropriate remedial action.                                                |
| 2 | Increase public awareness of shoreline access rights and responsibilities using educational brochures and signage directed at property owners                                                                    |
| 1 | Develop management guidelines and a program of regular maintenance based on the need for the bay's town landings and ways to the water to support existing uses.                                                 |
| 3 | Conduct parcel-by-parcel analysis to determine the exact extent of intertidal ownership within the bay, prioritizing to protect sensitive receptors, and examine the appropriateness of enhancing public access. |
| 2 | Study the impacts on ecology and existing uses of any proposed access improvements                                                                                                                               |
| 3 | Conduct bay use survey                                                                                                                                                                                           |
| 3 | Investigate the need to improve transportation options and support facilities at public access points where demand is evident; include sanitation facilities                                                     |
| 2 | Analyze sufficiency of emergency vessel access throughout bay.                                                                                                                                                   |
| 3 | Survey number of cars using access points                                                                                                                                                                        |
| 3 | Investigate installation of bike racks at town landings and ways to the water                                                                                                                                    |
| 2 | Provide education regarding access points for users and abutters                                                                                                                                                 |
| 2 | Assess appropriateness of recommendations for town landings and ways to the water from the 1975 Committee Report.                                                                                                |

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| 1 | Coordinate with the Conservation Commission and Public Works Department to develop guidelines for maintaining vegetation on public lands along the shoreline so that invasive species and rampant vegetative growth do not block scenic vistas. |
| 1 | Identify incentives to encourage private property owners to manage vegetation so as to allow views from public roadways.                                                                                                                        |
| 1 | Educate property owners regarding the value of scenic access and view sheds to community character.                                                                                                                                             |
| 2 | Explore innovative ways to permanently preserve visual access on private lands, including purchase of easements.                                                                                                                                |
| 1 | Coordinate with the Planning Board to include in development reviews consideration of how developments alter water views from public ways and encourage owners and developers to modify site plans to enhance and protect views.                |

## VI. Restoration

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|   | Establish a central resource of information and facilitate public education concerning the marsh ecosystem                                                                                                                              |
| 2 | Support acquisition efforts by the Conservation Commission, the Community Preservation Act Committee, and non-profit organizations to preserve scenic views of the bay and its estuaries                                                |
| 1 | Support funding of Conservation Commission storm water drainage plans                                                                                                                                                                   |
| 2 | Establish public education program to inform bay users of the importance of eelgrass beds and behaviors to minimize adverse impact                                                                                                      |
| 2 | Explore restoration of remaining potential herring runs at Stony Brook, Millbrook and Blue Fish River                                                                                                                                   |
| 2 | Establish a public education program to inform boaters about detection and avoidance of invasive species introduction                                                                                                                   |
| 2 | Identify selective areas where <i>Phragmites</i> eradication should be prioritized based on particularly sensitive ecological threat or scenic impact and establish a program to eradicate using means acceptable to other stakeholders |
| 1 | Establish dredging task force to optimize potential for funding, evaluate strategic options and implement contingency plan                                                                                                              |
| 2 | Establish a public awareness initiative about proper fertilization techniques and animal waste control                                                                                                                                  |
| 2 | Establish and implement best practices guidelines to minimize impact by amphibious vessels on marsh resource areas                                                                                                                      |

# **Chapter 1 INTRODUCTION/EXECUTIVE SUMMARY**

## **1.1 History and Structure of the Planning Process**

Duxbury Bay is an outstanding natural area with an estimated 1,200 acres of salt marsh and 4,650 acres surface water area (including Duxbury waters adjacent to Kingston). The Bay contains hundreds of acres of productive shellfish beds, eelgrass, a barrier beach, estuaries, and herring and rainbow smelt fish runs to name a few of its attributes. There is a great variety and population of resident and migratory birds supported by the Bay. This includes the largest tern colony in New England (5,000 pairs) and one of the largest heronries at Clarks Island. Additionally, Duxbury beach and the Bay flats support endangered species including piping plovers. The striped bass, bluefish and flounder fisheries are also thriving. Shellfish aquaculture is successful in the Bay. The variety and number of species indicate that this is a healthy example of a coastal ecosystem and it is enjoyed by a wide diversity of users. The health of this system is no accident. Although its fitness is attributed partially to its tidal flushing, which renews its waters every 24 hours, it is also due to the citizens of Duxbury who have taken action to preserve the Bay they love. Restoration of the barrier beach and innovative group septic and storm drain systems are a few examples of how the community acts to protect the Bay from severe storms and to maintain water quality. The health of the Bay demonstrates the success of the Town's tradition of caring for the Bay. This Draft Plan is a continuation of that tradition of Bay stewardship.

## **1.2 Initial Bay Management Plan and Establishment of The Duxbury Bay Management Commission**

In 2002, a twelve person Study Committee was appointed by the Board of Selectmen to develop a draft Duxbury Bay Management Plans, toward fulfilling a recommendation in the Town's Master Plan.

Over the following three years, the committee established a comprehensive document identifying and prioritizing initiatives the committee determined should be undertaken to achieve its mission. One of those priorities was establishment of the Duxbury Bay Management Commission (DBMC) to oversee implementation of the plan and advise the Board of Selectmen on issues relating to Duxbury Bay as they arose. That commission continues to meet regularly.

## **1.3 Definition of Study Area**

Using various local and regional plans as guidance, the Committee determined that the plan should cover the larger embayment that includes Plymouth and Kingston Bays and also its watersheds. The Committee believed that a collaborative approach including other towns would have been optimal, but was a too complicated and lengthy process to undertake initially. The Committee decided to proceed with a Bay management plan that the Town could begin in Duxbury waters and to recommend active pursuit of a future planning partnership with Plymouth and Kingston in future years.

In addition, the Committee did not want to create additional bureaucracy, especially where existing management appeared sufficient to protect the Bay. Therefore the Committee identified coastal wetlands within Duxbury Town boundaries as defined in the Wetlands Protection Act as the study area, including fish runs to their head waters, including but not limited to Island Creek, Stony Brook and Millbrook. In this way watershed impacts critical to Bay resources can be managed under the Plan.



## 1.4 Identification of Topics and Trends

Topics for the Draft Plan were initially developed from public workshops and the analysis and deliberations of the Study Committee. These major topics were (1) use of the Bay, (2) scientific understanding of the Bay and (3) management of the Bay. These were ultimately expanded into the following areas: public access, uses, protection of Bay ecology, and Bay management, water quality, shellfish/shellfishing and aquaculture, boating and safety, moorings, Bay access, restoration projects, and Bay management. Those topical areas are preserved in this plan update.

The broadest concern to the early committee was increased use of Bay resources and potential conflicts between uses. Although water quality was generally good, there had been an increasing trend of beach closures and shellfish bed closures and restrictions, a trend that continues presently. A moratorium had been declared on issuance of new aquaculture grants because of concern for potential use conflicts, exploitation of the existing system and lack of clear goals and policy for the activity. Increased boating was a clear source of potential use conflict that could lead to public safety issues. As a result, it was decided that no new mooring areas would be created without further study. Bay access was and continues to be important to the community. And, while improved Bay Access was identified as a primary objective, it was recognized that access must be controlled in order to protect the Bay's ecology and existing uses.

The concern for impact to Bay water quality and use conflict from increased usage remains. However, for multiple reasons, the number of moorings issued by the Harbormaster has not increased and few significant conflicts exist today. Nevertheless, the DBMC believes the priorities outlined in this plan remain important.

## 1.5 Progress since the Draft Bay Management Plan

The Bay Management Commission together with local community resources and a large number of local volunteers have made substantial progress toward achieving the goals established in the initial plan, though significant work remains to be achieved. Accomplishments include:

- a Creation of a central resource of available scientific data with completion of the first State of the Bay report
- b Supporting the Conservation department initiative to implement Best Management Practices for storm drainage to mitigate impact from surface runoff and protect the Bay from non-point source pollution
- c Support of the Harbormaster and Conservation department in establishing Duxbury Bay as a No Discharge Zone for marine sanitary devices
- d Support for creation of an Aquaculture Management Plan completed in 2009
- e Established an Emergency Storm Procedure to guide notification to Bay users and insure adequate storage space and access for Bay resources.
- f Begun a Dredging subcommittee to consider options to fund and implement dredging of Snug Harbor Basin and adjoining channels in case federal or state funding becomes unavailable.
- g Implemented a control procedure to insure new mooring fields are not established in Duxbury Bay without consideration by DBMC

- h Begun a comprehensive review of Town Landings and Ways to the Water to improve bay access.
- i Formally established the Duxbury Bay Management Commission as a permanent resource to implement this plan and advise Selectmen on Bay issues as they arise.
- j Restored the historic herring and smelt run at Island Creek.

## 1.6 Bay Management Plan Update

This updated Bay Management Plan is the first revision since the original Plan was published in 2005. The original study committee anticipated the need for periodic updates and established a goal of a new revision every five years. This plan is the product of several years of additional consideration and input from various Commission members who have participated since the first plan. While this updated plan preserves the structure of the initial plan, it has been extensively edited to address progress achieved against the initial plan as well as new priorities that have been identified since then. It is hoped that the resulting document will allow better focus on the current priorities and ultimately faster progress toward attaining the goals outlined.

## 1.7 Mission Statement and Strategic Priorities

Mission Statement: "To develop implement a management plan that will enhance and preserve the ecological health, pristine waters and natural beauty of Duxbury Bay while sustaining harmony among all its uses."

## 1.8 Strategic Priorities

- a Welcome and encourage community input;
- b Enlist, support and coordinate existing town regulatory structure;
- c Develop and maintain a baseline of uses;
- d Create and maintain a central source of available scientific data;
- e Create a mechanism to review and act on scientific and environmental data;
- f Establish a consensus for guidelines and parameters on Bay carrying capacity based on science;
- g Create a conflict mediation process;
- h Recommend and support restoration and conservation projects;
- i Provide and support education and training on the Bay;
- j Provide a structure for successful implementation of plans; and,
- k Establish a funding program to sustain and manage the plan.

## Chapter 2 HISTORY OF DUXBURY BAY<sup>1</sup>

### 2.1 Geology

#### a Geomorphology (shoreline change and stability)

Duxbury's shoreline is a case study in barrier systems. The geological story begins 15,000 years ago when the last New England glaciation, known as the Wisconsin stage of the Laurentide ice sheet, receded north and uncovered a new Duxbury. As the ice sheet retreated north, sea levels rose. At the height of ice sheet growth, sea level was approximately 350 feet lower than it is today. "You could walk out to Georgia's Bank," said Jim O'Connell, a geologist at the Woods Hole Oceanographic Institution. "The Gurnet and Saquish were much bigger than they are today because the sea level hadn't reached and begun to erode those areas," added O'Connell.

As the glaciers receded, they left boulders, sand, cobble, and clay behind. Clarks Island, Gurnet and Saquish are drumlins under the glaciers that remained as the ice melted. We know this today because of the extensive boulder platform fronting these land forms.

Duxbury Beach is known as a coastal barrier beach because the bay borders its landward side, and the open ocean is on its seaward side. A barrier island occurs when a coastal barrier detaches from the mainland. When it remains attached to the mainland, it is called a barrier spit. Duxbury Beach is a barrier spit pinned by a land form, and in this case that land form is Gurnet Point.

The barrier beach, however, is in constant motion. Storms accelerate this movement as storm waves carry and deposit sand into the sheltered waters of the bay, forming a new beach and dunes on the bay side. This process is known as landward migration. Storms erode the foreshore, the shore face, and the backshore, and strong waves break through the dunes bringing the sand into the bay. The backshore is where most people sit on the beach; the foreshore is the part extending from the berm out to the low waters; the shore face is the seaward-sloping surface that remains submerged except during very low tides.

Storm erosion combined with rising sea levels produces a more marked and rapid movement landward. Past storms, like the infamous Blizzard of 1978 and the No-Name storm of 1991, have carried sand from the frontside of the beach to the bay side in what geologists call an overwash. These storm overwashings have occurred in several locations along Duxbury Beach over the years. They allow sand to wash over and fill marsh and bay areas, thus moving the barrier landward.

"If you drew a line 2,000 to 3,000 years ago off the coast of Duxbury, the barrier extended from Howland Ledge off the coast of Marshfield southward," said O'Connell. Figure 1 from Hills & Fitzgerald (1992) illustrates the evolutionary model of the area showing cyclic barrier progradation and the anchor points for barriers and spits. Today, the Gurnet is a glacial drumlin holding Duxbury Beach, but this will change as future storms and climate variations impact the migration of this barrier beach. The International Panel on Climate Change has predicted that sea level will rise at an accelerated rate in the near future so that the current one-vertical foot in 100 years of sea level increase will almost double over the next 100 years. This scenario of increasing sea level rises combined with future storms will have a huge effect on how Duxbury Beach and the Bay look in the future. Some geologists predict that the barrier beach will migrate past Gurnet Point and attach itself to Clarks Island.

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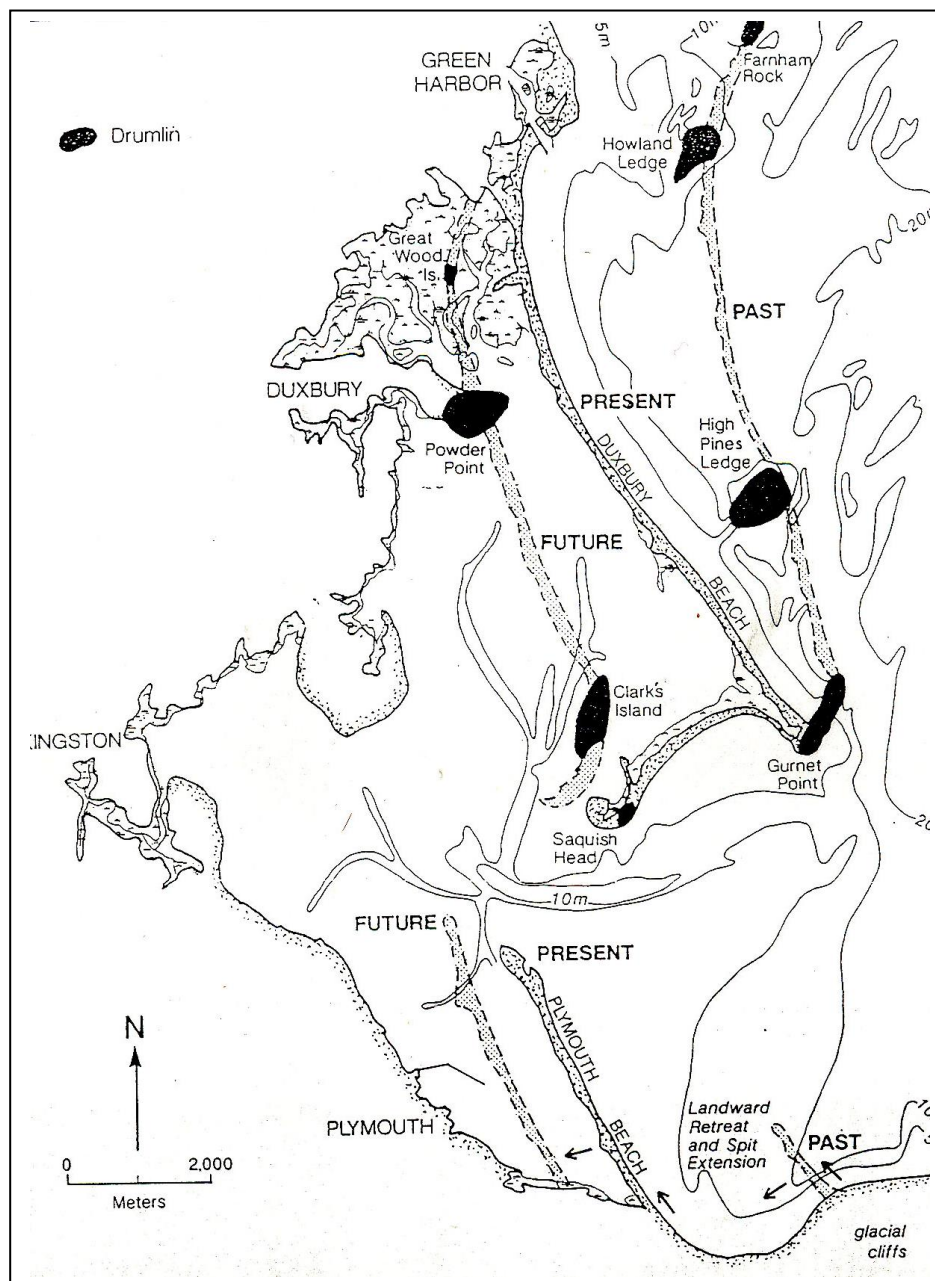
<sup>1</sup> Much of this section was provided by Debora Babin Katz and includes excerpts from her series on Duxbury Bay, called "Bay in the Balance," published in the Duxbury Clipper in 2002.

The history of Duxbury's barrier movement is seen in comparison studies of navigational charts over periods of years. O'Connell has been analyzing shoreline change from the mid-1800s to 1994 along Duxbury Beach. His study found that, in the mid-1800s, the width of the beach near the Powder Point Bridge is the same as it is today but it used to be 150 feet farther into the ocean.

Geologists have also determined that 3,700 years ago Duxbury's barrier was positioned approximately 500 meters offshore. That means Duxbury Bay is actually shrinking.

Other factors adding to the beach's front side erosion include the effects of the seawalls along Brant Rock and Marshfield's coastline. "Those coastal areas before the existence of seawalls provided a lot of the sand that created Duxbury Beach, which started as a small spit of sand," said O'Connell. Several thousands of years ago, the sand began to be added to the small spit, and then about 1,000 years ago, the rate of sea level rise slowed down and caused significant amounts of sand to be added to the spit. This process has made the beach longer and longer.

However, constant erosion, the lack of new sand feeding the beach, and rising sea levels will continue to impact Duxbury Beach and the Bay.



**Figure 1 Evolutionary Model of Area Showing Cyclic Barrier Propagation and the Anchor Points for the Barriers and Spits (from Hill and Fitzgerald, 1992)**



## 2.2 Archaeology and Recent Human History

In the spring of 1627, the colony of Duxbury was organized when numerous Pilgrim families were granted land titles to . Waterfront access was as prized then as it is today. The original grants, 20 acres for every man and 20 acres for each member of his household, fronted on the bay as much as possible and then extended inward. “The grants north of Plymouth ran to Jones River and on around the bay, some falling on the other side of the bay,” wrote Dorothy Wentworth, former town historian, in *Settlement and Growth of Duxbury, 1628-1870*.

These early settlers found the fields of the Native Americans, who came to Duxbury each summer from further inland to grow their crops along the shoreline and live off the abundant shellfish supply that Duxbury Bay provided them. They traveled the bay waters in their sturdy dugout canoes.

The pilgrims also relied on water transportation. The beginnings of shipbuilding are found in their need for sturdy, reliable vessels. “It started in a small way, with a boat built here and there along the coast, as need suggested and farm work allowed,” wrote Wentworth.

Duxbury’s economy, however, consisted mainly of farming, with the bay’s marshes providing hay for cattle and its mud flats supplying shellfish, until the late 1700s. By 1780, the shores of Duxbury had begun to experience the bustle of a new and thriving industry—shipbuilding. From 1820-1840, the town, with a population of only 2,500, “was thoroughly given to the sea and related industry,” noted Wentworth.

In 1874, this chapter of Bay history ended with the launch of the last vessel built in Duxbury near the shore of Shipyard Lane. Other bay-related activities continued, with new industries and interests arriving to these waters. Today, commercial and recreational fishing and shellfishing, aquaculture, sailing, boat racing and other maritime activities share a place in bay history and have become the catalyst for finding a balance in order to maintain and preserve the very resource they so depend on.

## 2.3 Water Quality

With the tides rushing in and out twice a day, the Bay has been the beneficiary of a natural cleansing that for centuries has kept its waters pristine. Statistics cited by environmentalists show that the bays—those in Duxbury, Kingston and Plymouth—have a tidal exchange of more than 66 percent. The advent of civilization and the unending pressure to develop the coastline, however, have produced an array of pollutants that have undermined nature’s ability to tidy up.

A 1974 study launched by the Mass. Division of Marine Resources in our bays resulted in water sampling stations, including three in Duxbury, being set up around the bays. The study cited pollution from several sources—factories, boat motors, surface drains, holding tank facilities, yacht clubs, and particularly the overflow at the Plymouth sewage treatment plant, which was overloaded due to storm drainage and ground water. Also discovered were chlorinated hydrocarbons such as Lindane, Aldrin, DDD and DDT. The evidence of these pesticides promoted pesticide-sampling stations to be established throughout the greater bay area and included one at Blue Fish River and another at Back River.

Federal legislation enacted in 1972 established a marine-protected-areas system and made the Plymouth, Kingston and Duxbury Bays part of the Cape Cod Bay Ocean Sanctuary. The new law banned the expansion of municipal effluent discharges into local waters.

In the 1980s, the town of Duxbury worked diligently to stop Plymouth from expanding its treatment plant, which was already in violation of the Ocean Sanctuaries Act. Plymouth was seeking approval from the Massachusetts Legislature to increase wastewater treatment from 2.7 million gallons per day to 4 million gallons per day under certain conditions (the plant had

been built to treat 1.7 million gallons per day). The Plymouth effort—facing strong opposition from Duxbury—failed as did a follow-up request. Duxbury resident, Robert Millar, chairman of Duxbury Ocean Sanctuaries Committee, spearheaded the drive to defeat both measures and pushed for regulations requiring that any ocean discharge from an expanded Plymouth treatment plant be sent east of a line from the Gurnet to Rocky Point in Manomet. Laws were finally enacted to ban any additional discharges into the Cape-Cod Bay Ocean Sanctuary.

In 1988, former state Senator William Golden, a Weymouth Democrat, proposed new legislation that would create strict environmental rules governing the Plymouth-Kingston-Duxbury Bay region by designating the bays as an Area of Critical Environmental Concern (ACEC). A committee of Duxbury residents considered the idea, but officials from Plymouth and Kingston refused to support the initiative and Golden's proposal fell short.

In 1992 the Division of Marine Fisheries closed the Blue Fish River to shellfishing due to high levels of pollutants from abutting residences. In 1993 the Bayswide Committee (Duxbury, Kingston and Plymouth) proposed the Bluefish River Project. As a result, the Town, with the help of Mass Coastal Zone Management and the Mass Bays Program, initiated a group septic system project with a remote leach field at the Ellison Center for the Arts. This project became a model for similar projects at Snug Harbor and Bay Road. This project was a unique partnership between residents, the Town, several government agencies and a non profit group. At the same time, many of the septic systems along the Duxbury portion of the Kingston Bay (off of Bay Road) were failing or stressed. Coliform bacteria levels were abnormally high, well above the acceptable levels. Kingston eventually solved its issue by hooking into a new treatment plant, which their town built.

In 2002 Duxbury tackled its pollution problems along Kingston Bay by identifying about 30 properties along Bay Road that had no solution for replacing failed systems. Another shared septic system was created based on the Bluefish River model. The system pumps sewage to a leaching field in a Town Park. While this shared system has made an impact, it hasn't solved all the pollution problems.

In Duxbury, bay pollution has historically come from individual disposal systems, storm drains, powerboats and other "nonpoint" sources of pollution such as lawn fertilizers, pesticides and herbicides. The town, however, has always taken a proactive stance on water quality issues.

The town continues to address pollution problems caused by storm drainage. One example is the Snug Harbor drainage system redesign to address issues in that area. This project has just recently received a construction grant. In 1998, the town obtained a State grant to acquire a pump-out boat and a pumping station on the town pier. Other anti-pollution programs to improve water quality include Duxbury's annual Beach Cleanup, which is part of the federal Coastal Sweep program. Battelle Labs, a not-for-profit organization in Duxbury, has sponsored the Beach Cleanup for the last twenty years.

Storm drainage control is also addressed as part of the Duxbury Open Space Plan published in 2008.

## 2.4 Shellfishing and Aquaculture

The history of shellfishing in Duxbury is as rich as its supply. Mussels and clams helped sustain the early colonists, who relied on the natural resource as a main staple in their diets. Long before the Pilgrims arrived, Native Americans harvested shellfish from the mud flats of these waters.

In the 1800s, a commercial use for the common soft shell clam took hold, and Duxbury became one of the leading suppliers of bait to fishing ports in Boston, Gloucester and Provincetown. "The harvest from Duxbury, Kingston and Plymouth was estimated to be as high as 100,000

bushels of clams per year,” said Frank Germano of the Massachusetts Division of Marine Fisheries. Today, that quantity would translate into more than \$9 million annually to the local economy, he noted.

In the 1860s, another market developed as steamers grew in popularity in New York and Chicago. Not surprisingly, soft shell clams in the flats of Duxbury, Kingston and Plymouth bays were close to extinction from over-harvesting. Most towns responded by racking and removing mussels, which threatened the soft shell clams. Duxbury went even further in its efforts to save the soft shell—instituting the first clam transplant program by taking clams from the productive flats along the Jones River and Kingston Bay and transplanting them to Duxbury’s beds. By the late 1800s, certain intertidal flats were restricted as part of the town’s effort to reverse the clam’s steady decline. Duxbury was also one of the first towns in the state to limit shellfishing to “residents only” (Essex was the first).

In 1920, the town initiated a relay program with the Commissioners of Inland Fisheries in an effort to boost the economy in which Quahogs from New Bedford and Fall River were transplanted to Duxbury Bay (an early attempt in 1869 proved unproductive and too labor intensive). Other efforts to restore the soft shell clam in Duxbury included Roosevelt’s Works Progress Administration that provided man-power to establish shellfish predator control (eliminating thousands of horseshoe crabs—a major enemy of the clam). The efforts paid off, and by 1936 clambers were digging record bushels of soft-shell clams compared to previous years.

It was felt that large beds of mussels prevented the soft shell clam from making the comeback in quantities seen in the harvesting of the 1800s. Duxbury, like many Cape towns, attempted to restore its clam beds by eradicating the mussels (there being a very limited commercial demand for mussels in the 1950s).

There have been other efforts to control over-harvesting of Duxbury’s shellfish species. In the 1980s, the town, under the advisement of the Shellfish Advisory Committee, a group of nine residents including ex-officio member Harbormaster Don Beers, introduced the Limited Entry Razor Clam license for commercial harvesting. This license is limited to 15 individuals per year, and licenses are issued via a lottery system. Commercial licenses for mussels are also limited, with only 10 licenses issued annually and a lottery system in place to fill open licenses.

Oyster farming is not new to Duxbury. Its history traces to the early 1900s and, of course, Native Americans and early colonists who relied on its cultivation as a primary food source. The earliest recorded commercial oyster grants in Duxbury date to 1901, when the town issued a grant to James Killion, who immediately transferred it to the Rocky Point Oyster Company in Providence. The grant authorized the company to plant, and later harvest, oysters along the full length of the beach channel south of the Powder Point Bridge. The seed came from Narragansett Bay. Unfortunately, Mother Nature played a major role in the failure of this enterprise, with cold winters causing the bay to freeze to more than a foot throughout the harvest season. After three consecutive winters of frozen waters, the project was abandoned.

In the 1930s and 1950s, additional attempts were made to seed oysters in the intertidal areas of the bay with little success. By the 1970s and 1980s, improved hatchery and grow-out techniques became more effective, and commercial oyster aquaculture grew in Massachusetts, although the state has lagged behind other states such as Connecticut.

In 1994, Massachusetts Governor William Weld, spurred on by the economic success of Connecticut’s oyster industry (which was contributing \$200 million annually to that state’s economy), directed a study known as the Massachusetts White Papers. This study outlined the factors constraining the state’s marine aquaculture industry. The findings included a highly developed coastline, multiple competing uses of Massachusetts’ waters, a redundant regulatory

system, overbearing legal issues, and the misunderstanding of aquaculture by the public and fishing industry. Historically, the primary barrier to commercial aquaculture has been the process of obtaining a license with so many regulatory levels to meet. Other obstacles in oyster farming include the upfront financial capital requirements to start such a venture, threat of disease, the learning curve for new technology, and genetic problems with seed stock. The work is also extremely labor intensive.

In Duxbury, oyster aquaculture returned as a commercially driven enterprise some 12 years ago. Grant owners established the Duxbury Grower's Association in 2000, in an effort to provide sound stewardship to the environment and address issues critical to the shellfish farming industry.

## 2.5 Maritime

The pilgrims were the pioneers of sailing in Duxbury Bay. They relied heavily on water transportation, as land travel was arduous and sometimes unsafe. The Bay played an important role in transportation because it was a well-protected water body with direct access between Plymouth and eventually Green's Harbor and Marshfield through the Cut River. Pilgrims set sail to get from here to there, and it is not too much of a stretch to suggest that they sailed to live, as would their heirs for at least two more centuries. Maritime use for survival in colonial times gave way to shipbuilding and commercial fishing in the 1800s. In 1840, Ezra Weston ("King Caesar") of Duxbury was considered the "largest ship owner in America." Over the years the community's maritime use of Duxbury Bay has changed both in character and intensity from its original use for sustenance to its present primary use for recreational boating. Commercial fishing and boat building are not as prominent as they once were. However, the skilled hands and keen eyes that landed fish and bent wood to frame also shaped the homes and churches that we still see as Duxbury. The rich and storied maritime history of Duxbury Bay is well told in two books by local authors. The Duxbury Bay Management Study Committee highly recommends, *Duxbury Bay*, by Frank Benson Lawson and Margaret Farnsworth Lawson<sup>2</sup> and, *Duxbury a Guide*, by Katherine H. Pillsbury.

## 2.6 History of Duxbury's Herring Run

In the first comers period of the early 1600s, Squanto, a Pautuxet Indian from the Plymouth Area,<sup>3</sup> is reported to have taught the Pilgrims how to plant corn in mounds fertilized with herring. Use of fish sustained native peoples and the Pilgrims. "Mattakeeset, meaning "place of many fish," is a Native American name still used in Duxbury."<sup>4</sup> Land ownership at Island Creek was established with land grants in the early 17th century. The Town regulated the Island Creek herring run at the town meeting in 1693, "...that if the fisher men of Duxbury, shall clear Island Creek brook tat said men shall have livery then to get herring for bait from time to time without moletstation; and to that end the town do agree, that if any, either English or Indian, do hinder tem herein, by makin of weirs, or hindering the fish from coming to the Pond or going down, they shall thereby forfit five shillings for every time they do so..."<sup>5</sup> The town meeting of 1702 shows "the town have given liberty to Ens. Seabury to make a dam upon Island Creek Pond brook, provided that he leaves a sufficient and free passage for the herrings up, and down, and also makes a sufficient cart way over the brook." This is the historic fish run which

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<sup>2</sup> Frank Benson Lawson and Margaret Fransworth, *Duxbury Bay*, Duxbury Rural and Historical Society, 1993

<sup>3</sup> Katherine H. Pillsbury, *Duxbury a Guide*, Duxbury Rural and Historical Society, 1999, p.18, Duxbury, Massachusetts

<sup>4</sup> Ibid. p.3

<sup>5</sup> Records of the Town of Duxbury, 1642-1770, Avery and Doten, 1893, 214-235.

was the basis for the restoration project completed a few years ago.

With access to the Bay, farm land and the Creek's herring run, the area had the resources to support a village. The name Island Creek eventually referred to a village with its "own school, post office, railroad station and social center, Island Creek Hall."<sup>6</sup> The growth in fishing in the early 18th century led Duxbury men to build ships or fish the grand banks. Island Creek provided herring as bait and sufficient deep water, and the area became important to ship building in the late 18th century. The shipping and ship building industry defined much of what we see today in Duxbury. In time, transportation and the economy changed, and the community turned away from the Bay and Island Creek. The herring run fell into disrepair until the 1940s when the present fish ladder was constructed. In recent history efforts have begun to restore these fish runs. Herring is important to many commercial and game fish including bluefish, striped bass, weakfish, and tuna. Herring is also important as bait for lobster. Towns that now celebrate the return of the herring include Bourndale, Middleboro, Weymouth, and Pembroke. The region has identified herring run restoration as important to the restoration of our environmental heritage and has included it in The Regional Open Space Plan, authored by the South Coastal Basin Team.

The Duxbury Bay Management Study Committee, in partnership with the Conservation Commission, the Department of Public Works, the Division of Marine Fisheries, the U.S. Fish and Wildlife Service, the Corporate Wetlands Restoration Program, Battelle Memorial Institute, and the Duxbury Bay Maritime School, implemented the Island Creek Rainbow Smelt and Herring Run restoration project. This restoration project was part of the Conservation Commission effort to restore the entire habitat of the Island Creek and Mill and Island Creek Pond (the only great pond in Duxbury). Both ponds are public open space. The Town has received a deeded easement to access the land where the upper control structure will be modified where Island Creek leaves Island Creek Pond. The Duxbury Rural and Historic Society recognizes the historic significance of the Pond and owns property where the Creek leaves Island Creek Pond. Previously, the Conservation Commission already restored water quality and removed invasive species from Mill and Island Creek Ponds. And recently, a multi-year restocking program was completed, due to the support of the Division of Marine Fisheries, and herring have been reintroduced to this waiting spawning and nursery habitat.

The Duxbury Bay Management Study Committee hopes that the other identified herring runs at Beaver Creek and Millbrook will eventually be restored. Debora Katz provided the following source notes from Duxbury historian Henry Fish, written about 1924, regarding herring runs:

From section on North Duxbury and Crooked Lane Neighborhood page 27 and on:

"Between the Cranberry Factory and the Marshfield line was a section of land that was settled on shortly before 1700, a part of the section was called Crooked Lane, from the fact through the place there was such a lane, and was said to have been crooked, but where it led to, no one seems to know.

Another old road entered the Crooked Lane clearing from the east by the way of the Herring Weir at the foot of Thomas Fishes swamp, now the cranberry bog of John Hadaway. This old path entered the clearing near the old tomb on Lincoln Street, and avoided the steep hill that is now used as part of the Temple St., which was not laid out as a road for many years after the first way was laid out into the clearing.

East of the Caleb Sampson farm was another lot that was bounded on the easterly side by the line between Marshfield and Duxbury, this farm was owned by Thomas Fish; one of the

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<sup>6</sup> Op.cit. p.51



prominent men of the town, who seems to have owned lands in several places, one being the swamp now commonly known as the Gifford Cranberry Bog, at the foot of which was his Herring Weir.”

From a section on Stoney Brook and Its Mills pages 95 and on

“In 1640 the town applied to the Court for permission to place the mill on this stream [Stoney Brook] and a committee was appointed to “View the same” and see if it was to be “Predjudicial.” The committee reporting that it was in no way prejudicial, and that the town should have the permission to turn the stream to the mill.

Standing near the site of the old mill at the present time, our thoughts wander from this yellow records, to that which is to be seen at the present time. The two ponds, the Spillway;(now the brook,) and the sites of the fulling mill, and the site of the last of the mills of the mill brook. Records may tell us of part of the history, yet there is more than they tell of. Tradition fails to tell much about the past, in regard to the early history of, and the many changes about the old mills....

Above the roadway may be seen two ponds, and between them a partition in which there may be seen a gateway, which was undoubtedly, the gateway of the first mill, and the small pond; the site of the first mill which stood at this place for about a hundred and five years, at which time it had become so out of repair that it was abandoned, and was replaced by another that was below the roadway, and a new dam constructed that flowed the site of the old mill, thus making what is now two ponds.

By the reading of the Colony Court records, one can readily see that there was a small group of settlers at this locality, as in other sections of the town, their lots varying from a mearstead, to ten acres, and their “Great lots” were elsewhere, two reasons can be advanced for this supposition, one is, the land near the brook was largely cleared by the Indians for their planting land, the Herring being their fertilizer for their corn fields, and this brook was once a famous resort for this lowly fish. The second reason is the same as in other localities, “In unity there is strength” and find these small groups scattered along the shore from Jones river to the Mill Brook.”

## Chapter 3 WATER QUALITY

### 3.1 Existing Conditions and Trends

Duxbury is fortunate to have some of the cleanest estuarine water on the Massachusetts coast. Duxbury Bay is the last major body of water and shellfish resource from Cape Cod to the New Hampshire border not seriously affected by coastal pollution. This situation is due in part to Duxbury's relatively low population density, to the concern of its citizens (who funded the State's first shared septic system at the Bluefish River and successfully fought increased discharge from the Plymouth Sewage treatment plant), and to the Bay's high tidal exchange rate of approximately 70 percent.

In the past years however, some warning signs have appeared that will require continued vigilance, and further action may be necessary to preserve the water quality of the bay. Sampling of bacteria by the state's Division of Marine Fisheries has resulted in the closing of part of the Back River to shellfishing, as well as changing the designation of an area at the mouth of the Bluefish River from "approved" to "conditionally approved."

Regular water quality measurements are taken in Duxbury Bay under two State programs. The Massachusetts Division of Marine Fisheries ("DMF") regularly samples stations in Duxbury Bay and adjacent waters for fecal coliform bacteria according to procedures in the National Shellfish Sanitation Program (NSSP) Model Ordinance promulgated by the U.S. Food and Drug Administration. Results of this sampling are used to determine whether designated areas are approved, conditionally approved, or prohibited for shellfishing. The other program involves regular testing of beaches during the bathing season by the Massachusetts Department of Health. Samples are taken weekly from June through August and tested for enterococci, as these bacteria are strongly associated with gastrointestinal symptoms reported by swimmers, and the results are used to determine the open or closed status of beaches.

More recently, the Provincetown Center for Coastal Studies has begun collecting general water quality data from several locations in Duxbury Bay as well as other estuaries of Cape Cod Bay. Those data have been published through the PCCS website and are also summarized in its annual report.

Also, in response to the priorities identified in the initial Bay Management Plan, published in 2005, the Duxbury Bay Management Commission has begun to tabulate and analyze numerous general water quality indicators. Those are presented in the State of the Bay report, which was initially published in 2010 and will be updated from time to time.

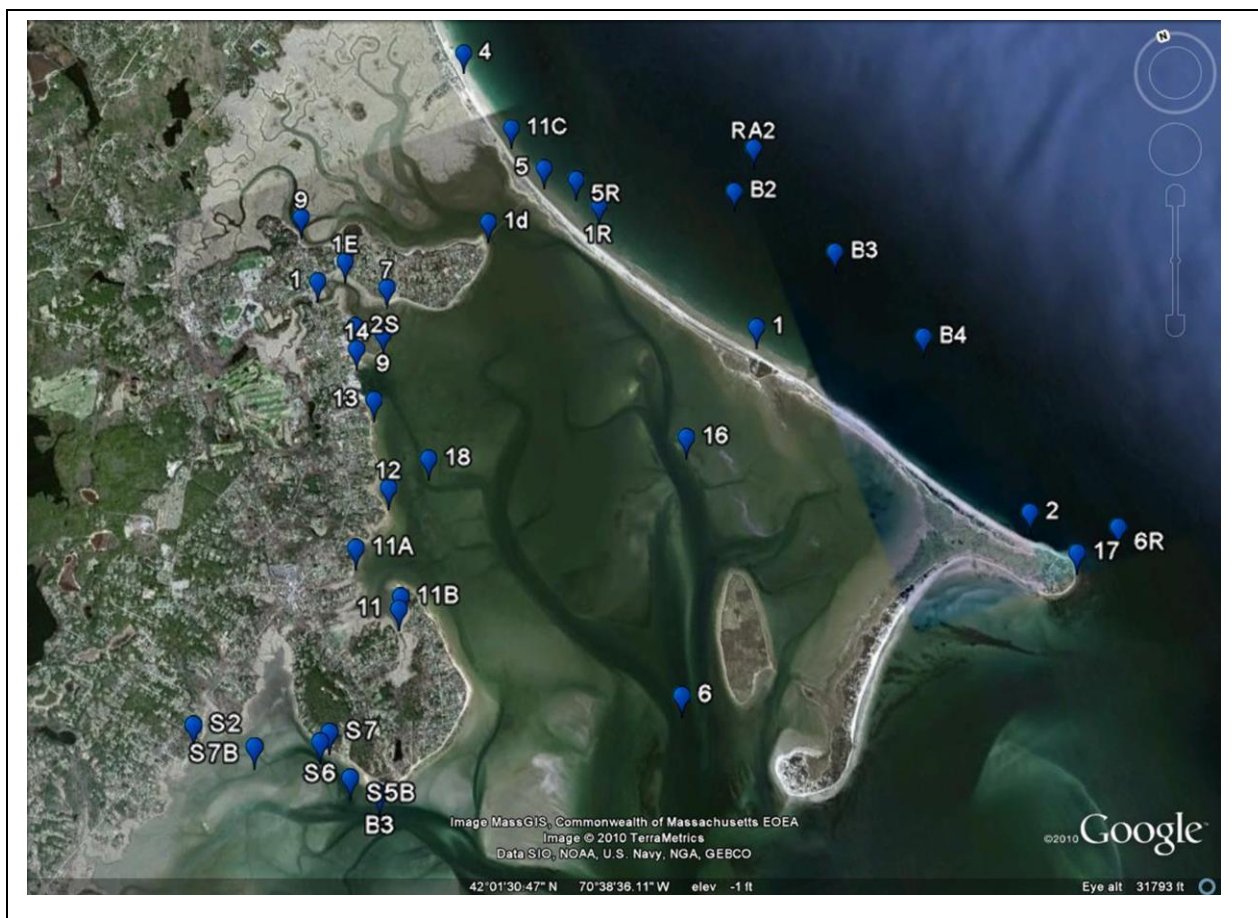
### 3.2 Physical Condition

Figure 2 shows the location of the fecal coliform stations sampled by the DMF. A total of 23 stations were regularly sampled in Duxbury Bay and adjacent waters in 2008 and another 13 stations were sampled in or around Duxbury Beach. Each of these areas is reported on separately by DMF, and recent reports contain a history of water quality related shellfish closings in these waters.

### 3.3 Division of Marine Fisheries Sanitary Surveys

- a **Duxbury Bay.** Duxbury Bay, with the exception of Eagle's Nest Bay and the Bayside Marine area, has historically been classified as approved for the taking of shellfish. Eagle's Nest Bay and Eagle's Nest Creek were classified as "approved" until 1983, when the Department of Environmental Quality Engineering (DEQE), now the Department of Environmental Protection (DEP), reclassified Eagle's Nest Creek

On April 6, 1994, the DMF closed a three acre area in Snug Harbor off Bayside Marine due to unacceptable water quality caused by three storm drains flowing into the area.



**Figure 2. Location of fecal coliform sampling stations**

High coliform levels found in surveys in 1987 and 1989 caused most of the Bluefish River to be prohibited to shellfishing for the entire year, with the exception of a triangular area at the mouth extending from Long Point to Thompson's Pier and Maxfield Point. Following completion of the 1993 survey report, a conditional, seasonally open area was approved north of a line from Thompson's Pier to Long

Point, and this area was enlarged in 1999, although the interior parts of the river remained closed to shellfishing.

The source of the high fecal coliform counts that caused the 1991 closing appeared to be from three failing septic systems near the Washington Street Bridge. In 1996, these properties were connected to an off-site shared septic system, resulting in significantly improved water quality in the abutting section of the river.

The Bluefish River's water quality problems continued to persist, however. In 2003, the formerly approved area at the mouth of the river was changed to conditionally approved for shellfishing. This means that no part of the river is open for shellfishing year-round, and only the part nearest the mouth is open on a seasonal basis, from November 1st to June 15th.

- c **Back River.** The Division of Marine Fisheries' classification area known as the Back River is actually a collection of small tidal rivers and passages extending throughout the Duxbury Marsh at the north end of the Bay. The largest of these are the Great Woods River, Pine Point River, Little Wood Island River, and Duck Hill River.

The March 29, 2002 triennial report for the Back River by the Division of Marine Fisheries stated that "there are few actual sources of pollution and the area has a long history of excellent water quality. The pollution sources identified during the shoreline survey do not appear to have a significant adverse impact on the water quality in the resource area." During the summer and fall of 2003, however, high fecal coliform counts at Station 9, in the Duck Hill River in the western part of the marsh, caused the river upstream of the sampling station to be closed to shellfishing. This was the first time that any part of the Back River system had not been approved for shellfishing. DMF personnel did not have an opinion as to the cause of the high coliform levels.

However, since then there have been no exceedences reported.

- d **Kingston Bay.** The northern shoreline and the northern half of Kingston Bay are in the Town of Duxbury, and four of Kingston's six DMF water quality sampling stations are in Duxbury. The Jones River empties into the west end of the Bay from Kingston, and just north of it, Island Creek enters the bay from Island Creek Pond in Duxbury. The shoreline along the south side of Kingston Bay in the Town of Kingston and extending south into Plymouth is densely populated and partially commercial. Kingston Bay has had more water quality problems than other waters within Duxbury.

In 1925, the National Shellfish Sanitation Program was established in response to a typhoid epidemic caused in part by the consumption of polluted shellfish, and in December of the same year the western part of Kingston Bay and an area along the Kingston and Plymouth shore extending southeast from Rocky Nook Point were closed to shellfishing. Among the main causes of pollution were discharges of untreated sewage from the Plymouth sewer system, and from 20 houses along the Jones River that did not have cesspools, but discharged their wastes through pipes directly into the river. Many of the houses in the Rocky Nook section of Kingston were seasonal and unoccupied in the winter months. The bay's water quality would fluctuate with the increase in population of the Rocky Nook area.

Between 1934 and 1978, the bay underwent a series of openings and closures, and from 1978 until 1993, the closed area was periodically enlarged until most of the bay was prohibited for shellfishing. Following a sanitary survey in 1997, however, the

closure line was moved to the west, allowing an area east of a line from Boundary Lane in Kingston to the pier at 146 Crescent Street in Duxbury to be opened to shellfishing. In 2003, the restriction on shellfishing in the bay was further lifted when a large part of the central bay was reclassified as conditionally approved and opened for shellfishing from November 1 to March 31, except for 48 hours after a rainfall of 0.3", leaving only the western most part of the bay prohibited for shellfishing (Resource Map). The DMF's Sanitary Survey Report attributed the improvement in water quality to the connection of more than 350 houses in Kingston, along the shore and the Jones River, to the new Plymouth wastewater collection system, and to the strict enforcement of septic regulations in Duxbury (which had taken 29 houses with failed or suspect septic systems and connected them to an upland community waste disposal system).

This area has been continually subject to conditional closures since then and remains the area associated with the highest number of coliform exceedences (16) between 1998 and 2008, the last year data were available by the time of this update.

### 3.4 Department of Public Health Beach Testing

Another ongoing water quality testing program in Duxbury Bay is the bathing beach monitoring program. During the swimming season from June through August, personnel from the Harbormaster's Office periodically collect water samples from public and semi-public beaches and send them to a laboratory where they are analyzed for Enterococci. The results are then sent to the State Department of Public Health, which compiles them and issues an annual report. In addition, Duxbury's Dept. of Health will issue beach closures based on the findings by posting signage.

Enterococci are bacteria that have been shown to be strongly correlated with swimming-associated disease, and if their concentration is over 104 colony forming units (CFU) per 100 ml for a single sample and 35 CFU per 100 ml for the geometric mean of at least five samples over a 30-day period, the beach is closed until further testing. Fortunately, the high flushing rate of Duxbury Bay means that high levels of bacteria are quickly dissipated and do not soon return.

There are five beaches sampled in Duxbury, two on the ocean side of Duxbury Beach, and the rest along the west side of Duxbury Bay and in Kingston Bay (see Figure 3).

**Error! Reference source not found.** below shows the number and location of exceedences that have occurred between 2003 and 2011.

### 3.5 Provincetown Center for Coastal Studies Data

The PCCS collects a variety of general water quality data from Duxbury Bay and other estuaries of Cape Cod Bay as part of its overall mission to preserve the quality of this ecosystem. Eight sites are monitored between Duxbury Bay and Plymouth Bay. Data are collected on temperature, salinity, water clarity, dissolved nitrates, phosphates, nitrogen, phosphorus and chlorophyll. These data are available through the PCCS website and are also summarized the State of the Bay Report.

Because the data have been collected only since 2006, long term trending isn't possible yet.

|                        | 2003 | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 |
|------------------------|------|------|------|------|------|------|------|------|
| <i>Gurnett</i>         |      |      | 0    | 5    | 0    | 0    | 0    | 0    |
| <i>Landing Road</i>    | 3    | 4    | 1    | 2    | 0    | 3    | 4    | 0    |
| <i>Residents Beach</i> | 0    | 0    | 0    | 0    | 0    | 0    | 2    | 0    |
| <i>Shipyard Lane</i>   | 2    | 1    | 0    | 0    | 0    | 0    | 0    | 0    |
| <i>West End</i>        | 2    | 1    | 0    | 0    | 0    | 2    | 0    | 0    |

Table 3-1 Coliform exceedences on Duxbury Beaches since 2003.

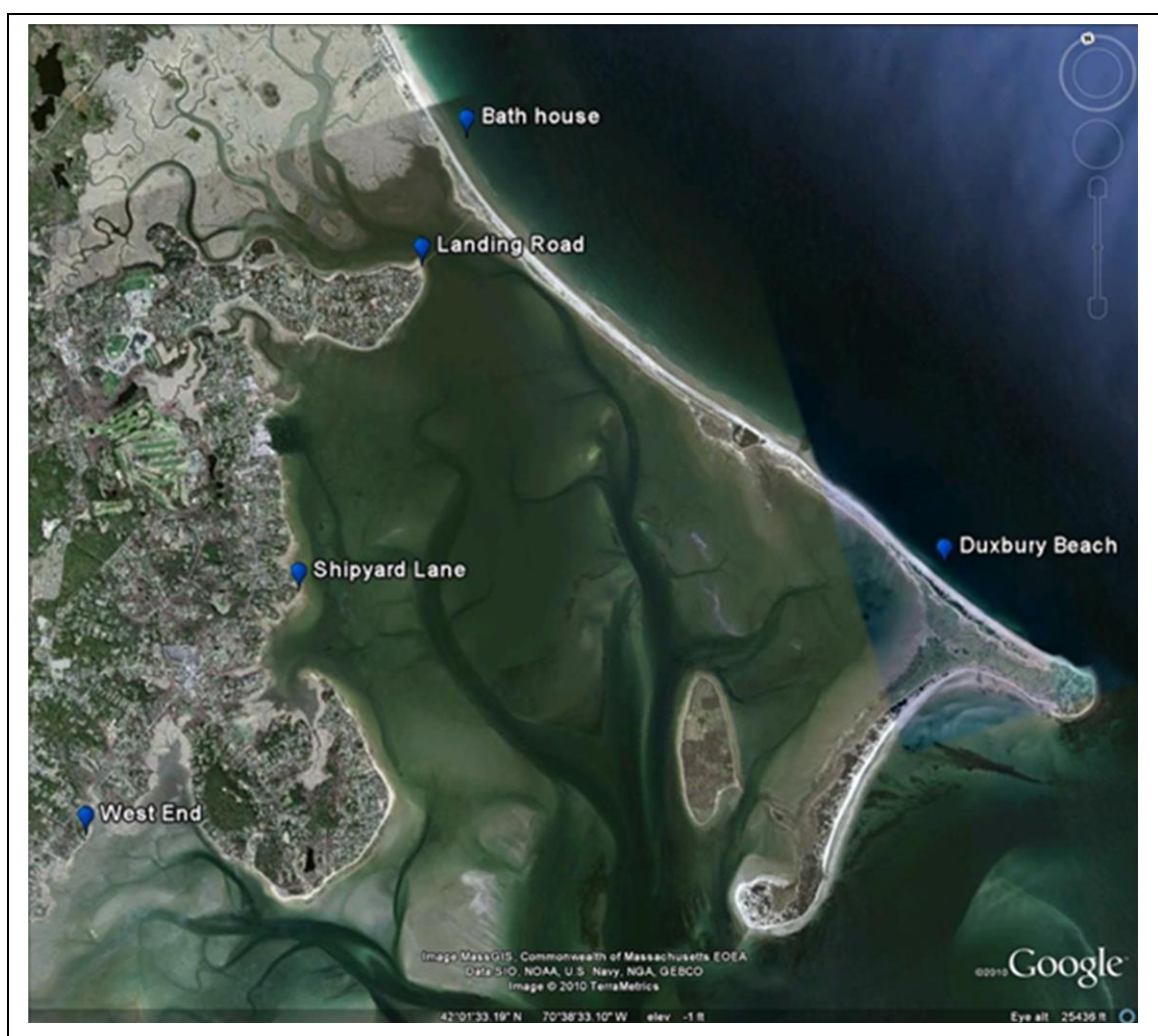


Figure 3. Beach sites monitored by Board of Health



### 3.6 Sources of Pollution

As part of the triennial sanitary surveys, DMF and the Town conduct shoreline surveys looking for and documenting any possible sources of pollution. The most recent such reports available are from 1999 for the Back and Bluefish Rivers, 2002 for Duxbury Bay, and 2002 for Kingston Bay.

Potential sources of pollution are individual septic systems, storm drains and stormwater runoff, and waterfowl. No significant sources of pollution were found in the Back River system, and at the time of the survey, there were no water quality problems in the system. In the Bluefish River, construction of a shared septic system for several houses with failed septic systems had improved conditions, but problems remain. Six properties along Washington, Harrison, and St. George Streets were listed as potential problems because of their close proximity to the marsh, and the DMF recommended that the Board of health inspect their septic systems. A storm drain at the A&P Plaza on Depot St. was cited as a problem, although plans were underway to correct this [what is the current status of this storm drain?] The Duxbury Yacht Club Golf Course and associated ponds contribute heavy runoff to the river during storm events, and may contribute nutrients, fertilizers, pesticides, and feces from waterfowl to the river.

In 2002, when DMF personnel conducted the shoreline survey of Duxbury Bay, they sampled several pipes which flow into the bay; none of these contained significant coliform numbers. The construction of a septic system shared by several properties in Snug Harbor has improved conditions in that area, and the Town is seeking a grant to install Best Management Practices (BMP's) and reconstruct the storm drains at Snug Harbor, which have been cited as a reason for closing the immediate area to shellfishing.

In Kingston Bay, as noted above, water quality has significantly improved since 350 houses in Kingston were connected the Plymouth wastewater collection system and 29 houses in Duxbury were connected to an upland community septic system. Storm drains still contribute fecal coliforms to the bay during wet weather, although there have been recent improvements because of the sewerage improvements and CPR grants to improve the drainage systems.

### 3.7 Current Bay Management

#### a Permitting Regulations and Structure

Several regulatory programs are intended to control water quality. These include the State's Title V septic system regulations (310 CMR 15.00) and Section 1 of the Duxbury Board of Health regulations. Also, the EPA's National Pollutant Discharge Elimination System (NPDES) General Permit for Storm Water Discharges from regulated Small Municipal Separate Storm Sewer Systems now applies to communities the size of Duxbury. Director of Public Works, Thomas Daley, submitted a Notice of Intent (NOI) for the Town to be covered under this permit on July 29, 2003. The NOI includes a list of all receiving waters and the number of storm drains discharging to them, and a summary of a Stormwater Management Program. The present structure for monitoring water quality and addressing water quality problems is fragmented between all the agencies cited above. This structure does not lead to effective proactive protection and restoration of the water quality of Duxbury Bay.

#### b Town By-Laws and Regulations

Section 1 of the Duxbury Board of Health Regulations supplements the State's Title V code for individual septic systems. Among its provisions is a setback distance of 150' from any freshwater wetland, salt, marsh, of coastal or inland bank,

considerably greater than the 50' setback required under Title V. Zoning to preserve the low density human impacts on natural systems in Duxbury is also vital to protection of water quality

c Harbor and Bay Preservation and Restoration Efforts

Duxbury's citizens and town officials have worked diligently to preserve the Bay's water quality, and the success of their efforts is measured by the fact that Duxbury has the most unrestricted shellfishing of any town from Cape Cod to New Hampshire. An important factor in the preservation of this quality has been Duxbury's strict enforcement of its septic codes, projects such as the shared systems at the Washington Street Bridge and Snug Harbor, and the construction of a community upland disposal system serving houses on Bay Road.

The Stormwater Management Program, as outlined in the NPDES NOI, is another program to maintain and improve Bay water quality. It includes the following elements:

Public Education - Put up posters in school and town buildings; stencil catch basins; pamphlets in water bills; educational material on web site; and seminars for community groups;

Public Participation - Comprehensive plan and zoning bylaw implementation committees; paint day collections; open space and recreation, community preservation, and bay management committees;

- d Illicit Discharge Detection and Elimination - Map drainage system; coordinate with DMF on testing outfalls; investigate pollution hot spots as needed via test results;
- e Construction Site Runoff Control - Propose changes re: drainage quality to zoning and conservation bylaws at Town Meeting; projects reviewed by Development Review Team (DRT); review and update subdivision rules and regulations biannually (example: drainage BMP's; enforce all approvals by regulatory authorities);
- f Post-Construction Runoff Control - Require and review BMP operation manuals as part of DRT and Con Com approval; Maintain drainage BMP's; Require annual maintenance and inspection reports for regulated projects; Submit bylaw to Town Meeting for BMP's for projects greater than one acre;
- g Municipal Good Housekeeping - Construct Snug Harbor BMP project when grant obtained from CZM; Implement maintenance schedule for town-owned BMP's; sweep streets and clean catch basins;
- h Existing Plans - Existing plans to maintain water quality include the septic regulations and Stormwater Management Program specified above.

### 3.8 Management Recommendations for Water Quality

Results show that Duxbury currently does an excellent job of maintaining the Bay's water quality, but that there is room for improvement in some areas. Management recommendations for water quality include:

- a Create and maintain a central source of available scientific data on water quality
- b Establish a program for monitoring and addressing water quality issues. Designate one responsible party for overseeing such programming. Establish a standard procedure to identify problem sources.
- c Establish a program to inform the officials of the schedule for triennial survey

- conducted by the Division of Marine Fisheries. Insure that appropriate town officials, (e.g., Board of Health and Conservation Agents) are provided with results promptly to enable further investigation of resulting problems.
- d Create a standing program to specifically identify potential pollution sources and recommended mitigation measures. Address water quality problems using existing authorities and Best Management Practices in areas with shellfish areas classified as conditionally open or prohibited and in swimming areas that have experienced closures. Explore and resolve where possible differences between water sampling protocols and thresholds for shellfish and swimming.
  - e Support the Open Space Plans call to “Investigate and develop an Integrated Pest Management Plan to reduce or eliminate the application of pesticides and herbicides on Town owned properties” and support education of private property owners regarding environmental impacts of pesticides and herbicides.
  - f Support the Stormwater Management Program as outlined in the National Pollution Discharge Elimination System Notice of Intent proposed for the Town drains in the Snug Harbor area by the Department of Public Works and the Conservation Commission. Establish a program to prioritize storm drains for Best Management Practices (“BMP”) proactively so that the Bay is protected from existing and future levels of non-point source pollution.
  - g Support the Harbormaster’s Office in the establishment and maintenance of a No Discharge Zone for marine sanitary devices in Duxbury waters.
    - Trends in dry/wet-weather bacteria indicators
    - Trends in beach closings
    - Nutrients and Eutrophication
    - Nutrient concentrations ( $\text{NO}_3$ ,  $\text{NO}_4$ ,  $\text{NH}_4$ ,  $\text{PO}_4$ )
    - Particulate concentrations
    - Biological oxygen demand (BOD)
    - Dissolved oxygen (DO)
    - Toxic Contaminants
    - Shellfish/benthic invertebrate tissue concentrations
    - Sediment contaminant levels
    - Shellfish Disease
    - Dermo
    - Neoplasia

## Chapter 4 SHELLFISHING & ACQUACULTURE

### 4.1 Shellfish Existing Conditions and Trends

Currently, Duxbury has approximately 17,000 acres of water open for shellfish harvesting.. Areas closed to shellfish harvesting include areas in western Kingston Bay (the Nook, south to Bay Farm), inside Eagles Nest, the inner harbor, inside Bluefish River, and the extreme northwestern areas of the Back River marsh system. The Massachusetts Division of Marine <sup>7</sup>

This chapter provides an overview of the types of shellfish grown in Duxbury coastal waters, existing conditions and recent trends in shellfish landings, a summary of current management of shellfish resources in Duxbury, and management recommendations.

#### a Soft-shell clams (*Mya arenaria*)

The soft-shell clam population has fluctuated greatly over the years in response to fishing pressure. The harvest from was as many as 100,000 bushels of clams per year from Plymouth, Kingston and Duxbury bays in the 1800's<sup>8</sup>, but the market for the human consumption of steamed clams that developed in the 1860's led to over harvesting of clams and caused a decline in production. A transplant program and "resident only" restrictions on clam harvesting led to another surge in the 1930's.

The spread of mussel beds and natural predation by horseshoe crabs eliminated areas of habitat for clams. The town sponsored programs to destroy horseshoe crabs and mussels, in response, and these programs were still underway as of 1963, when a bounty of 4 cents per live horseshoe crab was paid by the shellfish constable (12,000 crabs were harvested that year). In 1950, to address unfavorable market conditions, thousands of bushels of mussels were destroyed by the town.

Soft shell clam flats tend to be close to the shore. The 1974 DMF report (Iwanowicz *et al.* 1974) shows productive clam flats along Standish Shore on the west side of the bay south of Eagle's Nest Point, along the west side of the mouth of the Bluefish River just north of Long Point, and along the shore southwest of Powder Point, off King Caesar Road. In December 2003, a map available for shellfish license applicants at the Town Clerk's office showed a similar distribution, with soft-shell clams along the shore from Eagle's Nest Point to the Bluefish River, and from the Bluefish River to Powder Point, as well as, along the back side of Duxbury Beach on either side of the Powder Point Bridge. More recently, part of Kingston Bay has been opened to clamming and the large intertidal flats on the Duxbury side of the bay have been productively harvested.

#### b Quahogs (*Mercenaria mercenaria*)

The quahog or hard clam is another important Duxbury shellfish resource, although this has not always been the case. According to the 1974 DMF report, quahogs were relatively scarce at the time of the first European settlers, and little information appears before 1900. During the 20<sup>th</sup> century, there was an attempt to increase the value of this resource, and the Duxbury Town Reports state that quahogs were transplanted into the bay in fifteen of the 37 years between 1932 and 1969. In 1973, 50,000 seed quahogs were purchased and planted, and in 1982 the town

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<sup>7</sup> <http://www.mass.gov/dfwele/dmf/>

<sup>8</sup> Katz, D. 2002. The Real Dig: Duxbury Shellfishing. *Duxbury Clipper*, Sept, 25, 2002

purchased 80,000 quahog seed, cultured them, and broadcast them in 1983 in areas that were either fished out or ideal for propagation. In that same year, the Shellfish Department also purchased 200 bushels of contaminated quahogs from the Swansea area and seeded them in closed areas so that their spawn would repopulate other areas of the bay. Although no further relay programs were reported subsequently, quahog propagation efforts were apparently successful. In 1986, it was reported that the quahog resource had grown to a point where a small, strictly regulated commercial fishery could be instituted. In more recent years, the recreational harvest of quahogs has consistently exceeded that of soft-shell clams.

In the 1974 DMF report on Plymouth, Kingston, and Duxbury Bay, the only quahog beds in the three town area are shown in Duxbury along the shore south of Powder Point and on a flat just inside Duxbury Beach, to the southeast of Powder Point. In 2003, a map provided to applicants for shellfish licenses in the Town Hall showed quahog existence along the western shore of the bay from south of the mooring basin to the Bluefish River, and south of Powder Point, further offshore than the steamers.

#### c Mussels (*Mytilus edulis*)

Blue mussels have always been one of the most abundant shellfish in Duxbury Bay, and during the late 20<sup>th</sup> century went from being regarded as a pest requiring eradication to being a sought-after commercial species. Throughout most of Duxbury's history, there was little if any market for mussels and, because their spread choked out the softshell clams, there was an active campaign to eradicate them. Local fishermen would rake mussels into piles on the beach, pour oil on them, and burn them. As late as 1963, the shellfish constable's annual report stated that "Thousands of bushels of mussels have been removed from the bay in the past two years but they still remain a problem."<sup>9</sup> By 1971, however, a market for mussels had apparently developed, as 2,250 bushels of mussels were harvested that year by four commercial permit holders. By 1984, the commercial harvest of mussels had increased by over 16 times, to 37,037 bushels. In 1985, however, the mussel beds were hit by a hurricane and a hard freeze, both occurring at low tide, which devastated the resource. The harvest declined to 7,213 bushels in 1986, and the commercial mussel fishery was reported as being "almost non-existent" in 1987 and 1988. By 1992, however, the fishery had largely recovered, with 21,294 bushels harvested that year. In 1995, the harvest was reported as having been in decline for two years. The last estimate of mussel harvest available (1997) reported a commercial harvest of only 1,626 bushels<sup>10</sup>. Today the occurrence of mussels is highly cyclical, with their abundance being controlled primarily by natural events and possibly fishing pressure. Currently, only 10 commercial mussel licenses are issued annually. When an opening for a license to mussel becomes available, a publicized lottery occurs to fill the opening. A waiting list is also maintained and presently 10 residents are on the list.

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<sup>9</sup> Annual Reports, Town of Duxbury

<sup>10</sup> Division of Marine Fisheries Shellfish Landing Data, Neil Churchill, DMF, Personal communication

#### d Razor clams (*Ensis directus*)

Razor clams have long sustained a commercial fishery for bait, with 4,813 bushels marketed in 1937, and 3,242 bushels in 1938. As with mussels, the harvest has fluctuated, declining to 823 bushels by 1953, and only 75 bushels in 1983. In the early 1980's, however, a domestic Asian market for razor clams developed, and the commercial harvest increased to 2,320 bushels by 1997, the last year for which data was available. The Shellfish Warden cites cuts in Mass Division of Marine Fisheries funding as the reason data is not available past 1997. As with mussels, the commercial harvesting of razor clams is strictly regulated, with only 15 permits per year being issued to Duxbury residents only. There is also a waiting list for razor clam licenses.

#### e Oysters (*Crassostrea virginica*)

As noted by Debora Katz in her series, *Bay in the Balance*, “oyster farming is not new to Duxbury. Its history traces to the early 1900s and, of course, Native Americans and early colonists relied on its cultivation as a primary food source”. The first recorded commercial oyster grant dates to 1901 when seed from Narragansett Bay was planted along the beach channel south of the Powder Point Bridge. Repeated freezing throughout the harvest season, however, caused the project to be abandoned. From 1986 through 1990, 200 to 250 bushels of oysters were imported from the Cape and Mattapoissett, planted and allowed to cleanse for 30 to 60 days, then opened for recreational harvest. After 1990, this program could not be continued because of disease in other areas which could have spread into Duxbury Bay. In recent years, with improved techniques aquaculture of oysters on granted areas has become a successful enterprise and is discussed further in Section 4.3.

#### f Current Bay Management

Shellfish in Duxbury waters reside on private property and state land. Shellfish propagation and harvesting are managed through the Shellfish Division of the Duxbury Harbormaster's Office<sup>11</sup>, and by the state permitting agency DMF. State regulations are found in CMR Chapter 130.

- Clean Water Act, s. 401, U.S. Army Corps Rivers and Harbors Act;
- 310 CMR 10 Mass Wetland Protection Act Regulations;
- 314 CMR 9, (s.401 compliance) Mass Water Quality Certification; and,
- Town of Duxbury Wetlands Protection Bylaw, Ch. 9, Conservation Commission.

## 4.2 Shellfish Harvesting

#### a Current Conditions

Shellfish harvesting is an important activity associated with Duxbury Bay, and sustaining healthy shellfish populations should be one of the goals of a bay management plan. All persons engaging in shellfish harvesting within the town are required to have a town permit, and these are divided into two broad categories: recreational and commercial. A recreational permit, also known as a family permit, allows the digger to harvest shellfish for personal/family use and prohibits the selling of the catch. Recreational permits are open to both residents and non-residents of Duxbury. In recent years approximately 1400 to 1500 recreational permits per year have been issued. Commercial permits are restricted to Duxbury residents only, and about 40 to 60 permits per year have been issued in recent years. Ten permits per year are issued to harvest mussels and

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<sup>11</sup> <http://www.duxburyharbormaster.org/shellfish/shellfish.htm>



15 for razor clams (closed fisheries). There is a waiting list for both.

Trends observable in recreational shellfishing include the steady rise in permits granted each year after a decline in the early 1980's, the dominance of the fishery by quahogs and soft-shell clams, and the steady rise in clam harvests, with a large jump in 1997, although the yield did not nearly reach the levels in the 1930's, with a peak of 77,975 bushels in 1936<sup>12</sup>.

The commercial data indicate the surge in mussel harvests in the 1980s after they became more popular for human consumption, and the devastating effect on the entire commercial fishery, dominated by mussels, of Hurricane Gloria and a hard, below-zero freeze, both occurring in 1985 at low tide when the beds were exposed. It took several years for the fishery to recover from these impacts, and by 1997, mussel harvests had declined precipitously again, illustrating the cyclical nature of this resource. As of 2000, the shellfish constable reported that "Again the commercial landing of mussels and razor clam was low", attributing the reduction to natural events and cycles. The decline in mussels and razor clams is in contrast to the recent relative abundance of quahogs and soft-shell clams, enabling extended seasons and an increase in the commercial fishery for these species.

### 4.3 Current Bay Management

Shellfish are regulated by both the state Division of Marine Fisheries through Chapter 130 of the Massachusetts General Laws and Title 322 of the Code of Massachusetts Regulations, and by the Town of Duxbury, through Section 7.5 of the Town by-laws. The DMF oversees the status of the Commonwealth's shellfish growing areas. The DMF samples pre-determined monitoring stations on a regular, systematic basis for fecal coliform abundance. Based on this regular monitoring, shellfish growing areas are classified as approved, conditionally approved, or prohibited for shellfish harvesting.. The DMF also has the authority, and responsibility, to selectively close growing areas due to adverse environmental conditions such as extreme rainfall events and periodic red tide occurrences.

Shellfish harvesting is regulated through Town By-Laws Section 7.5 of the Town Rules and Regulations, last revised in 2001. The program is administered by the harbormaster, who also serves as the shellfish warden, with the guidance of the Town's Shellfish Advisory Committee, which meets on an as-needed basis. Some of the main provisions of the shellfish regulations are provided in the sections below.

The Town of Duxbury issues licenses on an annual basis, and determines the fees, harvest limits and the periods when bonus shellfish are open for harvest. Size limits are specific to each species of shellfish.

#### a Commercial Shellfish Licenses

Commercial licenses for razor clams and mussels are limited to 15 and 10 individuals per year, respectively. Harvest rates of these species is determined by the shellfish constable and approved by the Board of Selectmen.

Another category of commercial license is the combination commercial shellfish license and it allows the holder to take eels and sea worms as well as soft shell and hard clams when allowed. In recent years, the taking and marketing of quahogs and soft-shell clams has been permitted under this license. (Although the regulations define "shellfish" as soft-shell clams, quahogs,

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<sup>12</sup> source: D. Katz, *Bay in Balance*, published in the *Duxbury Clipper*

razor clams, sea clams, oysters, mussels, and scallops, “shellfish harvesting” includes the taking of sea worms and eels as well as shellfish.)

**b Recreational Shellfish Permits**

Quahogs, mussels, soft shell clams, and razors are limited to some specified volume per week per family.

Harvest seasons vary over the year. For example, at present soft-shell clams may only be taken during April, May, September, and October on Wednesdays and Saturdays. The season may be extended in years when the size of the resource is enough to warrant it. The 2004 season for soft-shell clams was extended through the end of January 2005, on Wednesdays only.

No shellfish taken on a recreational permit may be sold or bartered in any way.

#### **4.4 Harbor and Bay Preservation and Restoration Efforts**

The shellfish resource is a highly valued but exhaustible resource within Duxbury. The Town shellfish regulations recognize this. The town shellfish regulations and their interpretation and enforcement by the Shellfish Warden and the Selectmen therefore constitute an ongoing effort to preserve this resource.

Shellfish harvesting has been considered to be self-regulating. That is, when the number of legal size shellfish diminishes to a point when fishermen no longer consider the yield worth the effort, they either stop or move to another location. Shellfish abundance varies each year. Harvesting pressure on adult shellfish is one reason, but predation by gulls, crabs, and some species of fish impose constant pressure on shellfish populations. Annual recruitment of shellfish (the setting of juveniles) can vary dramatically over interannual periods. The reasons for this variability are typically beyond management control, though in some cases the establishment of spawning sanctuaries has been practiced to increase local recruitment rates. The recent expansion of shellfish aquaculture in the bay has dramatically increased the number of spawning adult oysters and clams. Though it is difficult to quantify, there is certainly a positive increase of wild shellfish setting in the bay as a result of aquaculture practices.

Preservation and restoration of the resource has been the goal of the transplanting and relay programs that have been implemented through the years. These efforts have included:

- In 1973, 50,000 seed quahogs and 10,000 oyster seed were purchased and planted;
- In 1983, 80,000 seed quahog were purchased and cultured, and 200 bushels of contaminated oysters were planted on closed areas for their spawn;
- In 1985, 100 bushels of clean oysters from the Cape were relayed and broadcast; and,
- From 1986-1990, 200 to 250 bushels of contaminated oysters were imported from the Cape and from Mattapoisett, allowed to cleanse for 30 to 60 days, and opened to recreational harvesting.

The relay of contaminated oysters had to be abandoned after 1990 because of disease in other areas which could have spread into Duxbury Bay.

In 1997, the town purchased 70,000 oyster seed, which were grown on trays within the Bay and sown them into the bay during 1998. The outcome of that effort is unknown. An active program of propagation may require additional commitment of Town staff and resources. It is unclear

whether the return to the Town justifies increased expenditure.

A citizen's water quality monitoring program was initiated by the DBMC in 2007 and a new task force has been formed to compile historical and existing monitoring information from outside entities (e.g., the MA DMF, MA DEP, UMASS, Center for Coastal Studies, Jones River Ecology Program).

#### 4.5 Management Recommendations for Shellfishing

There has been considerable fluctuation in the abundance and yield of most Duxbury Bay shellfish species over the years. This is partly due to natural causes but there are management measures which should be undertaken to prevent avoidable damage to the health and productivity of the shellfish resource; these include:

##### a Implement the Management Recommendations for Preserving Water Quality

Shellfish harvesting cannot be sustained without adequate water quality. The water quality management recommendations in Section 3.8 should be implemented and considered part of the shellfish recommendations as well. These recommendations include a proactive approach to resolving water quality problems that may contribute to periodic closures of shellfish beds.

##### b Establish a Map of Shellfish Areas

It should be determined whether certain activities that may pose risks to shellfish populations and human consumption should be managed more closely.

#### 4.6 Aquaculture

##### a Existing Conditions and Trends

###### i Physical Condition

Aquaculture in Duxbury Bay primarily consists of oyster farming, in which an individual secures the right to grow oysters over a specific area of bay bottom, consisting of up to three acres. To secure an aquaculture grant, an individual, who must be a Duxbury resident, must pick a site and submit an application to the Selectmen with a fee of \$100 and a description of how the farm will be operated. After a public hearing and an inspection of the site by the Division of Marine Fisheries, the individual may be granted permission to conduct aquaculture on the site for a period of three years.

Oysters are grown to marketable size on trays or in mesh bags, and are placed on the flats for the final stage of growth. It takes approximately two years to grow an oyster to marketable size (3" longest shell diameter), and the number of bags must constantly be increased and the mesh size increased as the oysters grow, a very labor-intensive process. Some farmers grow other species, such as bay scallops and clams.

###### ii Current Bay Management

Aquaculture is regulated by Massachusetts General Laws Chapter 130, Section 57. This state law simply authorizes cities and towns to grant shellfish aquaculture licenses, subject to certification by the Director of the Division of Marine Fisheries, if the project will cause no substantial adverse effect on shellfish or other natural resources, which they do if it is determined that existing shellfish are below a density of 1 quahog/ft<sup>2</sup>, 3 soft-shell clams/ft<sup>2</sup>, or 3 oysters/ft<sup>2</sup> at the site.

Aquaculture facilities are regarded as Category II (reporting) under the U.S. Army Corps of

Engineers Programmatic General Permit for Massachusetts, which requires written approval from the Corps.

The Town of Duxbury regulates aquaculture under by-law Section 7.3, Shellfish Aquaculture Grant Program. Applicants must also file a Request for Determination of Applicability with the Duxbury Conservation Commission.

b Management Recommendations for Aquaculture

On January 12, 2009 the Duxbury Board of Selectmen approved and accepted the Duxbury Aquaculture Management Plan<sup>13</sup>. This plan was developed and written by members of the DBMC, Duxbury Shellfish Advisory Committee, Duxbury Agriculture Commission, and the Duxbury Shellfish Growers Association. The purpose of this document is to:

- Provide recommendations that will promote the success and sustainability of the shellfish aquaculture industry;
- Explore and identify current and future shellfish aquaculture industry needs;
- Explore and provide recommendations on the basis for determining the area and/or number of aquaculture leases that can be supported in Duxbury waters.
- Explore any required modifications to the existing licensing and administration process and provide recommendations as appropriate.

Recommendations provided within that plan are summarized in Section II at the beginning of that document.

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<sup>13</sup> [http://www.town.duxbury.ma.us/Public\\_Documents/DuxburyMA\\_DuxBayCommMin/plan/damp.pdf](http://www.town.duxbury.ma.us/Public_Documents/DuxburyMA_DuxBayCommMin/plan/damp.pdf)

## **Chapter 5      BOATING SAFETY AND NAVIGATION**

### **5.1      Existing Conditions and Trends**

Recreational boating is one of the most popular uses of Duxbury Bay. The level of boating activity in the bay has steadily increased in recent year as demonstrated by the increased number of moorings documented in the chapter on Moorings, the founding of the Duxbury Bay Maritime School (DBMS) in 1998, the growth of sailing programs at the Duxbury Yacht Club, and the use of dry vessel storage at Bayside Marine, and by private boaters using public boat ramp facilities.

The Duxbury Harbor Master office moved forward and implemented several recommendations initially published in the 2005 Duxbury Bay Management Plan to improve boating bay safety and navigation in mixed use and high traffic bay areas. Regulatory and/or marker buoys have been deployed to identify swimming areas at Shipyard Lane and Landing Road beaches, and additional buoys posting no wake zones are now positioned near popular mooring fields. The Harbormaster office has worked with aquaculture grantees to more consistently mark and identify the locations of the use of cages resulting in increased navigation awareness.

In general, the increased commercial fishing vessel use of the Bay appears to have had little effect on boating, bay use and safety in the Bay. The connection between the availability of commercial launch service and the number of vessels in the Bay may need to be examined in the future if mooring growth continues based on the rate of the last 10 years.

While mixed use Bay boating activities have increased, the Bay remains safe and relatively harmonious. Continued growth in boating activity and the emergence of new types of personal watercraft and amphibious boats may pose new challenges to maintain a balance between the needs of the overall management of the Bay and small and large recreational vessels. For example, the large tidal range of the bay relative to its depth leaves extensive areas of exposed flats intersected by narrow channels at low tide, often resulting in more challenging navigation in congested narrow channels during peak summer boating periods. The Bay Management Plan goal is to encourage all users of the Bay to improve skills and to seek on going boating education and training that promotes safe navigation and equally important boating etiquette.

The following topics have been identified as areas of ongoing interest relative to boating safety and uses:

- a Excessive vessel speed and operation in close proximity to swimming areas;
- b Excessive vessel speed and wake impacts on other vessel types (power boat wakes impact power, sail and rowing craft);
- c Proliferation of moorings impinging on dock and float access, impediments to navigation, re-routing of vessels into areas of heavy vessel traffic in the inner harbor, and constriction of fairways to Snug Harbor docking facilities;
- d Communicating information on newly designated buoys marking aquaculture cages and the impact on boating navigation
- e Sail training locations and special sail regattas that impact other boating activities.

### **5.2      Current Bay Management**

At this time, there is no “plan” for the management of boats on Duxbury Bay. Instead boating activity is controlled through laws and regulations at the federal, state, and local levels. Existing regulations are aimed principally at boating safety.

### a Federal Management

The United States Coast Guard is the principal federal agency that regulates boating. Boating is regulated through safety standards that govern the construction of boats and require safety equipment. The Coast Guard also has rules that control navigation. Coast Guard regulations are authorized by the Federal Boat Safety Act of 1971.

Federal safety standards found in Title 46 of the Code of Federal Regulations set standards for construction, safe loading, flotation, fuel systems, navigation lights, the use of personal flotation devices etc.

Coast Guard Navigational Rules govern the operation of vessels and define which vessel has right of way when two vessels meet. There are specific rules applicable to sail boats and “vessels under oars.”

### b State Management

Massachusetts laws and regulations require the titling and registration of boats, regulate certain aspects of boating, and grant harbor masters authority over boat operators.

Chapter 90B requires that boats equipped with motors be registered with the state. Details concerning boat titling and registration are spelled out at Title 323 of the Code of Massachusetts Regulations (CMR). Chapter 40B also requires motorboats to be equipped with specified safety equipment and prohibits the operation of any vessel while under the influence of liquor or drugs. Detailed regulations addressing vessel safety equipment and operation appear at Title 323 of CMR. For example, 323 CMR 2.07 prohibits motor boats from being operated “within 150 feet of shoreline which is being used as a swimming area.” 323 CMR 2.08 requires that children who are at least 12 but under 16 years of age must complete a safe boating course and obtain a certificate before operating a motorboat. Chapter 90B, section 12 empowers harbor masters to enforce state regulations applicable to boating safety and operations. Additional harbor master authority comes from Chapters 91 and 102. Some authority is outdated. For example, under Chapter 102, section 23, the harbor master is authorized to order any vessel lying in a harbor to “cockbill the lower yards, brace the topsail yards fore and aft and rig in the jib-boom.”

### c Local Management

Boating is regulated at the local level by the Duxbury Harbor Rules and regulations that are enforced by that department. The regulations are adopted and may be amended by the Board of Selectmen. Although the Town has had a Waterfront Advisory Committee for many years, that committee has been inactive in recent years.

Harbormaster regulations govern the location and use of moorings and use of the Town pier and floats. The rules also establish reduced speed zones and define areas where water skiing is prohibited. The Harbormaster has authority to levy non-criminal penalties for violations.

The following areas have speed limitations: The Snug Harbor basin, the main channel south to Nun #18, Bluefish River, the Cove Street landing, Howland’s Landing, and Powder Point Bridge.

Water skiing is prohibited in these areas: Within 150 feet of bathers and bathing beaches, Bluefish River, the Back Marsh, the Cove Street landing, Howland’s Landing, and the Snug Harbor basin and the main channel south to Nun #18.

## 5.3 Management Recommendations for Boating and Safety

Boating safety is best achieved through Harbor Master visibility and the encouragement of all bay users to take personnel responsibility to seek education and training to improve navigation safety and etiquette on the water. The Bay Management Plan recommends the use of numerous websites that promote boating safety by explaining the rules of the road, the use of

running lights, aids to navigation, boating safety for kids, as well as the Coast Guard License tests and Massachusetts State requirements for junior boating operators.

Boating safety on the bay is further enhanced by programs at the Duxbury Bay Maritime School (DBMS) and the Duxbury Yacht Club (DYC). Each summer, DBMS provides sailing instruction for about 600 children and 175 adults and the DYC provides a junior sailing program for about 100 children. DBMS also offers a one-day class for women called “Women Underway” and classes in the winter on boating safety and coastal navigation. In addition, DBMS provides a series of safe boating classes for children that include the test for the certificate required by the state for children under the age of 16.

The following recommendations are presented for minimizing the impacts of boating on other boats and users of the Bay:

- a Encourage all boating Bay users to take personnel responsibility and to seek education to improve boating safety, operation, navigation and on the water etiquette.
- b Establish temporary alternative storage areas and locations for storm moorings and monitor and maintain Storm Management Plan relating capacity and facilities for safe vessel hauling.
- c While improvements have been made to utilize aquaculture buoys to identify cage structures and mark grant boundaries, more can be done to ensure consistency in the markings as well as educating the boating community on the buoys designation.
- d Explore the need to develop recommendations and define a methodology to address water craft conflicts of Bay use.
- e The charting of sensitive areas should be considered and displayed for public education access points to the bay.

## 5.4 Amphibious Rigid Inflatable Boats

The recent introduction of amphibious vessels in Duxbury presents a new challenge to managing the bay resource. These vessels are capable of navigating from water to land over virtually terrain.

Because essentially all of the tidal flats and shallow waters in Duxbury Bay are protected resource areas due to potential shellfishing and vegetation status and also because these vessels are unusually capable of damage to protected ground and vegetation, it is essential to the protection of the Bay resource that the use and impact of these vessels be monitored and managed.

The DBMC convened an ad hoc subcommittee to consider whether new regulations should be established to prevent adverse impact but decided that development of a Best Practices document coupled to an education program would be a useful first step.

The subcommittee recommends owners of these vessels be informed of the following best practices:

1. Aquaculture & Shellfish Avoidance
  - a. Before engaging in operations in ‘legs-down’ mode, make yourself familiar with the aquaculture locations provided in attachment (A).
  - b. Do not operate in ‘legs-down’ mode in aquaculture lease areas during periods of low tide.



- c. If in doubt about a possible aquaculture lease area, contact the Harbormaster or Conservation Administrator at the numbers below.
- d. Make yourself familiar with shellfish suitability areas as shown in attachment (A). Please note these maps are simply a guide; there may be additional shellfish resource areas.

## 2. Submerged Aquatic Vegetation (SAV) – Present and Historic Eelgrass

- a. Do not operate in 'legs-down' mode within 25 feet of known or visible SAV.
- b. When operating with 'legs-up', always monitor the water for signs of possible SAV in shallow depths. If you notice signs of SAV, trim the outboard up and transit away from the respective location.
- c. Make yourself familiar with known eel grass beds in attachment (A) for avoidance.
- d. Make yourself familiar with the visible characteristics of eel grass as shown in attachment

## 3. Private Property

Numerous sections of private property extend to mean low water during low tide, so it is imperative that all Sealegs owners be prudent when operating in 'legs-down' mode on exposed flats and beaches.

- a. If you own the waterfront property, be sure to operate directly in front of your respective location and avoid transit on adjacent property, until you're in 'legs-up' mode and the hull is fully displaced by the water.
- b. If you do not own the property, be sure to obtain written permission for land operations from the property owner before transiting in 'legs-down' mode.

## 4. Entry In and Out of the Water and Operation

- a. Driving Into the Water – upon entry into the water, shift to 'legs-up' mode as soon as you're able to obtain propulsion from the outboard.
- b. Driving Out of the Water – upon exiting the water, remain in 'legs-up' mode until sole outboard propulsion is no longer practical and would likely result in grounding.
- c. Minimize use on areas above mean high water, unless you are transiting on your respective property.
- d. Do not operate over any areas showing signs of vegetation, including eel grass and saltmarsh.
- e. Avoid transiting parallel along the shoreline to minimize impacts to sensitive wildlife (e.g., tern, plover, and horseshoe crab nesting sites).

## 5. Salt Marsh

- a. Do not traverse salt marsh in 'legs-down' mode at any time. Direct transit will result in compaction of the salt marsh platform and destruction of this important habitat. See picture below of salt marsh.

# Chapter 6      **MOORINGS**

## 6.1      Definitions

Moorings have been identified by the Duxbury Bay Management Study Committee and by

participants in several public workshops as an important topic for this Plan. Duxbury's *Rules and Regulations for Moorings, Permits and Waiting Lists*, section 7.6 § 2002, DEFINITIONS, defines "mooring" as the equipment and/or process prescribed by the Town of Duxbury used to temporarily secure a vessel to the bed of a body of water by means other than anchoring. Moorings in Duxbury waters are permitted by the Duxbury Harbormaster's Office under 310 CMR 9.07 (1) and (2) Annual Permits for Moorings, Floats and Rafts, and under local bylaw 7.6, *Rules and Regulations for Moorings, Permits and Waiting Lists*.

## 6.2 Existing Trends & Conditions

Moorings are traditionally grouped into areas with deep water and in other areas that provide public parking and access. The vast majority of moorings are located off of the Town Pier in the Snug Harbor area and at other town landings. These access points offer limited parking for boaters' cars and trailers and very limited storage areas for dinghies. The present size, location, and condition of these landside access points serve as a natural control of the number of vessels that can conveniently access the Bay.

The dramatic growth of boating activities, including recreational and competitive sailing, recreational and commercial power boating, and recreational and competitive rowing, underscores the need for more proactive management of the Bay, moorings, and landside support facilities.

If the current trend of increasing the quantity of active moorings continues, pressure to provide more vehicle and trailer parking, launch services, dinghy storage and dock space will also increase. Further, vehicular traffic congestion will increase, particularly in Mattakeesett Court and along Washington Street.

Current public perception is that traffic congestion and parking shortages already create problems during periods of high use. These issues must be considered as a fundamental component of the comprehensive management plan.

To date, there have been no specific studies to examine the environmental impact of enlarged mooring fields and increases in boat traffic. However, it is reasonable to suspect that the environmental problems affecting the Bay are linked in part to increases in boating activity.

The deterioration of water quality can be traced back to sewage discharge, excessive nutrient levels, hydrocarbons, and other toxic emissions. Fecal coliform counts increase along with increased vessel use when appropriate measures are not implemented to avoid direct discharge into Bay waters. Further, the physical deterioration of habitat results in displacement, disruption and mortality of essential organisms in all phases of life. One does not need to look far to see that the overall healthy balance in the Bay is a delicate proposition. The most visible marine ecological health indicators lie in the eelgrass beds and shellfish beds. Unfortunately, the eelgrass beds continue to shrink in size, and shellfish beds experience periodic closures.

A specific evaluation of the impacts of Bay usage is difficult due to a lack of monitoring of relevant ecological health indicators. However, enough is known about the general ecological impacts of boating to plan to limit those impacts as much as possible.

This section of the Duxbury Bay Management Plan addresses the impacts caused by moorings and vessels at rest on the moorings. The Boating section addresses potential impacts from increasing boat traffic from a variety of sources throughout the Bay.

## 6.3 General Summary of Mooring Field Areas

- a **Basin Deep Water:** The anchorage was engineered originally for 176 deep water mooring locations. Since August 22, 2002, 166 are being used, and the remaining

10 have been voided due to location conflicts.

Basin deep water moorings are located at Mattakeeset Court / Snug Harbor. The waiting list for deep water moorings is estimated to be 20 years. Deep water moorings cost \$8.00 per foot.

- b **Basin Flats:** The anchorage has 432 mooring locations on the flats. As of February 4, 2010, 25 were not yet assigned.

Basin flats moorings are located at Mattakeeset Court / Snug Harbor. Currently, there is no waiting list for these moorings. Basin flats moorings cost \$8.00 per foot.

- c **Howland's Landing:** There are 86 deep water moorings at Howland's Landing. As of February 4, 2010, all 86 moorings were being assigned.

Howland's Landing is a deep water anchorage that also has a waiting list. The waiting list was established in the spring of 2002 and currently has 45 names on it. The cost for moorings at Howland's Landing is \$7.00 per foot.

- d **Tidal Flats:** As of February 4, 2010, there were 220 tidal flats moorings registered with the Harbormaster Department.

Tidal flats moorings are located all over the Town's waterways, including the Back River, King Caesar Beach, Bluefish River, Winsor St., Water St., Josselyn Ave., Shipyard Lane, Eagle's Nest, Standish Shore, Bay Road, and Clark's Island. The cost for a tidal flats mooring \$25.00 flat rate.

- e **Two Rock:** There are 35 deep water moorings a Two Rock. As of February 4, 2010, 34 moorings were assigned.

Two Rock is a deep water anchorage that has a waiting list. The waiting list currently has 34 names on it. The cost for a mooring at Two Rock is \$7.00 per foot.

- f **North of Clark's Island:** There are 20 deep water moorings at Clarks Island. As of February 4, 2010, 20 were assigned.

Clarks Island is a deep water anchorage that has a waiting list. The waiting list currently has 27 names on it. The cost for a mooring at Clark's Island is \$7.00 per foot.

The Town received \$134,893.00 in mooring fees in 2009. The fee structure in the basin and basin flats is \$8.00 per foot. Howland's Landings, Two Rock, and Clark's Island is \$7.00 per foot. Tidal

**Table 6-1 Total Number of Moorings Assigned**

|                         | 2000       | 2001       | 2002             | 2003             | 2005       | 2006       | 2007       | 2008       | 2009       |
|-------------------------|------------|------------|------------------|------------------|------------|------------|------------|------------|------------|
| Basin Deep Water        | 165        | 165        | 166 <sup>a</sup> | 171 <sup>a</sup> | 166        | 166        | 166        | 166        | 166        |
| Basin Flats             | 398        | 424        | 381 <sup>b</sup> | 401              | 362        | 426        | 405        | 424        | 406        |
| Howland's Landing       | 85         | 85         | 86               | 86               | 74         | 84         | 80         | 80         | 68         |
| Tidal Flats             | 97         | 125        | 166              | 130              | 148        | 214        | 225        | 226        | 220        |
| Two Rock                |            |            |                  | 35               | 32         | 32         | 30         | 31         | 32         |
| North of Clark's Island |            |            |                  | 20               | 20         | 18         | 21         | 22         | 20         |
| <b>Total Moorings</b>   | <b>745</b> | <b>799</b> | <b>799</b>       | <b>858</b>       | <b>802</b> | <b>940</b> | <b>927</b> | <b>949</b> | <b>912</b> |

Source: Harbormaster's Department, February 4, 2010. These figures were compiled using the best available data from the Harbormaster's Department.

Flat moorings are a \$25.00 flat rate. The Town supports marine sewage pump out with a dockside pump and a pump-out boat. The current capacity for sewage is a 2000 gallon holding tank.

The number of new moorings and the size of boat the mooring fields will accommodate are dictated by the characteristically shallow water depths of the Bay. There is little room for expansion of "deep water" moorings in locations reasonably (although the definition of reasonable is subjective) reached by launch without further dredging of the Basin area. The U.S. Army Corps of Engineers was authorized in 1836 to conduct its first "improvement" dredging project. This project was completed in 1908 and consisted of a -6 MLW channel to South Duxbury Wharfs. In 1945 the Corps was authorized to expand this project to a 100 ft. wide channel -8 deep from the Two Rock Channel to the Town Wharf, where a 21-acre anchorage basin -8 ft. deep was dredged. Thus the area of this deep water anchorage at Snug Harbor (referred to as the Yacht Basin in Town Regulations) was established, and it has changed little over the years.

## 6.4 Landside Infrastructure

The Harbormaster's Office estimates that the current wait period for a deep water mooring is 20 years. There is no wait period for flats moorings. Local marina owners agree that the number of boats, their size, and the demand for mooring space is growing. The demand for moorings is placing increasing pressure on the harbormaster to increase the number of moorings. As stated above, there are concerns that an increase in the number of moorings in the Bay could damage natural resources and add to boating congestion. In addition, although local marinas feel they can service more moorings, they also admit they would not be able to provide emergency services for more vessels outside of their regular vessel customer lists.

The direct impacts from moorings would be those resulting from the physical placement and use of the moorings. Direct ecological impacts can include displacement of benthic organisms of

recreational and commercial value (shellfish) and impacts to the food chains that are built on benthic invertebrates and the animals that depend on them, especially migratory shorebirds. Direct impacts can also include mortality and behavioral impacts from toxic chemicals and heavy metals from internal combustion engines and antifouling paints and cleansers.

The indirect impacts from moorings would be those resulting from the use of moored vessels throughout the Bay.

## 6.5 Current Bay Management

### a Permitting Regulations and Structure

The number of mooring permits issued is determined by each local harbormaster in accordance with the following regulations:

- i Town of Duxbury Rules and Regulations For Mooring Permits and Waiting Lists, amended April 2010, which provide for, among other things, the requirements for all mooring permits issued by the Town of Duxbury, the mooring application process, mooring waiting list policies and procedures, minimum mooring requirements, guest moorings procedures, authorized mooring services and enforcement policies.
- ii Section 10A of M.G.L. Ch. 91 and its regulations, 310 CMR Section 9.00 (the Waterways Regulations) at section 9.07 Activities Subject to Annual
- iii Permit and 9.07 Section (1) General, states: "A written application for an annual permit must be submitted to the harbormaster of a city or town . . . , for the placement on a temporary basis of moorings, floats or rafts held by bottom-anchor, and ramps associated thereto, which are located within the territorial jurisdiction of the municipality. Within 9.07 there are additional stipulations regarding transfer of moorings, wait list and assignment requirements, etc. Under these regulations moorings cannot be limited to residents and residents cannot receive preferential treatment in mooring assignment." 31 CMR Section 9.07(1). These regulations allow transfer of mooring permits to immediate family. 31 CMR Section 9.07(2) (b) 2.d. However, this regulation contradicts Army Corps, *Open To All On Equal Terms Policy*, which only allows transfer of a mooring to a spouse.
- iv Rivers and Harbors Act of 1899, Section 10, Department of the Army Programmatic General Permit Commonwealth of Massachusetts, there are three levels of review depending on the mooring ownership, use, and location. Category I, non-reporting, applies to private, non-commercial, non-rental single-boat moorings authorized by the local harbormaster (not associated with any boating facility, not located in a Federal Navigation Project other than a Federal Anchorage, and not located in vegetated shallows). Category II, reporting, applies to moorings that do not meet the terms of Category I. Individual permits are required for moorings within the horizontal limits or with moored vessels that extend within the limits of a Federal Navigation Project, except those in Federal Anchorages authorized under Category I. Transfer of mooring permits is only allowed to a spouse under the *Open To All On Equal Terms Policy*. The Corps has various guidance documents for the location of moorings in relation to

Federal Projects. For private structures, including moorings, this means a set-back of three times the depth of the project unless the project is an anchorage or the mooring is municipally owned and complies with the open to all policy.

- v In general, individual moorings are permitted by the local harbormaster on an annual basis, although a renewal process is allowed. Municipal, commercial, or privately owned mooring fields must be approved by the harbormaster and the Army Corps of Engineers.

## 6.6 Harbor and Bay Preservation and Restoration Efforts

There are no formal efforts to manage moorings as a method to preserve and restore the Bay. The shallowness of the Bay, however, serves as a natural barrier to expanding deep water moorings while also providing the opportunity to expand inter-tidal flats moorings.

The Harbormaster's Department presently uses the following criteria to evaluate expanding the number and/or location of moorings in the Bay:

- a Public Safety
- b Riparian Issues – Ch.91- inter-tidal owner can object to the harbormaster approving a mooring in the inter-tidal zone. The harbormaster has letters from people who don't want moorings within 1,000 lf of MHW.
- c Congestion
- d Navigational Conflicts
- e Access and Use Conflicts – Property owners have mooring rights off their property if they can produce a deed to the harbormaster. Thirty-eight miles of shoreline are 99% privately owned. Many people object to moorings on their property (rights of inter-tidal ownership).
- f The Town of Duxbury's Storm Management Plan
- g Environmental Issues-Areas of critical marine habitat

## 6.7 Existing Engineering Plans

The Harbormaster's Department possesses and uses a plan (Figure 4) for basin deep water and basin flats designed by Amory Engineers (rev. 9/22/99).

## 6.8 Management Recommendations

The DBMC's Mooring Subcommittee is making the following management recommendations:

- a Prior to the creation of any additional designated anchorages or any increase in the number of authorized moorings for existing designated anchorages, the Harbormaster should consult with the DBMC, which should provide notice and an opportunity for public comment on the proposal.
- b The Harbormaster and the DBMC should evaluate all mooring field changes in the last 10 years and any future proposed changes
- c Encourage the Harbormaster to adopt regulations requiring that the review of all mooring applications consider the impact of proposed moorings on existing and historical eelgrass beds, aquaculture licenses, and natural shellfish populations.

- d Encourage the Harbormaster to evaluate the future use of helical moorings and other advances in mooring technology with Authorized Mooring Services.
- e Encourage the Harbormaster to evaluate a reconfiguration of the Basin mooring grid to minimize shoaling problems from gradual silting between dredging.
- f Charge the Duxbury Bay Management Commission with exploring its taking coordination responsibility for all public access to the bay
- g The Harbormaster should adopt regulations formalizing existing limits on the number of moorings in designated anchorages and adopting limits for any designated anchorages that lack mooring limits.
- h Prior to the creation of any additional designated anchorages or any increase in the number of authorized moorings for existing designated anchorages, the Harbormaster should consult with the DBMC, which should provide notice and an opportunity for public comment on the proposal.
- i The Harbormaster and the DBMC should evaluate all mooring field changes in the last 10 years and any future proposed changes in terms of impacts to:
  - i Public Safety
  - ii Ecology and Sensitive receptors
  - iii Existing Uses
  - iv Appropriateness of proposed use
  - v Levels of public service; and,
  - vi Landside infrastructure.
- j In order to better protect Duxbury Bay's resources and formalize existing practices, the Harbormaster should adopt regulations requiring that the review of all mooring applications consider the impact of proposed moorings on existing and historical eelgrass beds, aquaculture licenses, and natural shellfish populations.
- k Because mooring technology has progressed in recent years and will continue to develop, the Harbormaster should work with Authorized Mooring Services to evaluate the future use of helical moorings and other advances in mooring technology.
- l Because of tidal currents and the shoaling that occurs from northeast to southwest in the Snug Harbor Deep Water Basin, problems arise as the result of conflicts between moored boats. Accordingly, the Harbormaster should evaluate a reconfiguration of the Basin mooring grid to eliminate these problems. Any changes should be ready for implementation at the conclusion of the next dredging of the basin.





## Chapter 7 BAY ACCESS

### 7.1 Background

Historically, public access to Duxbury Bay was critical to the survival of early settlers who relied on the bay's abundant supply of shellfish and finfish. The bay also played an important role in transportation because it was a well-protected water body with direct access between Plymouth and eventually Green's Harbor and Marshfield through the Cut River. Public access for survival in colonial times gave way to access needed for shipbuilding and commercial fishing in the 1800s. In 1840, Ezra Weston ("King Caesar") of Duxbury was considered the "largest ship owner in America."

Today, Duxbury Bay is primarily used for a wide range of recreational activity from strolling along a beach, to bird watching, clamming, sailing, fishing, hunting, wind surfing, and water skiing. Commercial fishing and shellfishing exists on a moderate scale and aquaculture has developed into a major business. The number of users who want or need access to the bay continues to grow concurrently with the increasing population and affluence of residents.

A number of factors limit public access, among them shoreline structures, limited publicly owned shoreline and access points, and sensitive ecological receptors. Publicly owned "ways to the water" and town landings are residents' primary means of access to the bay. The perceived problems with these points of access include inappropriate use, lack of space to accommodate parking, lack of maintenance, encroachment by private abutters, significant numbers of dinghies along the shore, and lack of education and public awareness regarding public ways to the water.

The value of public access to the Bay was confirmed in a 2003 planning workshop held by the Duxbury Bay Management Study Committee when this topic received the highest number of votes as a topic to be explored in the Plan. The focus of bay access in this draft plan is thus based on feedback received from that public workshop. The topics discussed include:

- Rights to bay access;
- Scenic and visual access to the bay;
- Boat ramps/town landings/ways to the water;
- Parking related to bay access; and,
- Management recommendations.

Over the years, the demand for public access to Duxbury bay has changed both in character and intensity, from its original use for sustenance, to shipping and offshore fishing, to ship building to its present use for recreational boating. The bay's commercial use for fishing has also diminished, however, aquaculture has increased significantly. .

With South Shore population growth among the highest in the state, residents and visitors alike are looking for access to Duxbury Bay. Evidence of the increase in demand for recreation in the bay can be found in the increase in boating activity, including the increase in flats moorings (those outside of the harbor in the basin flats) of 186 percent since 1980. The number of deep-water moorings located in the dredged areas of the harbor and in the deep water at Two Rock (Two Rock is approximately 1 mile on a line from the Town Pier to the north end of Clark's Island, heading 132.72° SE) has only increased 10 percent since 1980 due to lack of bay depth during low tides. The increase in basin flats moorings and in the deep-water moorings at Two Rock (and in an area north of Clark's Island) has been made possible by and resulted in a seasonal full-time launch service with two vessels. The launch service has made these distant moorings accessible, which in turn has contributed to the overall increase in vessels in the bay. (See Chapter 6.0 Moorings, for a complete discussion of moorings.)

An increase in private facilities for boating is also evidence of the increasing demand for access to the bay. The popular and award-winning Duxbury Bay Maritime School, with 126 sailboats and 37 instructor boats, and 29 rowing shells and the increase in active rack storage of boats at Bayside Marine to 301 vessels since 1973 are examples of this increased demand. The Duxbury Yacht Club has also increased its fleet.

The increase in the number of permit requests to construct private piers has paralleled Duxbury's increase in population. In 2001, Duxbury's Annual Town Meeting approved the appointment of the Pier Access and Shoreline Study Committee (PASS). The committee focused on public safety and visual access impacts of pier construction. This committee's work led to the passage of a revised Zoning By-law regarding piers as well as the creation of a Waterfront Scenic Area in 2004.

At a time when public and private demand for access to the bay is increasing, some private property owners may not be aware of their responsibilities regarding public trust rights that allow others passage on their property. There is a perception that the growing tendency among private property owners is to limit public passage. Owners cite their desire for privacy, fears of liability, and loss of property rights as reasons for not allowing passage in areas that also provide public access. The following trends could impact public access to the bay:

Bay property owners who may not understand their obligations to preserve the public trust rights of access;

- Property owners who object to public moorings in tidelands;
- Lack of maintenance of town landings (overgrown vegetation that threatens visual access);
- Inappropriate use of town landings;
- Lack of space to accommodate demand at town landings;
- Lack of publicly owned shoreline and access points (about 10 percent of the total number of access points are public or non-profit owned<sup>14</sup>);
- Increasing private land development along the waterfront;
- Private encroachment on public access; and,
- Lack of landside facilities to accommodate demand, especially parking.
- Private use of landings for dinghy storage.
- Increased use of landings by the aquaculture industry.

Today, public access remains highly valued for recreational and commercial use and is protected for Commonwealth citizens regardless of residency, race, religion, sex, age disability or other illegal distinction under Massachusetts General Law c. 91.<sup>15</sup>

The historical public ownership of intertidal and subtidal coastal land dates back to long before the Massachusetts Colonial Ordinances of 1641-1647<sup>16</sup>. This legal tradition recognizes the fundamental rights and interests of all citizens to the sea and the shore. In addition, it establishes the government as the trustee of this public interest with a duty to conserve and restore these resources and the rights of the public to use them.

In colonial times, the value of these resources was not only critical to the economy but to the very survival of the colonists. By the 1630s the colonies were cash poor and sought a way to stimulate marine trade and coastal development. To encourage the construction of wharves

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<sup>14</sup> Bay Management Committee, *PASS Proposed Bylaw Presentation*, Duxbury, 2003.

<sup>15</sup> 310 CMR 9.07(1)General

<sup>16</sup> Department of Environmental Protection, *Regulations to Implement MGL Chapter 91*, Boston, June 1990.

and development of maritime commerce the Massachusetts Bay Colony gave upland owners property rights to the tidal flats, with easements for the general public for fishing, fowling and navigation. This legacy of private ownership to low water survives in the United States only in Maine and Massachusetts, both of which once belonged to the same colony.

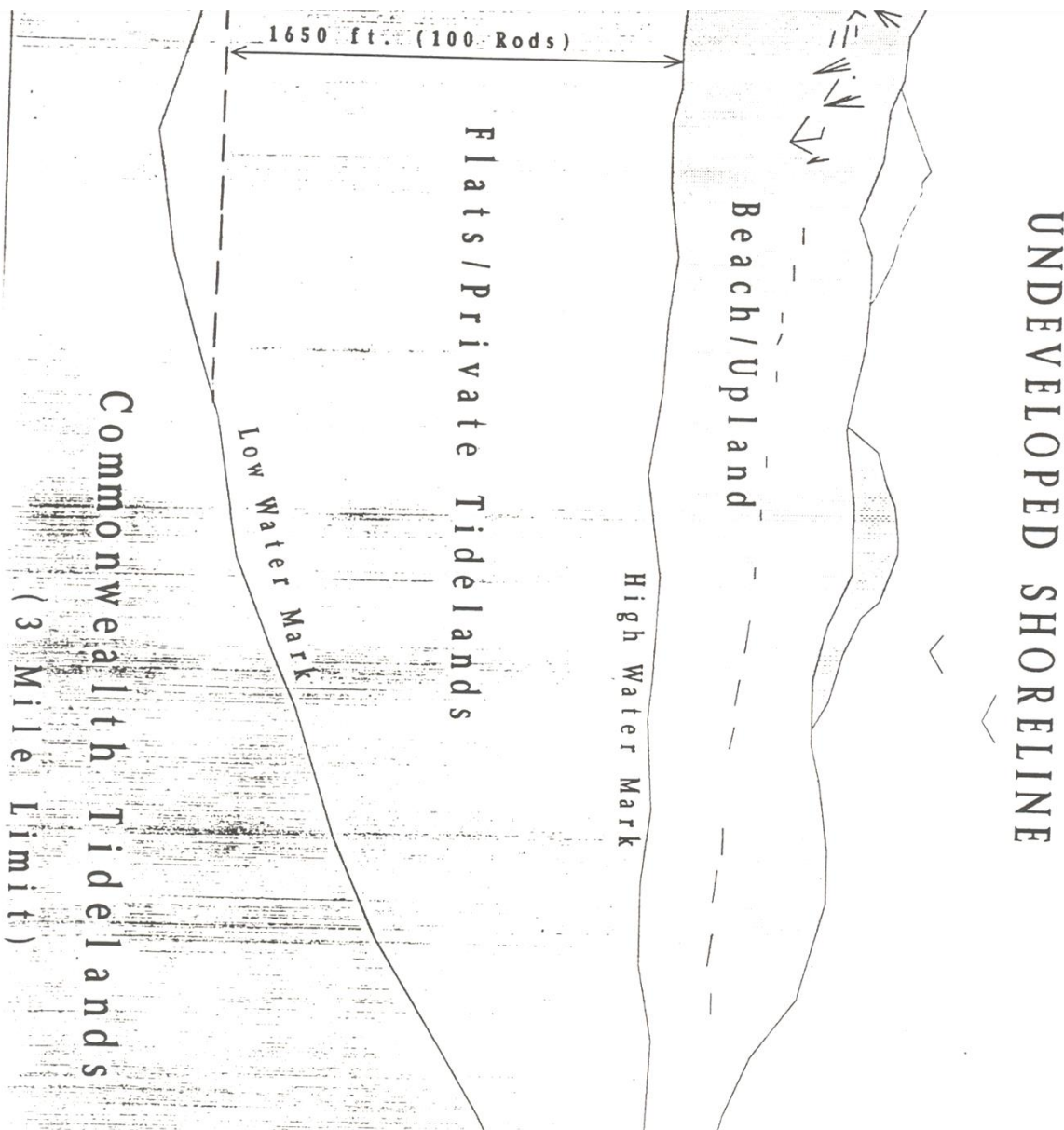
In 1866, the State of Massachusetts carried on the tradition of preserving the public's access rights on land between the historic high and historic low water lines (private tidelands) and lands extending from the historic low water line to the seaward 3 mile limit of State jurisdiction (submerged lands) by passing the Waterways Act, M.G.L. ch. 91. This Act preserves the public's rights in the intertidal area and submerged lands while allowing private ownership of land above historic low water. Landowners in Massachusetts own to the historic low water mark, unless their deeded property bounds exclude tidal flats. Because of this presumed private ownership the area between historic low and high water lines is called private tidelands in the State regulations, 310 CMR 9.00. By law, private tidelands extend on flats exposed at low water for a maximum distance of 1,650 ft. (100 rods) from the historic high water mark. Exposed flats beyond 1,650 ft. seaward considered tidelands that are publicly owned. Ch.91's jurisdiction also extends to filled (developed) tidelands from the existing mean high water line to the historic high waterline, a minimum distance of 250 ft. from the mean high waterline and thereafter to the first public way or the historic high waterline, whichever comes first (see graphic). On filled (developed) tidelands, Ch.91 preserves the public's rights for fishing, fowling and navigation by requiring setbacks and access rights for the general public.

Property owners who have bottom-anchored moorings above the low water mark are subject to harbormaster approval under 310 CMR 9.07 (1) and (2) (Figure 5). Other moorings in private tidelands are subject to property owner and harbormaster approval. Private owners retain their property rights to the low water mark by openly allowing public use through posting or other written or official notice.

In general, private property owners have rights as owners but also obligations to allow fishing, fowling, and navigation. That means that their activities cannot inhibit "fishing, fowling, or navigation" by the general public. All structures in the area between the historic high water mark and the historic low water mark must be approved through the regulatory process, which ensures public rights of passage for protected purposes. All structures such as piers, seawalls, and revetments, must be built to accommodate the public's ability to walk and therefore fish, fowl and navigate in the zone.

To allow public passage, Ch.91 regulations require 5 feet of clearance between the ground and the first structural member of a dock, pier, or walkway at low tide. Alternatively, the property owner can propose through the permitting process stairs up and over or around the structure to ensure public passage. In order to protect access, Ch.91 also requires that piers be set back a minimum of 25 ft. from property bounds, and in general they must not impact navigation to existing water-dependent structures or channels. In addition, Ch.91 often requires signage notifying the public of their passage rights at such private structures.

**Figure 5. Summary of Private Property Regulations**



## 7.2 Existing Conditions and Trends

### a Waterfront Access Points to Duxbury Bay

The majority of Duxbury Bay's 22 miles of shoreline is privately owned. There are 296 waterfront parcels, including streets. These parcels represent slightly less than 6 percent of the estimated 5,000 parcels in the town. Of the 95 waterfront access points, 16 are town landings, 10 are ways to the water, and 14 are non-profit access points. Duxbury Beach (which is leased to the Town of Duxbury by the not-for-profit Duxbury Beach Reservation, Inc.) offers over 4 miles of shoreline, but much of it is inaccessible without a four-wheel drive vehicle. The town sells both resident and non-resident permits but limits the number of vehicles allowed on the beach at a given time (250 resident and 250 non-resident vehicles). Access to Duxbury Beach is among the topics addressed in the Beach Management Plan and the Orders of Condition contained therein. The Duxbury Bay Management plan focuses on the town's boat ramps, town landings, and ways to the water.

Several of the town's access points are not well known to the public, offer no parking spaces, and are not regularly maintained. The physical condition of public access points, including town landings and ways to the water, vary depending on use and demand for maintenance. In some cases, private use may encroach upon public property and access rights due to inadequate boundary marking. Parking is inadequate at some locations. Overgrown vegetation at some locations obstructs public view sheds to the bay. All of these factors challenge the public's right to access the bay.

A number of factors could limit public access to the bay at boat ramps, town landings, and ways to the water. These include shoreline structures, private encroachment on public access points, limited publicly owned shoreline and access points, limited education of the public concerning access issues, lack of regular maintenance, limited parking, and sensitive ecological receptors. A fundamental limitation to use of boat ramps is the tidal cycle of the Bay. Just as for all other uses of the bay, use of the ramps ebbs and flows with the tides. The nearly complete tidal flushing of the bay allows vessel access for a few hours around each high tide. In this way the bay naturally regulates its use.

### b Town Landings and Ways to the Water

- i *Town landings* are defined by Protective Bylaw Article 300, Section 302 as "designated areas to which the Town has an undoubted right, which have been surveyed and recorded with the Plymouth County Registry of Deeds to the low water mark (mean)" and including, in north to south orientation along the shoreline, these include:

1. Ford Stand Landing;
2. Old Cove Landing;
3. Drew Salt works Landing;
4. Simeon Soule's Landing;
5. Peterson's Landing;
6. Powder Point Bridge (at the west end on both north and south sides);
7. Anchorage Lane Landing;
8. Bluefish River Landing;
9. Mattakeeset Town Pier;
10. Winsor Street Landing;
11. Water Street Landing;
12. Jocelyn Lane Landing;

- 13. Harden Hill Road Landing;
- 14. Landing Road Landing; and
- 15. Hicks Point Road Landing.

- ii *Ways to the water* are defined by Protective Bylaw Article 300, Section 302 as “designated areas to which the Town has rights as public ways to the water, by gift, or otherwise which may or may not be restricted as to their usage and which have not been specifically laid out and surveyed to the low water mark and recorded as a Town Landings.” Such ways and include, in north to south orientation along the shoreline, (1) Shipyard Lane Ellison Beach, (2) Elder Brewster Road, (3) Samoset Road, (4) Sagamore Road, (5) Massasoit Road, (6) Miles Standish Home Site, (7) Longview Road, (8) Elderberry Lane, and (9) Bay Farm.” Included in this right are roads properly laid out, improved and extended to the water by funds raised and appropriated by the Town and designated as public highways. Island Creek Pond (10) was not included in the by-law, but it is a way to an anadromous fish run, a resource dependent on the Bay and therefore it is in the Duxbury Bay Management Plan.

All town landings and ways to the water are open to parking for residents and nonresidents unless specifically restricted to residents by the Board of Selectmen.<sup>17</sup> The only restrictions on parking are the time restrictions at Mattakeeset Court, Myles Standish Cellar Hole, Island Creek Pond, and Elder Brewster Rd. The resident-only restriction applies at Shipyard Lane (Ellison Beach), and there is a fire lane parking restriction at Massasoit Road. Shipyard Lane has conflicts between swimming and boats, particularly sailboats with limited maneuverability.

Encroachment of private landowners exists to varying degrees at town landings and ways to the water. In some cases it is severe, giving the appearance that the access is private property; in other cases, the encroachment is caused by invasive vegetation. At some access points, private abutters assist the town in caring for access points.

The Town’s Department of Public Works (DPW) maintains the town landings and ways to the water. Maintenance consists of some trash removal and vegetation control. Although maintenance is routine at some landings, at others, maintenance occurs at the request of residents. The DPW keeps a record of this maintenance. As a result, all town landings and ways to the water have been maintained at one time or another. Abutting property owners and neighbors appear to want maintenance but not parking.

#### c Boat Ramps and Boat Launch Areas

There are eleven landings and ways to the water where boats can be launched. Ten of these boat launch areas are to the bay itself, and one—Shipyard Lane—is used seasonally. There is also a launch area at Island Creek Pond, which is included here because it provides access to a bay resource, a herring run. The vast majority of these areas consist of sand or gravel approaches to the water, and over the decades there has been little improvement to them. Only Mattakeeset Court has been significantly modified to accommodate boat launching.

Increased demand for the use of the Mattakeeset boat ramp has created increased demand for parking and apparent conflicts between some users including commercial and recreational users. State and Federal funding for this public ramp, as well as related pier and dredging projects, is partially based on commercial use of these facilities. However, the ramps and

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<sup>17</sup> Tom Daly, Director, Department of Public Works (DPW), Duxbury Bay Management Study Committee information meeting, March, 25, 2004.

access points that were predominantly used by commercial fishermen are now used more often by recreational users. This shift from commercial to recreational use is evident at Peterson's Landing (Bumpus).

d Vehicle Parking at Waterfront Access Points

There is limited parking at the waterfront access points to Duxbury Bay. The resident parking lots at Duxbury Beach offer the most vehicle spaces, with 388 spaces. Residents must purchase a resident parking permit from the town to use these lots. (Oversand parking on Duxbury Beach is covered in the Duxbury Beach Management Plan.) There are an additional 50 spaces in the parking lot at the west end of the Powder Point Bridge. These spaces are available to both residents and nonresidents and do not require a parking permit from the town. Mattakeeset Court has 65 car spaces. Additional parking occurs along Washington Street adjacent to Mattakeeset Court. None of these spaces require parking permits, although there are time limitations on some of the spaces at Mattakeeset Court. Completion of site development at the Duxbury Bay Maritime School adjacent to Mattakeeset Court has added 75 parking spaces to the Snug Harbor area. While these spaces are not available to the public, they have eliminated the need for DBMS students and staff to use Mattakeeset Court.

There are 27 spaces available at Shipyard Lane for residents only (town permit required) and 16 spaces at Howland's Landing. The only access point where parking is restricted to residents is Shipyard Lane. There are 10 handicapped access parking spaces among the designated parking spaces at the major waterfront access points.

Parking is also available at other landings as follows (those with no specific number have 3 or fewer): Bluefish River 8, Cove Street 5, Anchorage Lane 0, Drew Salt Works (Bay Pond Road) 1, Simeon Soule's Landing 2, Clark Peterson's Landing 5, Ford's Stand (Ocean Road North), Hick's Point, Landing Road, Hardin Hill 5, Josselyn Avenue, Winsor Street, and Water Street. The number of spaces at Bluefish River has recently been expanded to support the extensive recreational use of the area. Other public ways to the water with parking spaces include Bay Farm with about 25 spaces, and Miles Standish Cellar Hole (time restriction), Samoset Road, Sagamore Road, Elder Brewster Road (time restriction), and Massasoit Road, all with minimal parking (less than 2 or 3 spaces).

Only three (four if Shipyard Lane is included) of the ten boat ramps offer designated parking. There are a total of 180 vehicle parking spaces and 12 boat trailer spaces at these three places. Additional trailer parking occurs along Washington Street north of Mattakeeset Court. The remaining six boat ramps have no designated parking and also have no restrictions on parking, including residency requirements.

Table 11 identifies the town landings and ways to the water, including their location, whether they are also boat ramps, the availability of parking, and other information related to their accessibility and use.



**Table 7-1 Summary of Landings and Ways to the Water**

| Public Landing             | Resident Only      | Over Sand | Car Spaces | Time Restricted Car Space | Handicapped Car Spaces | Trailer Parking |
|----------------------------|--------------------|-----------|------------|---------------------------|------------------------|-----------------|
| Mattakeeset Ct.            |                    |           | 65         | 13                        | 4                      | 8               |
| Shipyards Lane             | Yes (not enforced) |           | 27         |                           | 2                      |                 |
| Howland's Landing          |                    |           | 18         |                           |                        | 4               |
| Powder Point               |                    |           | 50         |                           | 2                      |                 |
| Washington St.             |                    |           | 30         |                           |                        | 4               |
| Duxbury Beach <sup>a</sup> |                    | 575       | 388        |                           | 8                      |                 |
| Total                      |                    | 575       | 185        | 13                        | 10                     | 16              |

<sup>a</sup>There are 500 front road and 75 back road vehicles allowed

### 7.3 Scenic Access to Duxbury Bay

Duxbury's Open Space and Recreation Plan describes the Town's valuable bay views:

New England coastal towns are known worldwide for their scenic qualities. Duxbury is among the many South Shore towns which value their historic buildings, coastal character and scenic roadside views (Refer to Resource Map 6. Scenic Views of the Open Space Plan). Water views of Kingston Bay and Duxbury Bay are visible from many local streets, particularly Standish, Marshall, Crescent, Washington, Harrison, St. George Street, Bay and King Caesar Roads, and Powder Point Avenue. Many other smaller residential roads which connect to these feeder streets also offer exceptional harbor views. Mattakeeset Court, which leads to the Town Pier and the Duxbury Yacht Club, offers views of the boat basin and lively harbor activity.

The Pier Access and Shoreline Study (PAAS) Committee, appointed in 2001, confirmed the importance of scenic views of the bay to residents. While focusing on the visual impacts of new pier construction and pier repairs, the PASS findings identified six areas where the scenic view is affected by piers, "most notably the Blue Fish River/King Caesar Road area." The PASS Committee proposed, and after amendment, received the approval of the 2004 Annual Town Meeting to create a Waterfront Scenic Area (WSA) overlay district in the town's zoning by-law. The by-law defines the WSA district as "the abutting waterfront land viewed from a public road, along which there is an open, unobstructed view of the ocean, harbor, bay or estuary." The WSA district is further defined with a map approved by town meeting. The committee also proposed and received approval for a by-law change that governs pier construction in the overlay district and throughout Duxbury Bay.

The Town's Zoning By-law regarding piers addresses conflicts between public and private rights regarding pier construction and the public's right for visual access. Private piers require approval by the Conservation Commission and a special permit from the Zoning Board of Appeal (ZBA) approval. Requirements set forth at section 404.20 of the Zoning By-law include the following:

- 200 foot maximum length

- 4 foot maximum width
- Piles may not extend more than 2 feet beyond the edge of marsh
- Maximum float size of 200 square feet
- Setback of 25 feet from property sideline
- Setback of 50 feet from Town landing sideline.

Section 404.40 of the By-law encourages the use of shared piers and Section 404.50 imposes more restrictive specifications within the WSA.

Section 23.6 of the Duxbury Wetlands Protection Bylaw also addresses pier construction. That section provides that the Conservation Commission may allow, at its discretion, a pier (dock or walkway) no greater than 4 feet wide at any point; 200 feet long; a platform no greater than 6 feet by 8 feet including the walkway; and only one float no greater than 10 feet by 20 feet, if the Conservation Commission is satisfied that mitigation required in its order of conditions is sufficient to protect the resource areas at issue. Section 23.6(4) provides that piers shall not adversely affect the following: shoreline movement of sediments, shellfish habitat, fisheries and fish habitat, water quality, nor shall it obstruct shellfishing or obstruct the reserved public rights of fishing, fowling, navigation, or passage, or significantly destroy a public view or degrade aesthetic value.

The Open Space and Recreation Plan, the Comprehensive Plan, and the Pier Access and Shoreline Study Committee have all cited preserving visual access to the bay as being important to residents. The Duxbury Bay Management Study Committee also acknowledges that access to the scenic beauty of the bay may be the preferred way for some residents to enjoy the bay, and therefore preserving visual access is an important part of the Duxbury Bay Management Plan. In light of development trends cited earlier and encroaching vegetation, preserving visual access to the bay becomes more critical but difficult.

## 7.4 Current Bay Management

Currently, public/private access rights to Duxbury Bay are managed through the Duxbury Bay Management Commission, local by-laws and regulations and state and federal laws and regulations. These regulations also protect the functions of various wetlands, including coastal wetlands. By protecting the wetlands these laws protect the natural resources that motivate people to seek access to the Bay. In addition, local and state laws protect access and use rights on tidal flats where both private and public access occur; the Bay below low water is owned by the Commonwealth for all citizens. The regulations protecting wetland resources are based on assumptions regarding impacts on the wetlands and public uses from specific activities. The federal interests in regulating activities in navigable waterways and wetlands include general environmental health, wetlands, fisheries, flood hazards, navigation, recreation, water quality, economics, aesthetics, and land use. In addition, the harbor master under Mass General Law c.91, S10A approves annual permits for bottom-anchored floats in the interests of public access, navigation and public safety.

The approval of a by-law creating the Duxbury Bay Management Commission (DBMC) in 2005 established a new approach to managing Duxbury Bay. The DBMC consists of nine members appointed by the Board of Selectmen and serving three year staggered terms. Members must have an interest in the bay, including but not limited to aquaculture, boating, commercial and not for profit waterfront users, ecology, shellfish and finfish or have experience that enhances the diversity of the Commission. The Commission is not a regulatory body, but instead serves in an advisory role for the BOS and to other Bay related committees and groups.

The DBMC mission is to finalize, implement and maintain a management plan that will enhance

and preserve the ecological health, pristine waters, and natural beauty of Duxbury Bay for future generations while sustaining harmony among all its uses. A draft plan was published in June 2005 and updated in 2011.

The Commission evaluates and reviews any proposed change in Bay use and access, inclusive of structures, and all Bay related issues requiring the action of the Board of Selectman (BOS), Duxbury Conservation Commission (DCC), Zoning Board of Appeals (ZBA), Planning Board (PB), Harbormasters Office and Board of Health (BOH) involving Coastal Resource Areas as defined in M.G.L. c.131 S40 and the Town of Duxbury Wetlands Bylaw (Chapter 9 General By-Laws of the Town of Duxbury). The DBMC plays an advisory role and does not have regulatory powers.

Significant accomplishments by the DBMC include the following:

- Completion of the Duxbury Aquaculture Management Plan and approval of the plan by the Board of Selectmen. The report was prepared by a committee consisting of representatives of the DBMC, Shellfish Advisory Committees, Agriculture Commission, and the aquaculture industry. The plan recommended that the existing moratorium on new licenses be continued, and that licensees with less than three acres be permitted to seek approval for expansion to three acres. The plan also recommended that water quality monitoring continue, that a boaters guide to aquaculture be prepared, and that shellfish growers work with the Duxbury Shellfish Advisory Committee to explore winter storage for aquaculture equipment.
- Finalization of a Duxbury Storm Management Plan setting forth procedures to be implemented to haul boats in the event of a major storm.
- Initiated a study of issues relating to moorings in Duxbury Bay and worked with the Harbor master to update mooring regulations.
- Secured funding for the design and construction of a new fish ladder to allow herring to return to Island Creek Pond. The first returning herring are expected in the spring of 2011.
- Completion of the inaugural "State of the Bay" report. The report is intended to track data on water quality, boating activity and other recreational uses, fish and wildlife populations, invasive species and eel grass. In time, the reports will be used to identify trends and measure the health of the bay.
- Provided comments a variety of projects with the potential to affect the bay. These included several piers, proposed aquaculture license expansions (exempt from the moratorium), and a proposal to pump sea water from the bay.
- Secured a loan of water quality monitoring equipment from the federal EPA.

The Commission meets on the second and fourth Thursdays of each month. Meetings are at the Senior Center and begin at 7:00 PM.

## 7.5 Regulations for Waterfront Access Points: Town landings, Ways to the Water, Boat Ramps

### a State and Federal Regulations

State and Federal Regulation concerning waterfront access include the following:

- i Massachusetts General Law (M.G.L.) c. 131, S40, The Wetland Protection Act and its regulations at 310 CMR 10. This law and its regulations protect the natural functions of various coastal wetlands.
- ii M.G.L. c. 91, The Waterways Act, and its regulations at 310 CMR 9 and S.10A

provides for harbor master approval of annual permits for bottom-anchored floats. It is also the primary law and regulations to protect the public trust rights of fishing, fowling and navigation in tidelands out to the seaward limit of State jurisdiction, 3 miles.

- iii M.G.L. c. 111 s.17, Mass Environmental Policy Act, and its regulations at 310 CMR 15 (triggered if piers are >2,000 SF). This Act and its regulations ensure that necessary information is provided to regulatory agencies so they can evaluate a project's impacts on coastal wetlands and the public's rights of access.
  - iv M.G.L. c. 90B, and Regulations 323 CMR 1, 2 and 4, Dept. Fisheries Wildlife and Environmental Law Enforcement. These are the States boating laws which govern registration and use.
  - v M.G.L. c. 131A, the Massachusetts Endangered species Act and its regulations at 321 CMR 10.00 particularly as it pertains to public access and protecting endangered species such as the piping plover.
  - vi Federal Endangered Species Act S. 9 enforced by the U.S. Fish & Wildlife Service, specifically related to vehicle access in areas where Piping Plovers receive protection.
  - vii Federal Coastal Zone Management Act and its regulations at 301 CMR 20, 21 and 15 CFR Part 930 (if pier project exceeds MEPA threshold trigger). The purpose of these regulations is to ensure that federal projects or projects requiring a federal permit beyond a certain size are consistent with Mass Coastal Zone Management policies which both protect coastal wetlands and public use and access.
  - viii The Federal Rivers and Harbors Act, s. 9 & 10 structures, and its regulations at 33 CFR Parts 320-330 enforced by the U.S. Army Corps of Engineers. The state of Massachusetts is regulated under a Programmatic General Permit, with three levels of review for moorings, pile-supported structures, and floats. These levels of review are Category I, non-reporting; Category II, reporting-screening; and Category III, Individual Permit (see Appendix for Definition of Categories). The goal of these regulations is again to protect wetlands resources and public access.
- b Duxbury Town By-Laws and Regulations
- Town By-Laws and Regulations include the following:
- i Ch. 9 Wetlands Protection (Duxbury Wetlands Bylaw) protects all wetland areas mirroring 310 CMR 10 with local changes. Of particular interest is s. 9.6.0 Piers;
  - ii Protective Bylaw of the Town of Duxbury, Article 300, S. 302 Definitions, Waterfront Scenic Area (WSA), Article 404 Wetland Protection District Sections 404.10 through 404.40 (pier construction) and Article 900 S. 906.2 Special Permits, sub-section 3; and,
  - iii Protective Bylaw Article 300, Section 302.
- c Harbor master/Coastal Natural Resources Department, Town of Duxbury

The Harbormaster receives his authority from the following:

- i Municipal: Board of Selectmen Manual Section 7.7 Beach Rules and Regulations, S. 7.3,4 and 5 Shellfish Rules and Regulations, S. 7.6 Harbor Rules and Regulations; and,
- ii Mass General Laws: Ch. 102, 90, 90b, 131, 88, 130.

## 7.6 Regulations for Scenic Access

### a State and Federal Regulations

Scenic and visual access is often preserved as a result of historic and open space preservation, as well as preservation of agricultural and horticultural use.

- i Rivers and Harbors Act, Clean Water Act, 33 Code of Federal Regulations (CFR) Parts 320-330, U.S. Army Corps of Engineers requires project proponents receive sign-offs from the Mass Historic Preservation Officer and Tribal authorities under its Programmatic General and individual permits. This requirement can lead to the preservation of scenic and visual access;
- ii Historic Preservation Act, S. 106, projects must comply with this act to receive U.S. Army Corps of Engineers approval;
- iii National and State Registers of Historic Places, protects historic buildings, districts and sites;
- iv Open Space and Recreation Plan Requirements, Executive Office of Environmental Affairs, Division of Conservation Services;
- v M.G.L. Chapters 61, 61A or 61B, Chapter 61 is land under forest management. Chapter 61A applies to agricultural or horticultural land. Chapter 61B applies to land designated for recreational use; and,
- vi Conservation restrictions. Property owners place deed restrictions on land for scenic, open, or natural values; administered by the Plymouth Conservation District.

The Massachusetts Office of Coastal Zone Management produced a handbook, *Preserving Historic Rights of way to the Sea*, which can assist towns in maintaining and acquiring physical access, which can also result in preserving visual access.

### b Duxbury Town By-Laws and Regulations

- i Ch. 9 Wetlands Protection (Duxbury Wetlands By-Law) protects all wetland areas mirroring 310 CMR 10 with local changes. Of particular interest is s. 9.6.0 Piers. The Conservation Commission reviews new pier construction to protect wetland resources but also to protect public rights of fishing, fowling, navigation, passage and public views and aesthetic values;
- ii Protective Bylaw of the Town of Duxbury, Zoning Board, Design Review;
- iii Community Preservation Act (state legislation adopted by the Town of Duxbury in 2002), in which funds raised through a special tax of 3 percent are matched by

the state for projects, implemented by the Town Community Preservation Committee, provides funds for community historic, recreation and conservation purposes. The Committee provided \$70,000 for the restoration of the Island Creek rainbow smelt and herring fish run; and,

- iv Protective Bylaw of the Town of Duxbury, Article 300, S. 302 Definitions, Waterfront Scenic Area (WSA), Article 404 Wetland Protection District Sections 404.10 through 404.40 (pier construction) and Article 900 S. 906.2 Special Permits, sub-section 3.
- c Harbormaster/Coastal Natural Resources Department, Town of Duxbury
  - i Municipal: Board of Selectmen Manual Section 7.7 Beach Rules and Regulations, S. 7.3, 4 and 5 Shellfish Rules and Regulations, S. 7.6 Harbor Rules and Regulations
  - ii Mass General Laws: Ch. 102, 90, 90b, 131, 88, 130.

## 7.7 Recommendations for Bay Access

Population growth in Duxbury far exceeded that of the state from 1960 to 1980. This growth was attributed to the completion of Rt.3. The *1999 Duxbury Comprehensive Plan* projected that dwelling units could reach 10,082 units which could mean an 80% increase in population with the completion of the restoration of the Old Colony Railroad. The growth in town and the increase in use of Duxbury Bay suggest that additional management steps may need to be taken to preserve public access. While the Duxbury Bay Management Commission seeks to preserve existing uses on the bay, it also recognizes that improving access could lead to ecological impacts and impacts to existing uses. Therefore, any changes in access must take into account these potential impacts and be evaluated through the proposed comprehensive review process described below. The committee thus recommends that for the near future improvements to access points be limited to those that improve users' experience but do not increase the number of users until an analysis of the impacts of access improvements on ecology and existing use can be conducted.

The preliminary management recommendations outlined in this chapter are intended to preserve and enhance existing public access opportunities wherever there is a demand for such access and when the change is feasible and consistent with the mission of this plan. The management plan proposes a process for evaluating the consistency of proposed changes in bay access that is consistent with the plan's mission statement. This process is based on the assumption that changes in access, new access facilities and structures, or new services will have ecological and access-specific impacts. The evaluation process should review bay access changes based on an agreed-upon set of criteria. These criteria should include but not be limited to impacts to:

- Public safety, health and welfare;
- Ecology and sensitive receptors including wildlife;
- Uses and appropriateness of use and access;
- Scenic views and public aesthetic issues;
- Levels of public and private services; and,
- Supporting landside infrastructure

All of these criteria affect public access. (See Chapter 9.0, Bay Management) for further discussion of this evaluation process.)

Town By-laws do provide opportunities for public review of impacts on public access. For

example, the redevelopment of the Bayside Marine site, was subject to extensive special permit hearings before the ZBA, as well as review and approval by the Conservation Commission. In addition to expanding rack storage and thus decreasing the demand for moorings, the Bayside Marine project provided additional off-street parking and state-of-the-art storm water management. The DBMS redevelopment, completed in 2009, was voluntarily subjected to the Planning Board's site plan review process and was reviewed and approved by the Conservation Commission. That project included 75 off-street parking spaces in the Snug Harbor area and eliminated the need for school employees and students to occupy spaces at Mattakeeset Court. The project also included modern storm water management. The DBMS project was also reviewed by the Duxbury Bay Management Commission. Although the Commission lacks regulatory authority, its review of projects helps to insure that issues such as bay access are taken into account.

The Duxbury Bay Management Commission will continue to work with the Board of Selectmen, Town Meeting, Conservation Commission, Town Planner, Massachusetts Division of Waterways, Massachusetts Coastal Access Program, Community Preservation Committee, private open-space organizations, property owners, and any other interested parties to implement the various actions and recommendations.

- a Recommendations: The Duxbury Bay Management Commission makes the following recommendations to address public access to the bay:
  - i Charge the Duxbury Bay Management Commission with exploring its taking coordination responsibility for all public access to the bay;
  - ii Identify and reduce any obstructions to public passage along the shoreline and in the intertidal area through regulation, enforcement, and voluntary incentives to property owners.
  - iii Conduct a shoreline access survey to inspect structures that have been identified as causing obstructions to public passage. The survey would be a starting point for discussions with owners regarding voluntary compliance with public access rights. If voluntary action is not forthcoming, Ch.91 licenses could be checked for compliance. For structures that are not in compliance with Ch. 91 regulations, the State Department of Environmental Protection should require appropriate mitigation, such as signage alerting the public to its right of passage for fishing, fowling, and navigation.
  - iv Increase public awareness of shoreline access rights and responsibilities using educational brochures and signage directed at property owners.
  - v Develop management guidelines and a program of regular maintenance based on the need for the bay's town landings and ways to the water to support existing uses.
  - vi Conduct parcel-by-parcel analysis to determine the exact extent of intertidal ownership within the bay, prioritizing to protect sensitive receptors, and examine the appropriateness of enhancing public access.
  - vii Study the impacts on ecology and existing uses of any proposed access improvements.
  - viii Conduct bay use survey.

- ix Assess any increases in demand for public services.
- x Investigate the need to improve transportation options and support facilities at public access points where demand is evident; include sanitation facilities.
- xi Analyze sufficiency of emergency vessel access throughout bay.
- xii Survey number of cars using access points.
- xiii Investigate installation of bike racks at town landings and ways to the water;
- xiv Provide education regarding access points for users and abutters.
- xv Assess appropriateness of recommendations for town landings and ways to the water from the 1975 Committee Report.

## 7.8 Recommendations for Scenic Access to the Bay

Population growth, overgrown vegetation, and encroachment on public lands all could impact the community's goal to preserve its character by impairing scenic views and visual access to the bay. Scenic views of the bay and access to them define community character to many community members. Therefore, managing the preservation of scenic views and visual access to the bay preserves town character. Also, the Comprehensive Plan survey has shown single-mindedness on the part of residents to maintain a "rural, wooded community dedicated to environmental protection." Protection of scenic views and access to them can be achieved permanently through acquisition of view-shed properties and semi-permanently through regulations and temporary use restrictions.

- a Recommendation: The Duxbury Bay Management Commission makes the following recommendations to protect and enhance public views of the bay after impact review, with the long-term goal of achieving permanent protection when possible:
  - i Coordinate with the Conservation Commission and Public Works Department to develop guidelines for maintaining vegetation on public lands along the shoreline so that invasive species and rampant vegetative growth do not block scenic vistas.
  - ii Identify incentives to encourage private property owners to manage vegetation so as to allow views from public roadways.
  - iii Educate property owners regarding the value of scenic access and view sheds to community character.
  - iv Explore innovative ways to permanently preserve visual access on private lands, including purchase of easements.
  - v Coordinate with the Planning Board to include in development reviews consideration of how developments alter water views from public ways and encourage owners and developers to modify site plans to enhance and protect views.
  - vi Support acquisition efforts by the Conservation Commission, the Community Preservation Act Committee, and non-profit organizations to preserve scenic



views of the bay and its estuaries.

## 7.9 Funding

Funding for research, compliance education, and coordination activities would enhance the DBMC's ability to review existing public access, possible impediments at access points, and enable the Commission to work with Town officials to eliminate illegal obstructions. Funds for education and other action items would be sought from various granting agencies and non-profits. In light of the current constraints on the Town budget, the prospect of funding for the DBMC seems problematic.

## Chapter 8 RESTORATION PROJECTS

Restoration of Duxbury Bay's ecology and freshwater systems was considered fundamental to the Duxbury Bay Management Study Committee, who finalized the original Bay Management plan in 2005. That plan outlined a program to assess and address all ecological threats to Duxbury Bay in conjunction with the overall mission to "enhance and preserve the ecological health, pristine waters and natural beauty of Duxbury Bay for future generations". The plan also envisioned the Bay Management Commission leading planning and oversight of these projects in conjunction with and funded by a variety of town, state and federal government and various private organizations.

Selection priority was to be given to those projects that:

- focus on environmental health indicators (eg. Shellfish and water quality)
- are highly visible to the public or provide multiple benefits
- result in a measurable improvement in the ecology

Several specific programs were either underway or recommended at that time. Following is a summary of the programs and progress made since that plan was established:

- Reopen shellfishing beds closed by excessive coliform bacteria
- Assess and address eelgrass depletion
- Assist with and coordinate existing restoration projects and organize storm drain prioritization plan
- Re-establish herring and rainbow smelt in Island Creek, Mill Pond and Island Creek Pond

This plan refocuses efforts toward completion of the projects already underway as well as initiating and prioritizing new projects with a pragmatic view based on available funding, resourcing and regulatory limitations.

### 8.1 Updated Projects and Restoration

#### a Flora and Fauna Restoration

##### i Shellfish Beds and Beaches

Shellfishing habitat is a prominent health indicator of the bay and it remains a goal of this plan to increase unconditionally open areas that are now either permanently or conditionally closed to shellfishing due to excessive fecal coliform bacteria. The table below summarizes these:

|                                                                                  |
|----------------------------------------------------------------------------------|
| <i>Areas closed to shellfishing</i>                                              |
| Long Point Marine                                                                |
| Kingston Bay                                                                     |
| <i>Conditional (dependent on rainfall events washing pollution into the Bay)</i> |
| Back River                                                                       |
| Blue Fish River                                                                  |
| Eagles Nest                                                                      |
| Bay Road                                                                         |

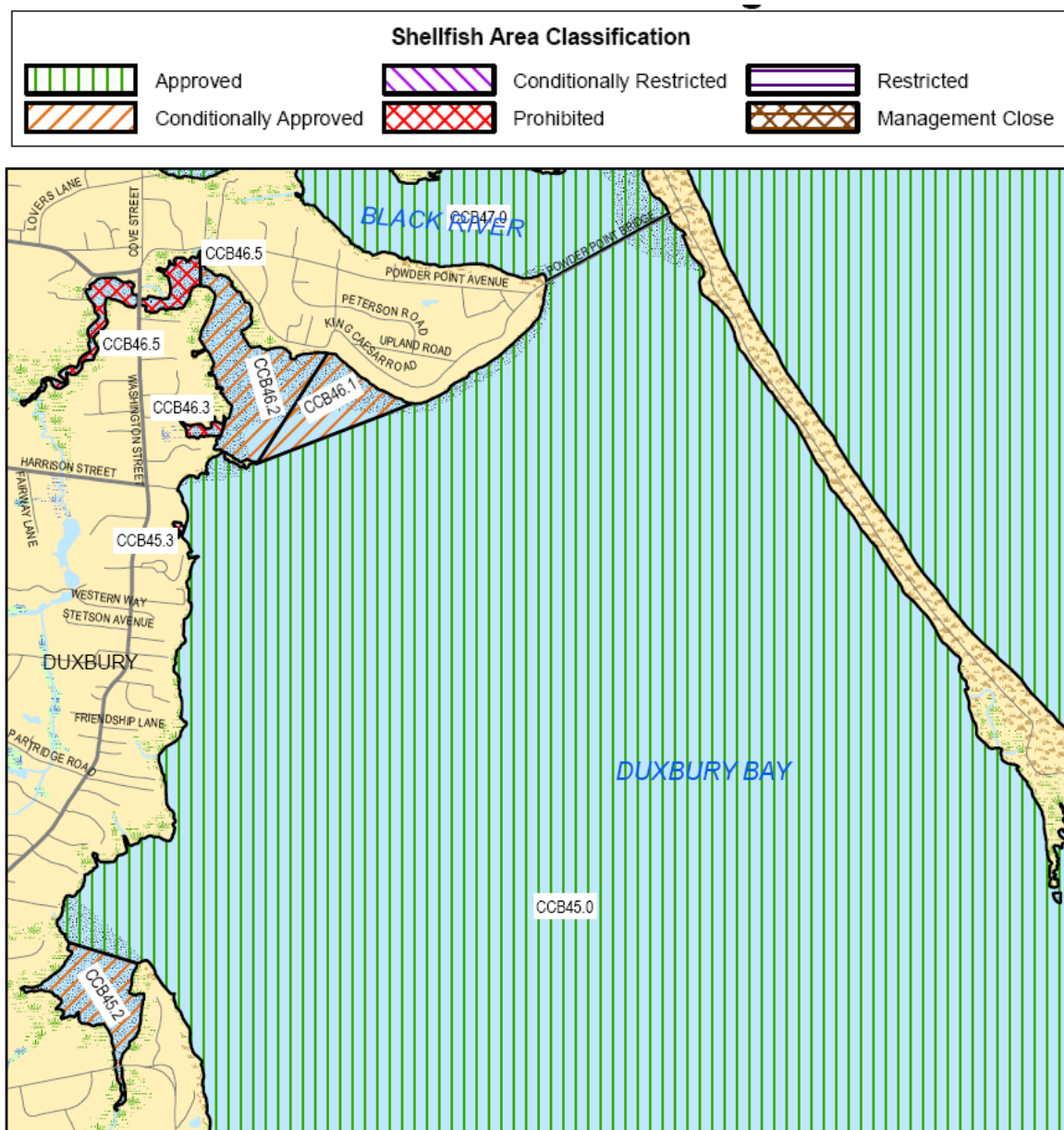
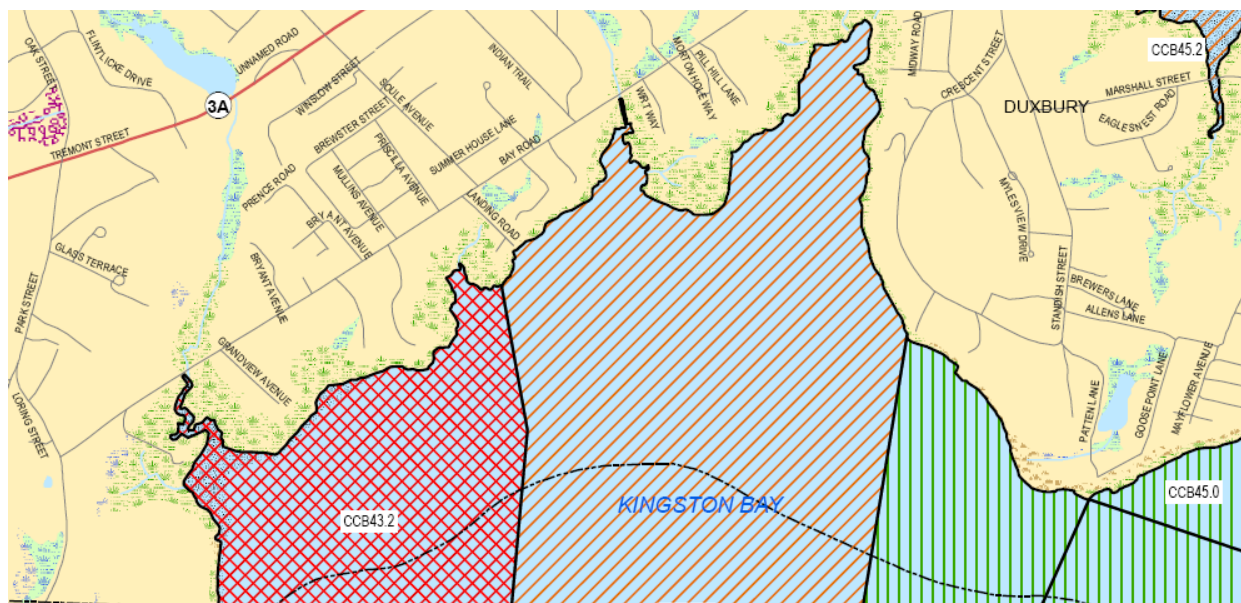


Figure 6 Restricted Shellfishing Areas in Duxbury Bay from MA DMF 2009



**Figure 7 Restricted Shellfishing Areas in Kingston Bay from MA DMF 2009**

There is significant evidence that excessive coliform levels are primarily due to stormwater runoff into the bay and subsequent contamination of the waters around the shellfishing beds. Accordingly, the problem may be addressed by focusing on stormwater drainage management using Best Practices developed and validated elsewhere.

The Duxbury Conservation Commission has been actively implementing a strategic plan to address stormwater drainage problems from areas draining into Kingston Bay. They monitored stormwater outfalls and mapped drainage areas from which they created a prioritized list of drainage projects. They have been applying for annual grants from CZM's coastal pollutant remediation fund to fund these priority projects.

Unfortunately, as a result of recent changes by DEP, Duxbury has been classified as an "urbanized watershed" and so is not a candidate for certain funding resources (e.g. 319, the federal stormwater grant program).

## ii Eelgrass Beds

Eelgrass grows only in shallow waters and is one of the most important habitats in Cape Cod Bay because it provides a refuge and nursery for juvenile fish and shellfish, and generally eelgrass beds support a higher abundance and diversity of marine life compared to surrounding un-vegetated areas. Many of these species are important to commercial and recreational interests in the area. Eelgrass is also important to resisting erosion caused by tides and normal currents.

There is significant evidence that eelgrass has been receding in our waters over the past several decades. This phenomenon is not fully understood but is believed to be caused by a number of factors:

- Winter ice formation scrubs the eelgrass beds during tidal motion, which is greater in Duxbury Bay than other nearby waters.
- Boating activity near low tide causes prop dredging that uproots eelgrass
- Wasting disease was recognized to have caused impact on the eel grass in Duxbury Bay and was probably exacerbated by elevated temperatures. (Fred Short- UNH) This

is not a current problem.

Water clarity is a primary stressor because it affects light penetration. Duxbury Bay is normally higher in turbidity than many embayments because of the predominance of salt marsh supplying organic sediment. The low residence time of our bay complicates this because of the resulting turbulence.

The Committee recommends establishment of an education effort to inform bay users of the importance of eelgrass beds, the impact to eelgrass from external factors, and what can be done to mitigate them (e.g. prop dredging, overuse of fertilizer).

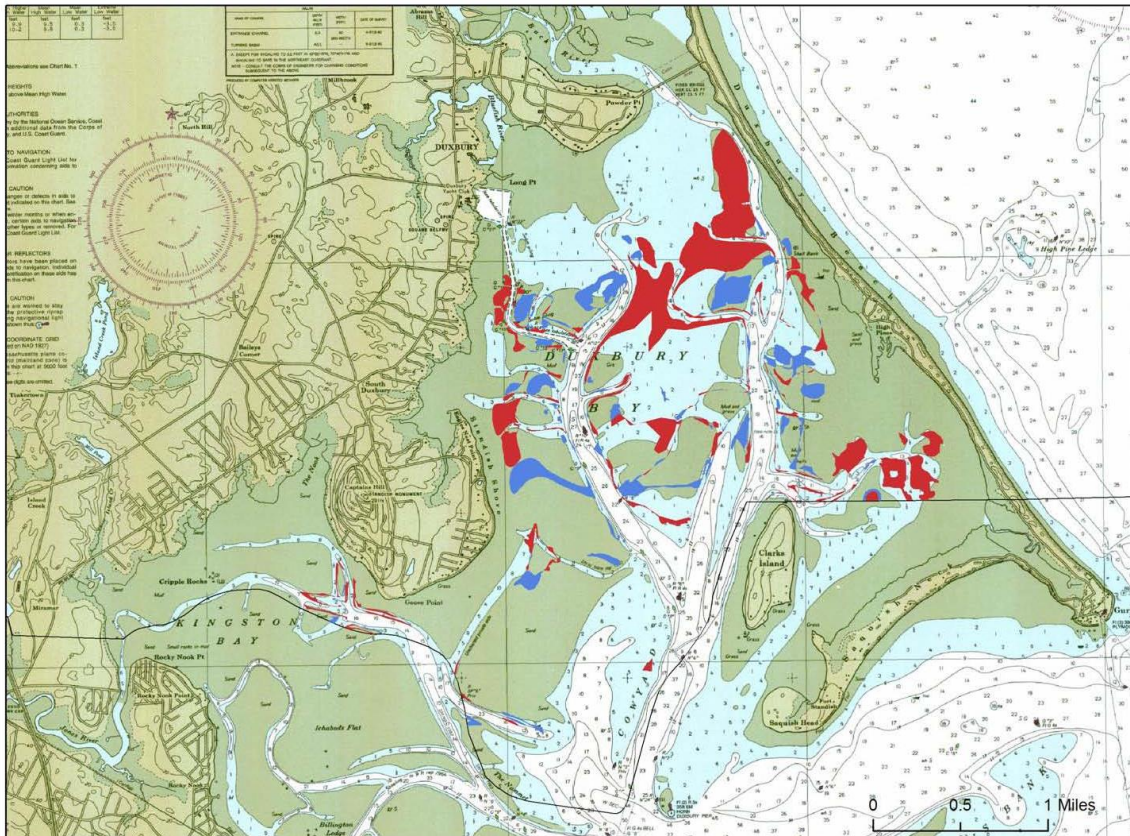


Figure 8 Eelgrass loss (red) and gain (blue) between 2001 and 2005



## b Water Quality

See Chapter 3.

## c Herring Run Restoration

Over the past five years, The Duxbury Bay Management Study Committee has organized and is now nearly finished with a demonstration project to restore the historic herring run in the Island Creek system.

Island Creek is the largest potential herring fishery in Duxbury. It consists of Island Creek (2 miles long), Mill Pond (7 acres) and Island Creek Pond (a great pond of 35 acres). The project involved cooperative contribution from the Duxbury Conservation Commission, DMF, USFWS, CWRP, The Gulf of Maine Council, Trout Unlimited, Fish America and the Duxbury Conservation Department and a private contribution from Battelle. Through this combined resource, over \$200,000 was raised for planning, construction and monitoring.

Over the two years following the receipt of these funds, planning and construction was accomplished to restore the original fish ladder at Tremont St. and construct a new control at Elm St. that allowed fish to pass into Island Creek Pond. Through community effort, including local Boy Scout Troop 62 in Duxbury and the Duxbury Conservation Department, the stream was restored and a maintenance program established. Also, an annual counting program is managed in accordance with MA DMF guidelines. With the help of the USFWS and DMF, the system was successfully stocked in three successive years.

Additional herring runs could be restored around the Bay including Stony Brook, Millbrook and the Blue Fish River.

## d Invasive Species

Waterborn: The presence of invasive species in Duxbury Bay is a relatively new phenomenon but represents a significant and largely unchecked threat whose impact cannot be fully predicted and probably cannot be completely controlled.

It is believed this has principally resulted from discharges of ballast water from vessels traveling to the US from abroad. At this time, the data are scarce and non-specific but the presence of various species of tunicate has been documented beneath floats at the Town Pier and on the cages used by the aquaculture industry. The presence of tunicates is of concern because they have the potential to cover the bay bottom and harm both natural and cultivated shellfish and eelgrass. The aquaculture industry removes and destroys tunicates found on oyster cages, a practice that helps to control their abundance.

Other species, most notably green crabs, have been observed by shellfishermen and their numbers appear to be growing rapidly. Green crabs have the ability to feed on young oysters.

Invasive species have the ability to multiply rapidly because of the absence of natural predators. Because they represent such a pervasive hazard, the DBMC believes that continued vigilance of invasive species are needed.

Since the presence and impact of these invasives is probably not preventable, the DBMC recommends a program be established to slow the spread of invasives through education about practices that affect spread, early detection, and avoidance of release of additional non-native organisms into the system.

For the past several years, a monitoring initiative has begun to evaluate the presence of numerous invasives in Duxbury Bay, conducted through Coastal Zone Management by the Massachusetts Bays Program and the North and South Rivers Watershed Association. For more information, please see <http://www.mass.gov/czm/invasives/index.htm>. The following

table provides qualitative trending of the monitoring program underway.

**Table 8-1. Duxbury Bay Invasive Species Data, 2006-2011.**

|                                        | 2006     | 2007   | 2008 | 2009     | 2010   | 2011   |
|----------------------------------------|----------|--------|------|----------|--------|--------|
| <b>Solitary club tunicate</b>          | common   | common | few  | abundant | common | common |
| <b>Colonial orange sheath tunicate</b> | common   | common | rare | common   | none   | few    |
| <b>Colonial golden star tunicate</b>   | abundant | common | rare | abundant | few    | rare   |
| <b>Colonial “mystery” tunicate</b>     | none     | none   | none | few      | few    | few    |
| <b>Orange striped anemone</b>          | few      | none   | none | none     | none   | none   |

Plants: *Phragmites australis*, the common reed is among the most prevalent sights along our coastal waterways, but are a non-native species that dominate estuarine upland areas. Once established, they choke out other flora and diminish biodiversity, in these areas. Local eradication is possible but usually requires use of herbicidal materials that are tedious to apply without adversely affecting life downstream. Although our bay is benefitted by a large natural tidal flow, the aquaculture community is rightly sensitive to broadscale use of typical methods to control *Phragmites*.

DBMC recommends pursuit of a selective approach to addressing *Phragmites* to reduce the impact of this invasive on especially sensitive areas (e.g. Island Creek) and on the most important aesthetic vistas (e.g. along the Bluefish River inlet).

e Dredging:

Duxbury Bay has historically required dredging to preserve access to deepwater from Snug Harbor and Howlands Landing and to preserve the depth of Snug Harbor basin. Generally, dredging is required about every twenty years and, since dredging last occurred in 1996, we are overdue at this time.

In addition to dredging the main basin and approaching channels, several private interests require dredging (e.g. local marinas, Duxbury Bay Maritime School, Battelle and the yacht club). A variety of permits and test data must be assembled each time before dredging can be accomplished. With DBMC's guidance, a task force has been established to organize communication between the stakeholders and facilitate the numerous peripheral tasks that must be completed prior to dredging.

Because we are a federal waterway, funding has historically been provided by federal and state sources and the project has been executed by the Army Corps of Engineers. Currently, there is significant concern that these funding sources cannot be relied on for the foreseeable future. For this reason, DBMC recommends establishment of a task force to lobby for available government funding, evaluate the strategic options forward in the event that adequate outside funding cannot be secured, determine the best option to maintain acceptable basin depth and deepwater access, and facilitate execution of the selected option.

f Marsh Management and Restoration:

Duxbury's marsh areas are an extremely rich and diverse ecosystem that provide sustenance for a extensive variety of organisms, a vital buffer between the land and our bay, as well as a

unique aesthetic backdrop to our beautiful bay vistas.

Extensive pressure is being exerted on this vital resource from all directions. Unbridled development of coastal properties with increasing impact from insensitive fertilization techniques, infiltration by invasive species, increased effluent from wild and domestic animals, and impact from recreational activities from the landside as well as the gradual but unescapable rise in sea level prompted by global warming, threatens to deplete this fragile ribbon.

DBMC recommends an initiative be started to address the impact from manageable stressors to the marsh and to establish and implement a plan to increase the open space along upper marsh edges.

- Central resource of information concerning the marsh ecosystem
- Public education initiative about fertilization and animal waste control
- Best practices guidelines on use of amphibious marine vessels
- Support funding of Conservation Commission stormwater drainage plan
- Establish education effort to inform bay users of the importance of eelgrass beds and procedures to minimize impact
- Explore restoration of remaining potential herring runs including Stony Brook, Millbrook and Blue Fish River
- Establish an education program to inform boaters about detection and avoidance of invasive species introduction.
- Identify selective areas where *Phragmites* eradication should be prioritized based on particularly sensitive ecological threat or scenic impact and establish a program to eradicate using means acceptable to other stakeholders
- Establish dredging task force to optimize potential for funding, evaluate strategic options and implement contingency plan
- Establish central resource of information concerning marsh ecosystem
- Establish a public awareness initiative about proper fertilization techniques and animal waste control
- Establish and implement best practices guidelines to minimize impact by amphibious vessels on marsh resource areas



## **Chapter 9      BAY MANAGEMENT**

### **9.1    Background**

The Duxbury Bay Management Commission was established in 2005, following a recommendation by the Duxbury Bay Study Committee who surveyed the existing environmental conditions and management of the Bay and concluded a permanent commission should be established. DBMC's charter and responsibilities are defined by two bylaws, ratified at Duxbury's annual Town Meeting on March 12 of that same year.

Prior to DBMC's establishment, numerous monitoring and management activities had long been established to insure public safety. There had also been numerous community initiatives undertaken by ad-hoc groups to improve water quality, such as the shared septic systems at the Bluefish River, Landing Road and Snug Harbor, and to restore or preserve other sensitive aspects of the environment and its esthetics.

But the lack of a central organized body to analyze and communicate collected data or to prioritize and coordinate initiatives and activities affecting bay use or water quality was identified as a significant gap in the overall management of the bay resource. Establishment of the DBMC was envisioned to address this gap.

The study committee wisely recognized the potential for added bureaucracy, hence DBMC has no regulatory authority; rather, its role is to assess and advise the Selectmen.

### **9.2    Laws, Regulations and Planning Efforts Related to Bay Management**

Table 9.1 summarizes the local town committees and town boards that have responsibilities for the Bay. Table 9.2 summarizes the laws and regulations related to management of Duxbury Bay.

**Table 9.1 Summary of Town Committees and Boards with Bay Responsibilities**

| <b>ORGANIZATION</b>           | <b>CURRENT ROLE</b>                                                                                                                                                                                                                                  | <b>VESTED AUTHORITY</b>                                                                                                                                                                           |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bay Management Commission     | <ul style="list-style-type: none"> <li>•Coordinate collection and communication of bay related data</li> <li>•Advise selectmen on initiatives that could impact bay resource</li> <li>•Implement Bay Management Plan</li> </ul>                      | Town Bylaws Art's 23, 24                                                                                                                                                                          |
| Board of Selectmen            | <ul style="list-style-type: none"> <li>•Shellfish rules and regulations</li> <li>•Approve aquaculture grants</li> <li>•Beach regulations</li> <li>•Harbor regulations and fees</li> </ul>                                                            | Various Bylaws                                                                                                                                                                                    |
| Conservation Commission       | Regulate activities wetlands and aquaculture                                                                                                                                                                                                         | MGL C.131 S 40<br>Bylaw C. 9, and related regulations.                                                                                                                                            |
| Beach Committee               | <ul style="list-style-type: none"> <li>•Liaison to Beach Reservation</li> <li>•Powder point bridge</li> <li>•Event planning</li> <li>•Disaster/oil spill</li> </ul>                                                                                  | <ul style="list-style-type: none"> <li>• Advisory to Board of Selectmen</li> <li>• Bylaw</li> <li>• Rep. Town Interest</li> </ul>                                                                 |
| Board of Health               | <ul style="list-style-type: none"> <li>•Beach water quality</li> <li>•Septic systems</li> </ul>                                                                                                                                                      | <ul style="list-style-type: none"> <li>• Title 5</li> <li>• MGL c. 111 s.17</li> </ul>                                                                                                            |
| Fire Department               | <ul style="list-style-type: none"> <li>•Emergency response</li> <li>•Parking restrictions</li> </ul>                                                                                                                                                 |                                                                                                                                                                                                   |
| Harbormaster's Department     | <ul style="list-style-type: none"> <li>•Public safety</li> <li>•Health and welfare management of waterfront activities</li> <li>•Management of natural resources</li> <li>•Mooring</li> <li>•Shellfish Constable</li> <li>•Herring Warden</li> </ul> | <ul style="list-style-type: none"> <li>• MGL C. 102,90,90B, 91, 131, 88, 130</li> <li>• State Regulations</li> <li>• Board of Selectmen</li> <li>• employer</li> </ul>                            |
| Department of Public Works    | <ul style="list-style-type: none"> <li>•Landing maintenance</li> <li>•Landing projects</li> </ul>                                                                                                                                                    |                                                                                                                                                                                                   |
| Shellfish Advisory Committee  | <ul style="list-style-type: none"> <li>•Assist Harbormaster</li> <li>•Advise Board of Selectmen</li> <li>•Review aquaculture grants</li> <li>•Shellfish Management Plan</li> </ul>                                                                   | <ul style="list-style-type: none"> <li>• Bylaw 7.5 Shellfish</li> <li>• Bylaw 7.3,7.4 Aquaculture</li> <li>• Advisory to Board of Selectmen</li> <li>• Appointed by Board of Selectmen</li> </ul> |
| Waterfront Advisory Committee | <ul style="list-style-type: none"> <li>• Advise Board of Selectmen</li> <li>• Assist Harbormaster</li> <li>• Review Fee structure, access/landings, rules and regulations</li> </ul>                                                                 | <ul style="list-style-type: none"> <li>• Advisory to Board of Selectmen</li> <li>• Appointed by Board of Selectmen</li> </ul>                                                                     |
| Zoning Board of Appeals       | Special permits                                                                                                                                                                                                                                      | Protective Bylaw                                                                                                                                                                                  |

**Table 9-2. Summary of State and Federal Laws and Regulations Related to Duxbury Bay**

| <b>ORGANIZATION</b>                                                        | <b>CURRENT ROLE</b>                                                                                           | <b>VESTED AUTHORITY</b>                                                                                                                                                |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mass Coastal Zone Management Office                                        | All activities affecting coastal resources as stated in Program Policies, regulatory and management interests | <ul style="list-style-type: none"> <li>• FED CZM ACT</li> <li>• 301 CMR 21.00, 20.00, 22.00, 23.00, 25, 26</li> <li>• 15 CFR Par 930</li> </ul>                        |
| Mass Environmental Policy Act Unit                                         | Activities affecting coastal resources above stated thresholds, information gathering for regulatory process  | M.G.L.C30, S.61-62H<br>301 CMR 11.00                                                                                                                                   |
| Mass Department of Environmental Protection (DEP), Water Pollution Control | Surface and groundwater discharges, dredging, wetland filling, regulatory                                     | MGL C. 21 S.26-53<br>314 CMR 9.00<br>314 CMR 3.00                                                                                                                      |
| DEP, Wetlands Protection Program                                           | Protect the functions of the wetland resource areas (see Wetlands Protection Act), regulatory                 | M.G.L. C.131 S.40<br>310 CMR 10                                                                                                                                        |
| DEP, Waterways Regulation Program                                          | Protect public trust access rights of fishing, fowling and navigation, structural integrity, regulatory       | M.G.L. C.91<br>310 CMR 9                                                                                                                                               |
| Mass Division of Marine Fisheries                                          | Protect and restore marine fisheries and related habitats                                                     | National Shellfish Sanitation Program<br>M.G.L. C. 130<br>22 CMR 1 - 14                                                                                                |
| U.S. Food and Drug Administration                                          | Human consumption of shellfish and shellfish sanitation                                                       | National Shellfish Sanitation Program                                                                                                                                  |
| U.S. Army Corps of Engineers                                               | Wetlands resources, public access, historic assets, navigation, and flood hazards                             | <ul style="list-style-type: none"> <li>• Rivers and Harbors Act</li> <li>• Clean Water Act 1977</li> <li>• Marine Sanctuaries Act</li> <li>• 33 CFR 320-330</li> </ul> |
| U.S. Environmental Protection Agency                                       | Protection of water quality and wetlands resources, regulatory, grants                                        | <ul style="list-style-type: none"> <li>• Clean Water Act, S.301, 402</li> <li>• 40 CFR Parts 12-125</li> </ul>                                                         |

**Table 9-3. Summary of State and Federal Laws and Regulations Related to Duxbury Bay**

| <b>ORGANIZATION</b>                              | <b>CURRENT ROLE</b>                                                                | <b>VESTED AUTHORITY</b>                                                         |
|--------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| U.S. Coast Guard                                 | Navigation safety, bridge permits, boating safety education, Auxiliary             | -General Bridge Act<br>-Title 33 U.S. Code Parts 525-533<br>-33 CFR Parts 62-66 |
| National Marine Fisheries Service                | Marine fisheries protection and habitat restoration, advisory to Corps Permits     | -Rivers and Harbors Act<br>-Clean Water Act 1977<br>-Marine Sanctuaries Act     |
| U.S. Department of Agriculture                   | Public health, shellfish cleanliness, habitat restoration, regulatory, restoration |                                                                                 |
| National Oceanic and Atmospheric Administration. | -regulatory and restoration ocean resources                                        |                                                                                 |
| U.S. Fish and Wildlife Service                   | -wildlife and habitat protection and restoration, includes fisheries               |                                                                                 |

### 9.3 Existing Bay Management Plans

Presently the following Plans affect management of Duxbury Bay:

- Duxbury 2008 Open Space and Recreation Plan
- Duxbury Comprehensive Plan, 1999
- Pier Access and Shoreline Study Committee surveyed policies in 27 waterfront towns
- Report of the Duxbury Town Landing Study Committee, 1975
- Aquaculture Management Plan 2010

### 9.4 Existing Conditions and Trends in Bay Management

The Duxbury Bay Management Commission has established a State of the Bay report to document existing conditions with respect to the Bays ecology. Also, this Bay Management Plan contains a significant body of historical reference information about activities in and around the bay, as well as initiatives the DBMC recommends be undertaken to preserve the quality of the Bay and preserve harmony between its many existing uses.

There are no significant, chronic water quality issues at the present time. The relative lack of commercial or industrial development along the waterfront coupled with Duxbury's historically conservative philosophy has resulted in a shoreline that is remarkably and naturally beautiful.

Water quality testing data shows that the Bay has generally excellent water quality and meets or exceeds State requirements for shellfishing and swimming. But the Bay does have water pollution challenges causing shellfish bed and beach closures. These closures are often rain-event-related but water quality is affected by pollution from humans and other natural<sup>18</sup> sources within the Bay and its watershed.

Restoration and regulatory efforts to eliminate or prevent Bay pollution are undertaken by several Boards and Committees. For instance, the Mass Division of Marine Fisheries (DMF) regulates shellfishing by closing and opening beds for fishing based on their sampling and measuring of bacteria in marine waters. Fresh water tributaries may carry bacteria from failed septic systems to the Bay but DMF has no authority in fresh water tributaries or over septic systems which are under the jurisdiction of the Board of Health. To optimize mitigation of pollution problems these agencies and levels of government must be coordinated, because the type and origin of the pollution vary. This example illustrates the need for coordinating one agency's information and authority with another's to identify and fix water quality problems.

This type of coordination is well demonstrated by the Town's implementation of the Blue Fish River, Snug Harbor and Landing Road shared septic systems projects and the current storm drain remediation project in Snug Harbor. All the Boards and staff involved in these projects are to be congratulated and supported in their efforts.

Education continues to be a key component of Bay management. Through education, users learn their rights and responsibilities both in terms of other users and the Bay's ecology. Presently, there are education efforts from the Harbormaster's Office, the U.S. Coast Guard and the Duxbury Bay Maritime School regarding boating safety and Bay ecology.

Management trends show a growing and successful effort to coordinate and provide comprehensive Bay management. These efforts are primarily related to either project regulation or specific restoration projects. No particular group has a standing charge to coordinate Bay management efforts or provide a comprehensive view of Bay issues. Currently, Bay management consists of efforts to:

- Regulate Bay activities;
- Restore Bay ecology and use (projects);

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<sup>18</sup> Waterfowl, especially in areas of limited tidal flow such as Bluefish River, are believed to be a significant source of bacterial contamination.

- Conduct specialized planning (shellfish, beach);
- Develop policy (Board of Selectmen);
- Educate Bay users regarding use impacts on other users and the ecology; and,
- Conduct scientific inquiries (presently this is limited to support of projects).

## **9.5 Recommendations for Future Bay Management**

The establishment of DBMC has addressed the gap identified in the overall management of Bay resources and activities. Since 2005, the Commission has met regularly to review requests from the Selectmen and other committees, make recommendations to the selectmen based on potential impact to Duxbury Bay, as well as to implement other initiatives in this plan.

Two primary dynamics were anticipated. One is that conflict among bay users would increase. The concern from many about the impact of additional piers on the esthetic beauty of the bay was a prominent concern when the DBMC was established and this Commission reviewed and advised about number pier applications. The growth of Duxbury Bay Maritime School and of aquaculture in the Bay have significantly increased activity in the bay, especially in Snug Harbor, over the past 5 years, but there have been numerous, effective corrective actions implemented to remedy problems as they arose.

The second dynamic anticipated is that water quality would be affected by increased population, water-front development and insensitive management practices (e.g. synthetic fertilizer use). Although beaches and shellfishing beds have undergone closure for coliform exceedences, there is no evidence that the frequency of these closures is increasing over time.

A third dynamic has become evident: the changes to our Bay habitat with respect to eel grass and the balance of organisms in the benthic zone, near the bottom of our bay. Because of limited retrospective data, it is difficult to determine whether the current trend is historically significant but the decrease of eel grass habitat in the bay poses a significant threat to bay health, as it is an interdependent link for much of the marine life here. Similarly, there is little historic data to evaluate for comparison, but the numbers and quantity of invasive organisms, such as green crabs and various tunicate species, are increasing steadily. While it is not clear what, if anything, can be done to effectively address either threat, it seems self-evident that vigilance for signs of impact from these dynamics is important.

In general, DBMC believes the oversight it has provided has been effective in improving overall bay management and recommends the Commission continue functioning in its present form for the foreseeable future.

## **9.6 Emergency Disaster Response Planning**

Emergency disaster response planning includes natural disasters such as storms, hazardous materials response, nuclear accidents and terrorist attacks. The Town's response organization to these disasters is the Duxbury Emergency Management Agency (DEMA)

which has a specific plan for each of these disasters. The DEMA is located in the Central Fire Station at 668 Tremont Street (Rte.3A) in Duxbury. The agency's role is defined to include man-made and natural disasters. The Duxbury Bay Management Study Committee felt that additional planning was needed to address the specific needs of the Bay when confronted with natural disasters such as hurricanes, northeasters and other major storm events.

### **9.6.1 Harbor Evacuation Planning**

The Duxbury Bay Management Commission and Harbormaster have been concerned about emergency response planning as it is a public health and safety issue. Over the past two years, DMBC together with the Harbormaster and other stakeholders in Snug Harbor have developed a new Storm Management Plan to establish a comprehensive plan in the event the Harbormaster determines evacuation is prudent due to storm risk.

### **9.6.2 Oil Spill Planning**

By federal and state law every oil spill must be reported to the proper agencies. This includes the U.S. Coast Guard national response center (800-424-8802), the Town Harbormaster (781-934-2866), and the Mass Department of Environmental Protection and the Environmental Police. The Harbormaster will notify the appropriate first responders in Town including the Duxbury Emergency Management Agency (DEMA) and Fire and Police Departments. Because the speed of containment is critical to limiting spill impacts on the Bay's ecology, the Town keeps a spill response equipment trailer at the Fire Station and the Harbormaster has been trained in the use and deployment of the equipment as has the Town's Hazardous Materials Team (HAZMAT). In addition the Town has maps of sensitive receptors in the Bay ("Sensitivity of Coastal Environments and Wildlife to Spilled Oil, National Oceanographic Atmospheric Administration (NOAA), Office of Response and Restoration, Hazardous Materials Response Division Seattle, Washington, Coastal Services Center, Charleston, South Carolina, MCZM, Maps, 3/1999"). The US Coast Guard will respond as will the DEP. After an evaluation of the spill, these agencies will direct private contractors to respond as needed endeavor to determine responsible parties, and evaluate impacts and determine required cleanup and loss compensation.

## **9.7 Education**

Nearly every aspect of this plan mentions education as an important component of Bay management. This includes any type of vessel use, aquaculture, wildlife and habitat protection. The Duxbury Bay Management Commission maintains the recommendations of the original Bay Study Committee about the importance of education.

As stated in the bylaw, the Commission will support existing and develop new education and outreach programs to:

- Educate the next generation of Bay stewards;
- Inform users of their impacts and responsibilities; and,
- Support Bay projects.

The Duxbury Bay Management Commission recommends education programs for the general public as well as more formal class room programs. Some of the recommendations for education include:

1. Public Education

- Gather and document the regulatory bodies and regulations regarding bay uses to serve as a "Users Guide"
- Prepare a public/private access rights brochure
- Develop a restoration and user project related education, including:
  1. signage
  2. brochures
  3. event displays
  4. volunteer field work and education
  5. greenscapes education
  6. green boating
  7. green marinas

2. School and Private Non-Profit "formal education" on topics such as:

- Boating
- General ecology
- Bay specific ecology
- Environmental protection and restoration project ecology
- Natural system based management
- Habitat restoration
  1. anadromous fish
  2. eelgrass beds
  3. salt marsh
  4. nonpoint source pollution
  5. invasive species

## 9.8 Future Studies

Future studies and ongoing monitoring are important to the effective management of the Bay as new ecological information and use impacts information are generated. Studies are recommended in the areas of management and natural sciences. The monitoring of the health of the Bay and use impacts is important as the basis of management decisions. Changes in use are now subject to the Commission's evaluation, and therefore monitoring of use on a periodic or continual basis is needed. Specific activity-based studies will be identified by the Commission or can be referred to the Commission by existing boards and committees. The Commission will then appoint subcommittees or work groups to further study the issues identified in original Plan and by the DBMC. The Commission has established a central repository of environmental health indicators and uses and provide or support the provision of ongoing monitoring of environmental health indicators. Finally, the Commission will convene integrated work groups to identify, prioritize, and address Bay ecology and use issues. Some studies identified in the initial planning phase include:



1. Dredge disposal alternatives;
2. Bay wide organization (Kingston, Plymouth, Marshfield);
3. Eelgrass health and distribution;
4. Bacterial source identification for closed shellfish areas and periodically closed beaches; and,
5. Define and identify areas of Critical Marine Habitat such as eelgrass and shellfish beds.
6. Assist in the development of various plans as defined in the recommendations table (mooring, dredge, aquaculture).
7. Explore need for and utility of “Watersheet Zoning”

## 9.9 Funding Program

A funding program is necessary to execute many of the recommendations of this Bay Management Plan. The Bay Commission would benefit greatly from staff to support its major functions of reviewing projects in coastal wetlands and its program to maintain and restore Bay ecology uses. The bylaws of the Commission anticipated this. It is authorized in Art. 23 Sect. 6.14.4, “With the approval of Town Meeting . . . to hire staff and/or consultants using private and public grants and other appropriations”. Project-related hires supported by grants should also be authorized by the Town Manager and Board of Selectmen as long as any related Town commitments are in place. Presently, the Commission needs one-full time equivalent person, which could be a contract position, and additional funds related to specific Bay projects.

The Harbormaster/Coastal Resources Department is in need of a consistent and reliable revenue stream and a plan for operations, maintenance and capital expenses. Currently, there is no dedicated waterways fund as required by law, to be funded with a minimum of one-half of boat excise taxes (MGL C 50B, S.2 (b), to support the Harbormaster/Coastal Resources Department functions. The rate of excise tax collection for trailered, moored or rack storage vessels is unknown.

The Harbormaster/Coastal Resources Department is funded out of the general fund. This Department generates significant funds from various fees, including fishing licenses and moorings. A comprehensive inventory of capital needs for this Department should be conducted and be matched with a revenue plan to meet those needs. This plan would identify all costs related to maintenance, operations, and capital projects of the Department. The plan would also show revenues generated from waterways activities for which the Department is responsible. The plan would be for a term of 5 to 10 years. This plan would support the Town’s efforts to manage its budget and the functions of the Department. The revenue plan should include fee structure analysis and analysis for the need of a user fee for rack and otherwise stored vessels whose use requires Town services and infrastructure.

In general, the funding plan for the Bay could combine the Harbormaster/Coastal Natural Resources Department and the Duxbury Bay Management Commission. Whether or not a combined plan is developed, both the Commission and the Department need a funding plan with the following components:

1. A 5 to 10 year Cash Flow Proforma plan including all maintenance, operations and capital project costs and revenues from waterways fees
2. Fee Structure analysis of current revenues generated from waterways activities and their uses within the Town for:
  - a. Compliance with tax law (boat excise taxes, MGL C 50B, S2(b))
  - b. Rates (sufficient to meet costs?), compliance, structure and equity (fair allocation of costs of services)
3. Multiple Resources for generating funds including:
  - a. Grants
  - b. Annual Town Meeting,
  - c. Fees, including special fees for Bay related activities,
  - d. Gifts and,
  - e. Other Government Funding, including the Community Preservation Act.

## **Article 23**

### **DUXBURY BAY MANAGEMENT COMMISSION<sup>19</sup>**

Proposed Articles for Approval at the March 12, 2005 Town Meeting

## **Article 23**

To see if the town will vote to amend the General By-Laws of The Town of Duxbury by adding a new Chapter 6.15 Duxbury Bay Management Commission (DBMC) to read as follows:

### **6.15. Duxbury Bay Management Commission**

6.15.1 The Duxbury Bay Management Commission shall consist of nine members appointed by the Selectmen, to three year staggered terms. In appointing members, the selectmen shall consider individuals representing varied interests of the Bay including but not limited to aquaculture, boating, commercial and not for profit waterfront users, ecology, shellfish and finfish, and other residents at large whose experience will enhance the diversity of the commission.

6.15.2 The mission of the Duxbury Bay Management Commission is: *“To implement and maintain a management plan that will enhance and preserve the ecological health, pristine waters and natural beauty of Duxbury Bay for future generations while sustaining harmony among all its uses.”*

6.15.3 The Duxbury Bay Management Commission shall have the following powers and duties:

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<sup>19</sup> Minor changes were made to this article at Town Meeting

- a. Evaluate and review any proposed change in Bay use and access, inclusive of structures and all Bay related issues requiring the action of the Board of Selectman (BOS), Duxbury Conservation Commission (DCC), Zoning Board of Appeals (ZBA), Planning Board (PB), Harbormasters Office and Board of Health (BOH) involving Coastal Resource Areas as defined in M.G.L. c.131 S40 and the Town of Duxbury Wetlands Bylaw (Chapter 9 General By-Laws of the Town of Duxbury).
  1. Bay related issues and proposed changes will be evaluated at a minimum in terms of impacts on the following;
    - a) public safety, health and welfare
    - b) ecology and sensitive receptors including wildlife
    - c) uses and appropriateness of use and access
    - d) scenic views and aesthetic issues
    - e) levels of public and private services
    - f) supporting landside infrastructure
  2. Based on the above evaluation, the commission shall make recommendations and comments to the responsible regulatory or policy setting authorities.
- b. Recommend changes to regulations and policies. Submit recommendations and comments to the Board of Selectmen, Town Meeting and other regulatory authorities to change Town Bylaws, regulations and policies as needed to achieve the plan mission.
- c. Provide a forum for dispute resolution for projects and use impacts and conflicts on the Bay.
- d. Develop and implement a restoration, enhancement and management program to maintain and restore the Bay ecology and harmony among its uses by:
  1. Maintaining a central repository of environmental health indicators and uses.
  2. Providing ongoing monitoring of environmental health indicators.
  3. Convening integrated work groups to identify prioritize and address Bay ecology and use issues.
  4. Developing program to install best management practices for Town storm drain system and other non-point sources of pollution.
  5. Providing project management to supplement existing Town staff for ecological and other Bay related projects.
  6. Seeking funds and grants for implementation of projects.
  7. Supporting existing and developing new education and outreach programs to:
    - a. Educate the next generation of Bay stewards
    - b. Inform users of their impacts and responsibilities
    - c. Support Bay projects.
- e. Maintain the plan as a living document to guide the work of the DBMC by:
  1. Using the Plan as a guide in all the commission's recommendations, plan updates and activities.
  2. Reviewing and updating the Plan at a minimum every five years.

3. Incorporating proposed Plan changes on an ongoing basis as needed.
  4. Appointing subcommittees or work groups to further study issues identified in original Plan and by the DBMC.
- f. Recognize the Bays' interdependence and explore cooperative Bay management with the towns of Kingston and Plymouth.

6.15.4 With the approval of Town Meeting the DBMC shall have the authority to hire staff and/or consultants using private and public grants and other appropriations.

And to see if the town will vote to amend Chapter 9 Wetlands Protection of the General By-Laws of The Town of Duxbury by inserting the following in 9.1.6., after the last sentence in this section:

**“The Duxbury Conservation Commission shall within 3 days of receiving a request for a determination or a notice of intent for proposed activities within a coastal resource area, as defined in Chapter 131, section 40 of the Massachusetts General Laws or as determined by the Duxbury Conservation Commission, request that the Duxbury Bay Management Commission provide written comments and recommendations before taking final action on said request for determination or notice of intent.”**

### **Article 23**

#### **DUXBURY BAY MANAGEMENT COMMISSION**

Proposed Article Approved as Amended by the March 12, 2005 Town Meeting

### **Article 24**

To see if the town will vote to amend the Duxbury Protective Bylaw, Section 404.8 Special Permit Procedures, item 1, by inserting the following language, **“the Duxbury Bay Management Commission,”** in the first sentence after “Duxbury Conservation Commission,” so the section reads:

*The Board of Appeals shall refer a special permit application to the Conservation Commission, **the Duxbury Bay Management Commission**, the Board of Health, and the Planning Board for written comments and recommendations before taking final action on said special permit application. In addition to the above noted boards, the Board of Appeals may refer a special permit application to any other Town agency/board/department for comments and recommendations if it so desires before taking final action on said special permit application.*

And amend Duxbury Protective Bylaw, Section 906.4 Referral, by inserting the following language, **“Duxbury Bay Management Commission,”** in the first sentence after “Duxbury Conservation Commission,” so the first sentence reads:

*In addition to those applications for a Special Permit which require site plan approval under Section 615, the Special Permit Granting Authority shall refer a Special Permit application to the Board of Appeals, Board of Health, Conservation Commission, **Duxbury Bay Management Commission**, Planning Board, Water Advisory Board and Design Review Board for written comments and recommendations before taking final action on said Special Permit application.*

And amend Duxbury Protective Bylaw, Section 615.7 Required Procedures for Site Plan Review, by inserting the following language, “, **the Duxbury Bay Management Commission**,” in 615.7 (2.) after “the Conservation Commission” so that 615.7 (2.) reads:

Within five (5) working days of receiving a Site Plan, the Planning Director or his/her designee shall distribute copies of the Site Plan to the Planning Board, the Department of Public Works, the Police Department, the Fire Department, Highway Safety Committee, the Conservation Commission, **the Duxbury Bay Management Commission** and the Board of Health. If the proposed activity requires a special permit, the special permit granting authority shall receive a copy of the Site Plan.

## **APPENDICES**

**APPENDIX A: RESOURCES MAPS**

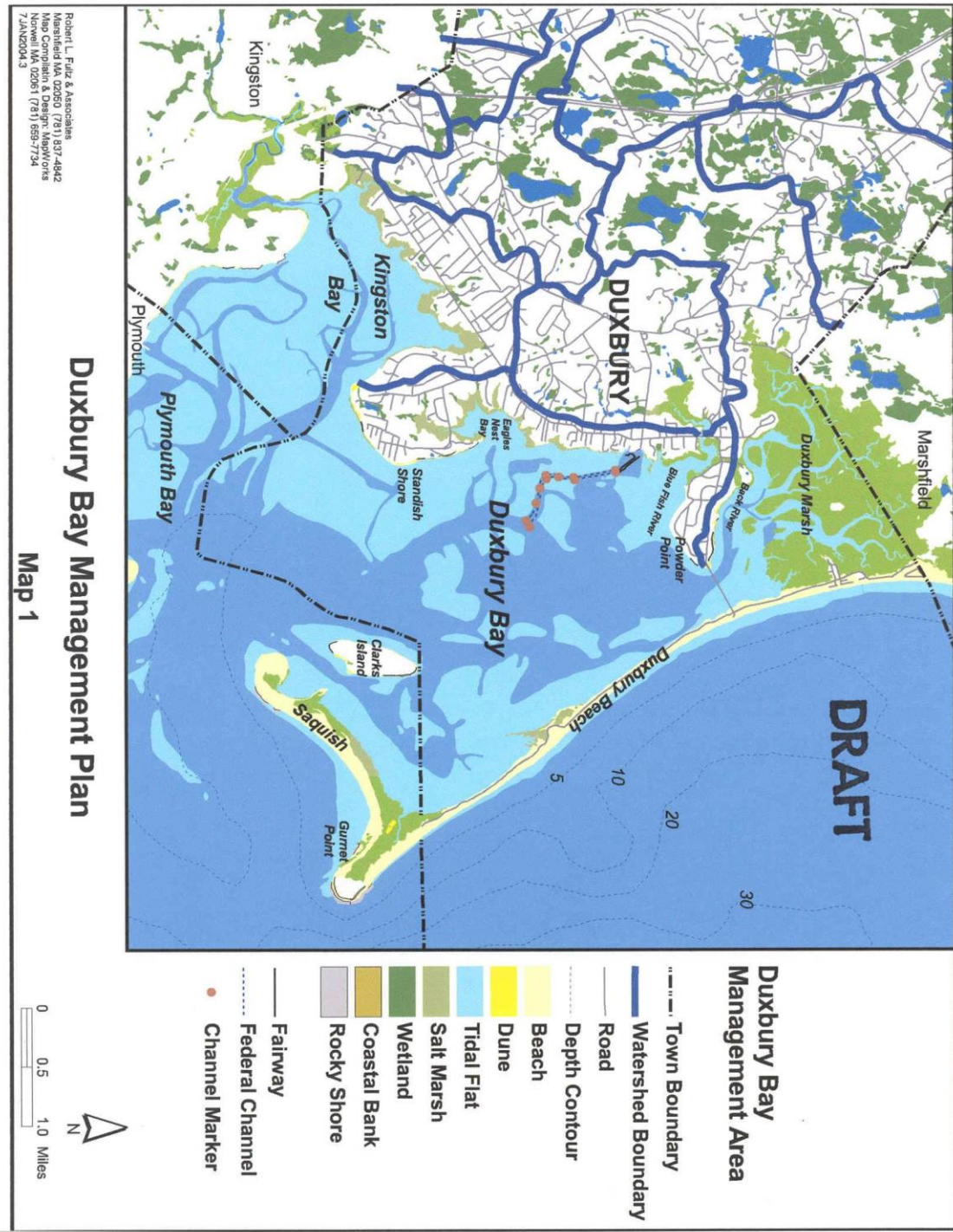
**APPENDIX B: WATERWAYS RULES AND REGULATIONS**

**APPENDIX C: MOORING RULES AND REGULATIONS**

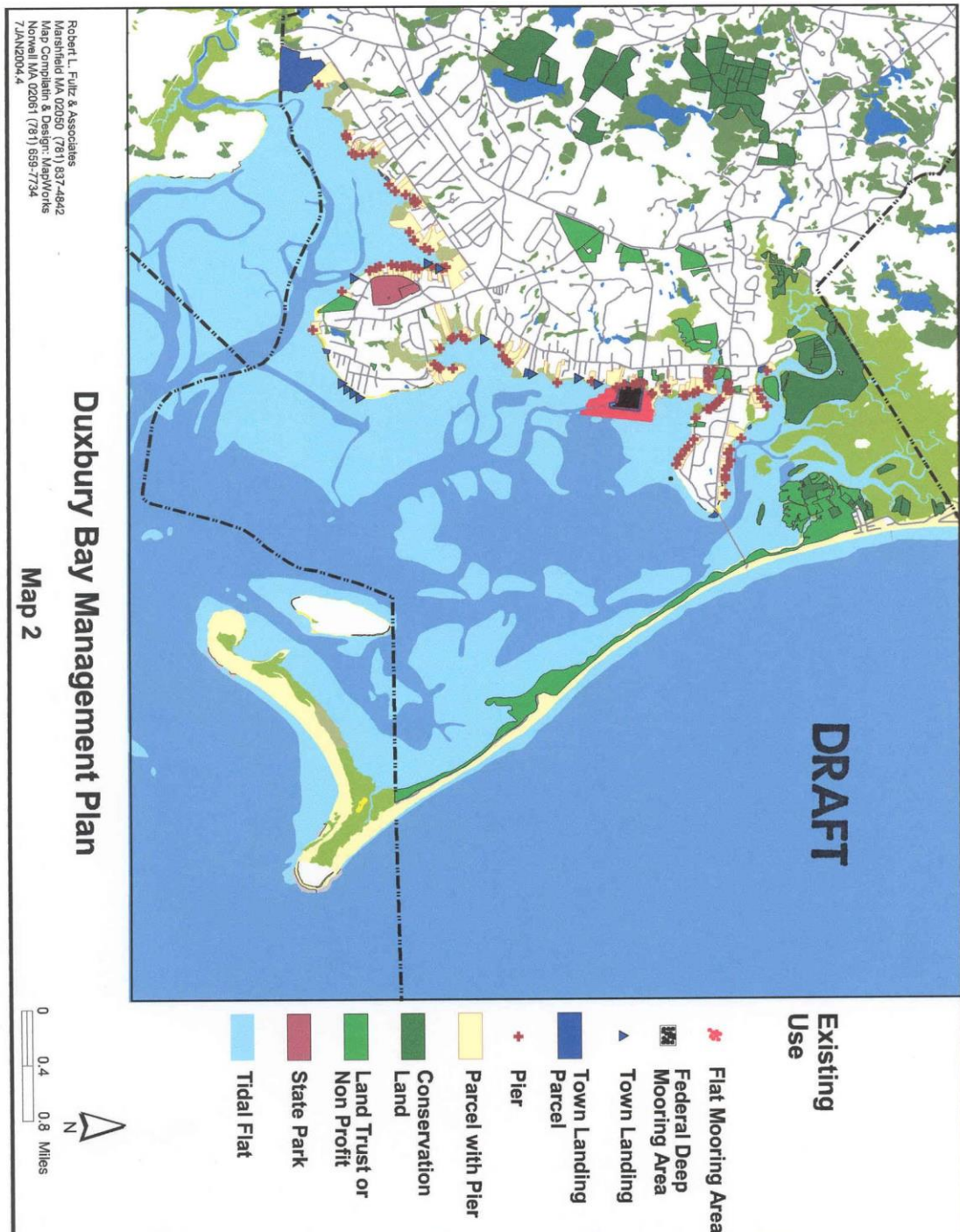
**APPENDIX D: INPUT FROM PUBLIC WORK GROUPS**

**APPENDIX E: CHARACTERISTICS OF TOWN LANDINGS  
AND WAYS TO THE WATER, DUXBURY,  
MASSACHUSETTS**

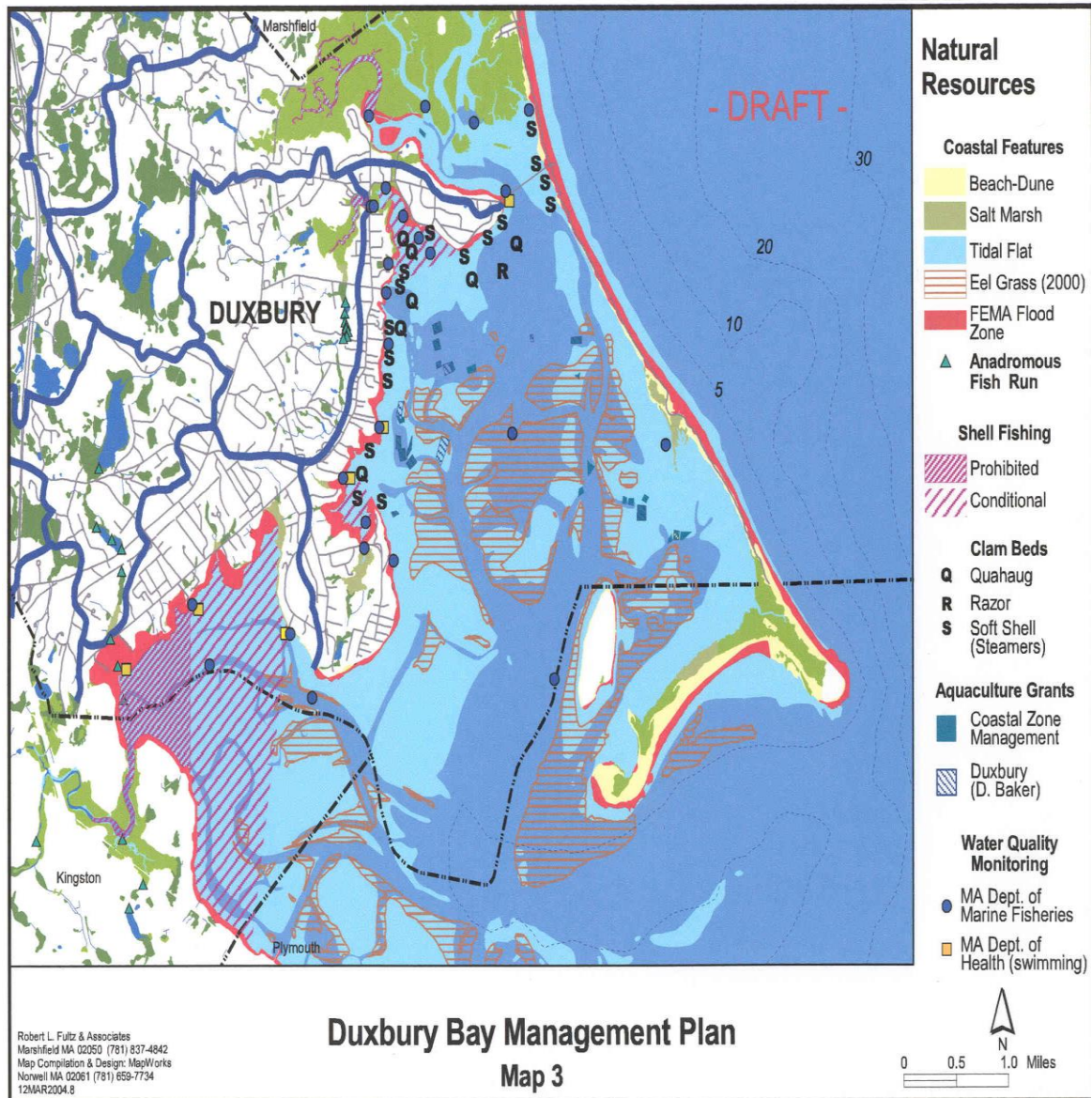
# APPENDIX A: RESOURCE MAPS

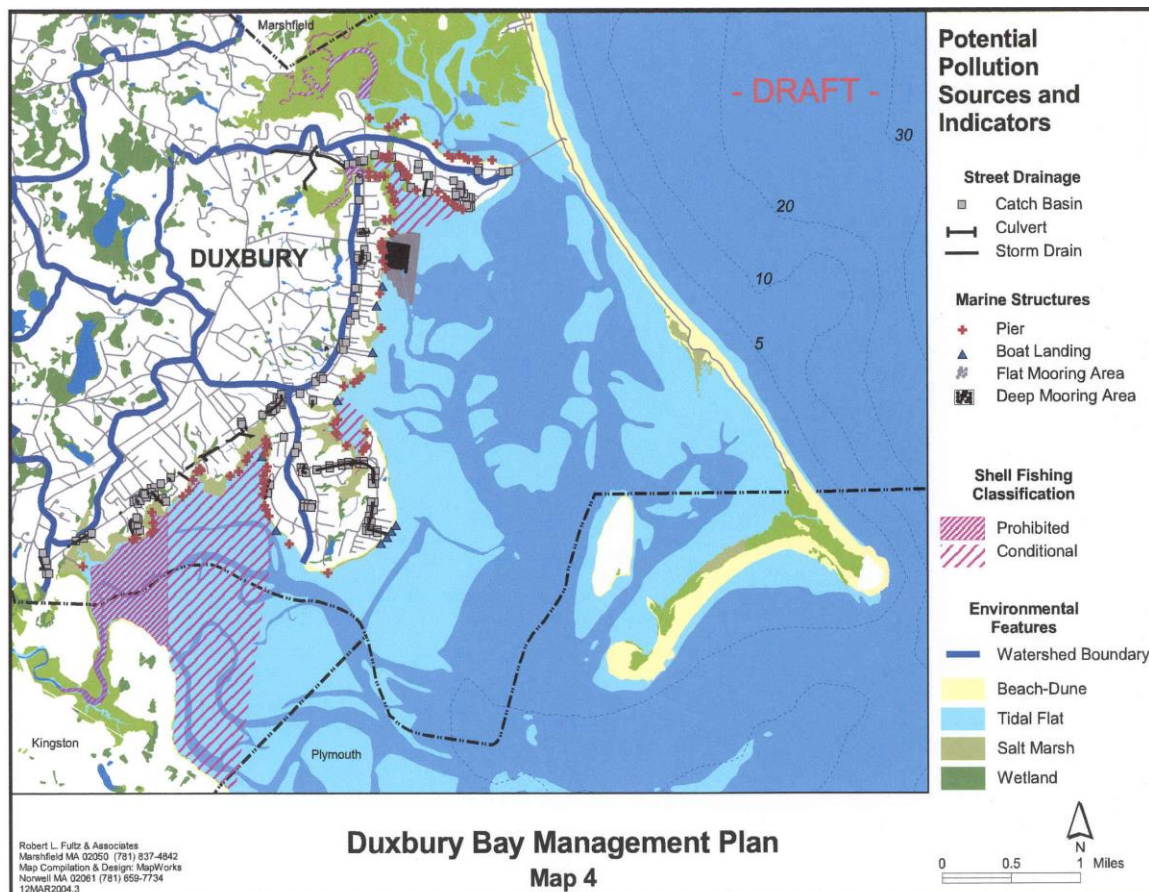












## **APPENDIX B: WATERWAYS RULES AND REGULATIONS**

### **TOWN OF DUXBURY**

### **HARBORMASTER**

#### **Rules and Regulations**

#### **for Duxbury Waterways**

**Effective date: 02 May 2005**

#### 7.6 § 1001: AUTHORITY AND ENFORCEMENT

- (a) Under the authority of Massachusetts General Laws, Chapter 90B § 15B; Chapter 88 § 19; Chapter 102 § 19-26; Chapter 91; Chapter 40; 310 CMR 9.07 and all other applicable laws and regulations, the following rules and regulations are promulgated effective 02 May 2005.
- (b) The Harbormaster and Assistant Harbormasters have the authority to enforce all rules and regulations set forth herein.

#### 7.6 § 1002: DEFINITIONS

The following words, for the purpose of these regulations, unless another meaning is clearly apparent for the way the word is used, shall have the following meanings:

- (a) “anchoring” means to secure a vessel for a short period of time, not to exceed twenty-four (24) consecutive hours, to the bottom of a body of water by dropping an anchor or anchors or other ground tackle from a vessel.
- (b) “berth” means any space wherein a vessel is confined by wet slip, float, mooring or any other type of docking facility.
- (c) “channel” means a navigable route for the passage of vessels, established by customary use or under the authority of federal, state or municipal law.

- (d) “designated anchorages” means established mooring fields at Snug Harbor Deep Water (dredged) Basin, Snug Harbor Basin Flats, Howland’s Landing, Two Rock Channel and Clark’s Island.
- (e) “Duxbury waterways” means all waters within the embayment, harbors, rivers and open waters within the boundaries and jurisdiction of the Town of Duxbury, over which the Town of Duxbury may exercise its powers and authority.
- (f) “down skier” means any person(s) in the water for whatever reason while being towed.
- (g) “fairway” means any locally designated and / or maintained water areas reserved for the unobstructed movement of vessels through mooring fields or designated anchorages.
- (h) “flag display” means a regulation flag held upright, above the spotter’s head, at a sufficient height to be visible to passing vessels.
- (i) “Harbormaster” means the duly appointed Harbormaster or any duly appointed Assistant Harbormaster.
- (j) “mooring” means the equipment and / or process used to secure a vessel, other than by anchoring, that consists with the gear as prescribed by the Town of Duxbury, which is placed on the bed of a body of water and attached to a buoy.
- (k) “other object” means any floating object, including but not limited to floats, rafts or barges, temporarily secured to the bed of any body of water by anchor or mooring.
- (l) “person” means any individual, partnership, trust, firm, corporation, association, commission, district, department, board, municipality, public or quasi-public agency or authority.
- (m) “race” means any scheduled competition of vessels, rowing sculls / gigs, wind surfers or other, organized to race within the waterways of Duxbury.
- (n) “regulation flag” means a 12” by 12” bright orange flag secured appropriately to 2 ½ foot dowel ½” in diameter used to advise surrounding vessels of a down skier.
- (o) “spotter” means as prescribed by M.G.L. Chapter 90B, a competent person at least twelve (12) years old, in addition to the operator, in a position to observe at all times the person(s) being towed.
- (p) “transient guest” means any vessel that does not have a valid mooring permit issued by the Town of Duxbury.
- (q) “vessel” means every description of watercraft, other than a seaplane on the water, used or capable of being used as a means of transportation on the water.
- (r) “water skiing” means the towing or manipulating of a surfboard, wake board, ski, tube or any similar device or person(s) whether or not on a device.

#### 7.6 § 1003: GENERAL RULES AND REGULATIONS

- (a) All vessels entering Duxbury waterways are subject to these rules and regulations and to the direction of the Harbormaster, who shall be consulted before anchoring for over twenty-four (24) hours or tying up to any town facilities or moorings.
- (b) The Town of Duxbury shall not be responsible for any loss or damage to boats or vessels.
- (c) Boat owners / operators shall be held responsible for damage caused by them or their vessels to surrounding vessels, moorings, structures, pilings or related facilities owned privately or by the Town of Duxbury.
- (d) No unauthorized person shall board any boat within Duxbury waterways.

- (e) No person shall molest or damage any vessel, tender, etc., anywhere along the shoreline of Duxbury waterways.
- (f) No person shall use or borrow another's tender without first gaining permission from the owner or agent.

7.6 § 1004: TENDERS

- (a) All tenders stored on or at Town tender floats shall display name and phone number of owner and have annual tender permit adhered to port bow.
- (b) No tender over 10' in length shall be stored on its assigned tender spot.  
Tenders spots assigned to tenders over 10' before 21 March 2005 will be grandfathered. No tender over 12' in length shall be tied to tender tie-up float.
- (c) All tenders assigned to a tender spot or to the tender tie up floats shall be free of debris and dewatered at all times.

7.6 § 1005: WATERFRONT ACCESS AND USE

- (a) The Town of Duxbury shall not be liable for any injury, damage or loss sustained during the use of any Town owned or controlled waterfront facilities. Facility users shall be responsible for any injury, damage or loss caused by them.
- (b) Vehicle access on the Mattakeesett Court Town Pier will be allowed only with the permission of the Harbormaster.
- (c) Town Pier vehicular weight (2 axle) shall be limited to 4 ton.
- (d) Vehicle speed on pier shall not exceed 2 MPH.
- (e) Parking on the Pier shall be limited to fifteen (15) minutes only.
- (f) Placement of gear of any kind on the Town Pier, float(s) or waterfront facility will be allowed only with the permission of the Harbormaster. Gear so placed shall be neatly stacked and shall not hinder or obstruct access to the Town Pier, float(s) or waterfront facility at anytime.
- (g) The waterfront facility shall be free of any debris, trash, etc. at all times. It shall be the responsibility of the user to be in compliance at all times.
- (h) All pier barriers and gates shall remain closed and locked in place except during authorized access.
- (i) The use of marine engine power to assist in launching or hauling at any public launching / landing ramp is prohibited.
- (j) All trailers parked on the property of the Town of Duxbury shall be parked in designated trailer areas in a manner that does not block, restrict or inhibit access or traffic flow to a landing, facility, roadway, public way, parking lot or private property.
- (k) Trailers parked at Mattakeesett Court shall be disconnected from the tow vehicle and be parked in designated trailer areas, seasonally from May 1<sup>st</sup> thru November 1<sup>st</sup>, unless authorized by the Harbormaster.
- (l) Trailer parking on the property of the Town of Duxbury for a period of time longer than twenty-four (24) hours without approval from the Harbormaster is

strictly prohibited. Trailers violating this may be cited and removed at the owner's expense.

- (m) All trailers parked on the property of the Town of Duxbury shall be legally registered and be equipped with a valid license plate.
- (n) Any unauthorized access or use of any Town of Duxbury waterfront facility shall be strictly prohibited.
- (o) There shall be no parking of any motor vehicle at any Town of Duxbury waterfront facility after 12:00 A.M. (midnight) unless authorized by the Harbormaster.

7.6 § 1006: BERTHING

- (a) No vessel will be allowed to remain at the Town Floats or the Town Pier Facility for over thirty (30) minutes. There will be no overnight docking or tie-up allowed at the Town Floats or the Town Pier. Vessels with emergency or exigent circumstances will be limited at the discretion of the Harbormaster.
- (b) All transient guests entering Duxbury waterways shall receive the permission of the Harbormaster before anchoring for a period of more than twenty-four (24) hours.
- (c) All transient guests entering Duxbury waterways shall receive the permission of the Harbormaster before tying up to any Town of Duxbury waterfront facility.
- (d) All transient guests entering Duxbury waterways shall receive the permission of the Harbormaster before tying up to a mooring.
- (e) Mooring permit holders shall not rent out their assigned mooring. Any mooring not being used by its owner may be designated as a "guest mooring" and temporarily assigned at the discretion of the Harbormaster.

7.6 § 1007: SPEED

- (a) All vessel operators are responsible for their wakes at all times. M.G.L. Chapter 90B and 323 CMR 2.07 operation shall apply.
- (b) No vessel is to exceed any posted speed limits. Posted areas are as follows:
  - 1. Snug Harbor Deep Water (dredged) Basin and Snug Harbor Basin Flats as designated by regulation speed buoy – NO WAKE, headway speed only.
  - 2. Bluefish River as designated by regulation speed buoy, 6 MPH / NO WAKE.
  - 3. Duck Hill River as designated by regulation speed buoy, 6 MPH / NO WAKE.
  - 4. Howland's Landing as designated by regulation speed buoy – NO WAKE, headway speed only.
  - 5. Speeds in any anchorage not posted shall be NO WAKE, headway speed only.
  - 6. Under the Powder Point Bridge as posted, 6 MPH / NO WAKE.

7.6 § 1008: CONDUCT



- (a) Disorderly conduct is forbidden at all times anywhere on Duxbury waterways or at or on the Town Pier, waterfront facility or any other property of the Town of Duxbury.
- (b) Negligent operation is prohibited. No person shall operate any vessel in a reckless or negligent manner so as to endanger the life, safety or property of any person.

7.6 § 1009: WATER SKIING

- (a) Water skiing as hereinafter permitted is subject to all the provisions as set forth in M.G.L. Chapter 90B. Strict attention shall be made to M.G.L. Chapter 90B, Section 8 (A-D) and to the Rules and Regulations as adopted by the Commonwealth under the authority of M.G.L., Chapter 90B, Section 11, Chapter 275, Section 2, act of 1960 and / or other applicable law.
- (b) Water skiing is prohibited in the following areas:
  - 1. Within one hundred fifty (150) feet of bathers, bathing beach, shore, piers, bridges, floats and other boats including those anchored or underway, and of anchorages.
  - 2. The Bluefish River - inside of designated regulation speed buoy.
  - 3. Duck Hill River - inside of designated regulation speed buoy.
  - 4. Within one hundred fifty (150) feet of any designated anchorage or other mooring field.
- (c) Down Skier Flag Rule:

No persons shall operate or cause to be operated any motor boat on Duxbury waterways towing a person or persons on water skis, surfboard or any similar device, or a person or persons whether or not on such a device, unless there is in such motor boat a spotter in addition to the operator, whose responsibility and duties include, but not limited to, the following:

- 1. Relays - The signal and condition of the person or persons being towed to operator.
  - 2. Assists the skier when necessary and possible.
  - 3. Implements the DOWN SKIER FLAG RULE. The spotter is required to display the Regulation Flag whenever the person or persons being towed are down in the water. The Regulation Flag is only to be displayed at the start or immediate completion of a run, including all wipeouts or falls. When the Regulation flag is displayed it shall be visible to all vessels within the vicinity of the down skier.
  - 4. Whenever the operator's vision is obscured for whatever reason, i.e. (river bends, etc.) or the spotter's view of the skier is obscured for whatever reason, the towing shall be immediately terminated, at which time, proper operating procedures shall be followed M.G.L. Ch. 90B.
- (d) The Rules and Regulations / Operating Procedures and Rules of the Road as adopted by the Commonwealth shall be strictly abided by.

7.6 § 1010: SWIMMING / SCUBA DIVING

- (a) Swimming and / or scuba diving are prohibited in or across any marked channel except in emergency situations or as approved by the Harbormaster.
- (b) There will be no swimming or diving allowed from any pier, float or vessel within any designated anchorage.
- (c) Skin Diving or scuba diving within any designated anchorage shall be permitted only at the discretion of the Harbormaster. M.G.L. Ch. 90B shall apply, and or other applicable law.

7.6 § 1011: HARBOR POLLUTION

The follow are strictly prohibited within Duxbury waterways:

- (a) Pumping or dumping overboard any treated or untreated vessel sewage.
- (b) Dumping of refuse of any kind.
- (c) Dumping of fish or fish carcasses.
- (d) Pumping or dumping overboard of hazardous waste to include but not limited to gas, oil and anti-freeze.

7.6 § 1012: RACES

- (a) There will be no races held within the boundaries of any designated anchorage from May 1<sup>st</sup> through November 15<sup>th</sup> unless authorized by the Harbormaster.
- (b) The Harbormaster shall be notified in writing of all races at least forty-eight (48) hours in advance of the day of the race.
- (c) Race notification forms will be available at the Harbormaster's office.
- (d) The Harbormaster may impose terms and conditions, within twenty-four (24) hours of receipt of a race notification, based on but not limited to the following:
  - 1. Impacts on public safety.
  - 2. Impacts on Duxbury waterways.
  - 3. Impacts on any Town of Duxbury waterfront facility.
  - 4. Impacts on any Town service.
- (e) This does not relinquish the responsibility of the applicant to make applications to all appropriate agencies as required by federal and / or state law for comment and / or approval.
- (f) Organizations sponsoring races shall be responsible to provide crash boats and personal in sufficient quantities to assure the safety of the race participants.

7.6 § 1013: ABANDONMENT

- (a) No vessel, mooring or other object shall be abandoned, sunk or otherwise placed where it may constitute a danger or obstruction to navigation or a public nuisance.
- (b) Any vessel improperly secured, swamped, sunk or washed ashore may be removed or relocated by the direction of the Harbormaster if the owner does not take corrective action after appropriate notice or within 24 hours.



- (c) The expense of such removal or relocation, and any liability incurred, shall be the responsibility of the owner.
- (d) Nothing in the above shall be deemed to restrict earlier action by the Harbormaster with or without notice to the owner if, in their judgment, such action is necessary to protect life or property.

7.6 § 1014: RULES AND REGULATIONS OF THE HARBORMASTER

The Harbormaster may from time to time promulgate Rules and Regulations relating to matters within his powers and jurisdiction under Massachusetts General Laws Chapter 102 § 19 through 26, Chapter 90B, Chapter 91 and all other applicable laws and rules and regulations.

7.6 § 3001 NON CRIMINAL DISPOSITION PENALTIES

- (a) The penalty for violations of any of the Town of Duxbury Rules and Regulations for Duxbury Waterways and Town of Duxbury Rules and Regulations for Moorings, Permits and Waiting Lists shall be \$25.00 for the first offense; \$50.00 for the second offense; \$100.00 for the third offense and \$200.00 for the fourth and subsequent offenses.
- (b) Violations of the Town of Duxbury Rules and Regulations for Duxbury Waterways and Town of Duxbury Rules and Regulations for Moorings, Permits and Waiting Lists may be sufficient cause for the Harbormaster to refuse an individual or his vessel the use of town properties or any town waterfront facility for a period of time as determined by the Harbormaster and Board of Selectmen.
- (c) Offenders will be prosecuted by the Harbormaster and all other enforcement agents.
- (d) In the event that any provisions, sections or clauses of the Town of Duxbury Rules and Regulations for Duxbury Waterways and Town of Duxbury Rules and Regulations for Moorings, Permits and Waiting Lists are found to be invalid, such invalidity shall not affect the validity of the remaining portions of those Rules and Regulations.
- (e) The revenue from any said fines, after payment of fees and expenses, shall be paid to the Town of Duxbury, Massachusetts.
- (f) The Town of Duxbury Rules and Regulations for Duxbury Waterways and the Town of Duxbury Rules and Regulations for Moorings, Permits and Waiting Lists may be amended by the Board of Selectmen.

## **APPENDIX C: MOORING RULES AND REGULATIONS**

### **TOWN OF DUXBURY**

### **HARBORMASTER**

#### **Rules and Regulations**

#### **for Moorings, Permits and Waiting Lists**

**Effective date: 02 May 2005**

#### **7.6 § 2001: AUTHORITY AND ENFORCEMENT**

- (c) Under the authority of Massachusetts General Laws, Chapter 90B § 15B; Chapter 88 § 19; Chapter 102 § 19-26; Chapter 91; 310 CMR 9.07 and all other applicable laws and regulations, the following rules and regulations are promulgated effective 02 May 2005.
- (d) The Harbormaster and Assistant Harbormasters have the authority to enforce all rules and regulations set forth herein.

#### **7.6 § 2002: DEFINITIONS**

The following words, for the purpose of these regulations, unless another meaning is clearly apparent for the way the word is used, shall have the following meanings:

- (s) “anchoring” means to secure a vessel for a short period of time, not to exceed twenty-four (24) consecutive hours, to the bottom of a body of water by dropping an anchor or anchors or other ground tackle from a vessel.
- (t) “Authorized Mooring Service” means a certified service approved by the Harbormaster for the commercial building, maintenance, repair (service), placement and hauling of mooring gear and equipment in designated anchorage areas.
- (u) “berth” means any space wherein a vessel is confined by wet slip, float, mooring or any other type of docking facility.

- (v) “boatyard” means a facility whose function is the construction, repair, sale, or maintenance of vessels, which may include provisions for vessel storage, launching, hauling and docking while awaiting service or use.
- (w) “designated anchorages” means established mooring fields at Snug Harbor Deep Water (dredged) Basin, Snug Harbor Basin Flats, Howland’s Landing, Two Rock Channel and Clark’s Island.
- (x) “draft” means the measurement from the water line to the deepest part of the hull.
- (y) “Duxbury waterways” means all waters within the embayment, harbors, rivers and open waters within the boundaries and jurisdiction of the Town of Duxbury, over which the Town of Duxbury may exercise its powers and authority.
- (z) “emergency boat yard” means a boatyard that a mooring permit holder prefers the Harbormaster contact first in the event that the permit holder cannot be reached to address an issue that may arise with their mooring or vessel. This shall not require the boat yard to act on behalf of the permit holder.
- (aa) “fairway” means any locally designated and / or maintained water areas reserved for the unobstructed movement of vessels through mooring fields or designated anchorages.
- (bb) “grace period” means that a permit holder of a mooring location shall be extended a one (1) year grace period (one boating season) to retain their designated mooring, if for whatever reason the ownership of their vessel has been terminated or is in a state of disrepair and / or extraordinary circumstances exist, at the discretion of the Harbormaster.
- (cc) “guest mooring” means any mooring that is not being used by the permit holder may be used as a berth for a transient or other vessel.
- (dd) “Harbormaster” means the duly appointed Harbormaster or any duly appointed Assistant Harbormaster.
- (ee) “length” means the straight line measurement of the overall length on deck from the bow to the stern, measured parallel to the centerline, including rudder, bumpkins, outboard motor brackets, engines, platforms, or any substantial object or gear that requires additional mooring space consideration.
- (ff) “mooring” means the equipment and / or process used to temporarily secure a vessel, other than by anchoring, that consists with the gear as prescribed by the Town of Duxbury, which is placed on the bed of a body of water attached to a buoy.
- (gg) “mooring buoy” means a buoy designed to float a mooring chain.
- (hh) “mooring location” means an assigned location where a buoyant vessel is secured other than a pier, float, wharf, bulkhead, pile or other.
- (ii) “other object” means any floating object, including but not limited to floats, rafts or barges, temporarily secured to the bed of any body of water by anchor or mooring.
- (jj) “pendant” means a length of thimble line attached to a mooring chain used to secure a vessel to the mooring.
- (kk) “permitted vessel” means any vessel with a valid mooring permit in Duxbury waterways issued by the Harbormaster.
- (ll) “person” means any individual, partnership, trust, firm, corporation, association, commission, district, department, board, municipality, public or quasi-public agency or authority.
- (mm) “pick-up buoy” means a buoy that is attached to a mooring pendant.
- (nn) “tidal flats” means all mooring fields located within Duxbury waterways, other than any designated anchorage.

- (oo) “transient guest” means any vessel requesting a guest mooring that does not have a valid mooring permit issued by the Town of Duxbury.
- (pp) “vessel” means every description of watercraft, other than a seaplane on the water, used or capable of being used as a means of transportation on the water.
- (qq) “winter buoy” means an unsinkable buoy that is attached to a mooring that is not hauled for the winter.
- (rr) “winter stick” means a block of wood not to exceed four feet (4’) in length and is attached to a mooring that is not hauled for the winter.

#### 7.6 § 2002 APPLICABILITY

These rules and regulations apply to all of the following:

- (a) Mooring equipment and mooring locations within the waterways of the Town of Duxbury including all fresh water bodies of water.
- (b) Vessels or other objects anchored or moored within the waterways of the Town of Duxbury to include all fresh water bodies of water.

#### 7.6 § 2003 MOORING PERMITS (*General*)

- (a) All mooring permits are numbered in a manner that designates location.
- (b) It shall be the responsibility of the mooring applicant to annually file a complete mooring application with the Harbormaster for each mooring location they are assigned.
- (c) All vessels moored within the waterways of Duxbury, with the exception of transient vessels, shall have a valid Duxbury mooring permit adhered to the port bow or another conspicuous location as approved by the Harbormaster.
- (d) All other objects shall have a valid Duxbury Mooring Permit or be actively licensed pursuant to M.G.L. Chapter 91.
- (e) All moorings shall have a valid “Mooring Tag”, issued by the Town of Duxbury, secured to it above the waterline in a clearly visible and easily accessible manner.
- (f) Mooring permits are not transferable except by the Harbormaster.
- (g) Mooring permits and locations are not transferable to a new owner if the vessel is sold.
- (h) Mooring permits shall not be “passed down” or transferred to another family member. The only exception of this rule is that if a permit holder is deceased, the mooring permit may be transferred to their legal spouse.
- (i) A mooring permit and location given up for any reason by any vessel owner immediately voids all rights and privileges to that mooring permit and location.
- (j) The Harbormaster may inspect all moorings periodically throughout the boating season to eliminate any hazardous conditions.
- (k) Mooring locations are subject to change at the discretion of the Harbormaster.
- (l) Mooring permit holders are prohibited to change the size of a vessel on their assigned mooring without the advanced approval of the Harbormaster.
- (m) It shall be the responsibility of the mooring permit holder to provide a minimum of twenty-four (24) hours advanced notice to the Harbormaster prior to launching their vessel and occupying their assigned mooring location.

#### 7.6 § 2004 SCHEDULE FOR ISSUING MOORING PERMITS

- (a) All mooring permits issued annually shall expire on December 31<sup>st</sup> of the calendar year they were issued.
- (b) Mooring applications are available February 15<sup>th</sup> of each calendar year unless otherwise amended by the Harbormaster.
- (c) Renewal of an existing mooring permit shall be completed before April 15<sup>th</sup> of each calendar year, unless amended by the Harbormaster.
- (d) Mooring permits that are not renewed before May 15<sup>th</sup> of each calendar year may be voided and the location may be reassigned pursuant to these rules and regulations.

#### 7.6 § 2005     APPLICATION PROCESS

- (a) A person desiring assignment of a new mooring location and issuance of a new mooring permit shall:
  - 1. Complete a valid “Town of Duxbury Mooring Application” provided by the Harbormaster Department.
  - 2. Complete applications shall be submitted to the Harbormaster Department in person.
  - 3. The accepting officer will date stamp and initial the application as received.
  - 4. It is the responsibility of the applicant to contact the Harbormaster within seven (7) to fifteen (15) days of the date the application was received to check on the status of the application.
  - 5. Any application not processed within fifteen (15) days of the date it was received by the Harbormaster shall be considered denied.
  - 6. As mooring locations become available, mooring permits will be granted by the Harbormaster in the order that the application is processed.
- (b) A person wishing to renew an existing mooring permit located in one of the Town of Duxbury’s designated anchorages shall:
  - 1. Follow all written instructions of the Harbormaster.
  - 2. Complete a valid mooring application and attachments they receive.
  - 3. Return the completed valid mooring application, all attachments and payment to the Town of Duxbury as prescribed by the Harbormaster.
    - i. If the applicant chooses to return the completed mooring application and the attachments in person, they will receive a mooring permit at that time.
    - ii. If the applicant chooses to return the completed mooring application and the attachments by mail, they shall include a self addressed stamped envelope so the mooring permit can be mailed to them.

#### 7.6 § 2006     WATERFRONT FEES

The waterfront fees for the Town of Duxbury shall be as followed:

- (a) Snug Harbor Deep Water (dredged) Basin Mooring Permit - \$7.00 per foot annually.
- (b) Snug Harbor Basin Flats Mooring Permit - \$7.00 per foot annually.

- (c) Howland's Landing Mooring Permit - \$6.00 per foot annually.
- (d) Two Rock Channel Mooring Permit - \$15.00 annual administrative application registration fee.
- (e) Clark's Island Mooring Permit - \$15.00 annual administrative application registration fee.
- (f) Tidal Flats Mooring Permit - \$15.00 annual administrative application registration fee.
- (g) Tender Float Spot Permit - \$110.00 annual administrative registration fee.
- (h) Tender Float Tie-Up Permit - \$90.00 annual administrative registration fee.
- (i) Waiting List Application - \$15.00 per application, initial one time administrative registration fee.
- (j) Permitted Vessel Guest Mooring - \$10.00 per vessel, per night administrative registration fee from 15 June through 15 September. Vessel owners shall complete and sign a Guest Mooring Invoice at the Harbormasters Office.
- (k) Transient Guest Mooring - \$30.00 per vessel, per night, administrative registration fee from 15 June through 15 September. Vessel owners shall complete and sign a Guest Mooring Invoice at the Harbormasters Office.
- (l) Transient Vessel Guest Mooring – no charge on or before 14 June and on or after 16 September if use of guest mooring is approved by the Harbormaster.
- (m) Any vessel while requesting and / or requiring (local) boat yard service may be exempt from said transient guest mooring administrative registration fee at the discretion of the Harbormaster.

## 7.6 § 2007 MOORING REQUIREMENTS

The following shall be considered as minimum requirements for mooring gear in the respective anchorage:

- (a) Snug Harbor Deep Water (dredged) Basin
  - 1. Boats under 25' - 300 lb. mushroom or 300 lb. Pyramid - 30' 1/2 chain - 12' of 5/8" rope to chock or (2) 1/2" lines as pendants.
  - 2. Boats 26' to 35' - 300 lb. Mushroom or 300 lb. Pyramid - 30' 1/2 chain - 12' of 3/4" rope to chock or (2) 5/8" lines as pendants.
  - 3. Boats 36' to 45' - 500 lb. mushroom or 500 lb. Pyramid - 35' 5/8 chain - 15' of 1" rope to chock or (2) 3/4" lines as a pendants.
- (b) Snug Harbor Basin Flats
  - 1. Under 16' - 50-75 lb. mushroom or 70 lb. Pyramid - 25' 5/16 chain 12' of 7/16" min. rope to chock or (2) 7/16" lines as pendants.
  - 2. 17' to 18' - 100 lb. Mushroom or 135 lb. Pyramid - 25' 3/8 chain 12' of 1/2" min. rope to chock or (2) 1/2" lines as pendants.
  - 3. 19' to 20' - 150 lb. Mushroom or 135 lb. Pyramid - 25' 3/8 chain 12' of 5/8" min rope to chock or (2) 1/2" lines as a pendants.
  - 4. 21' to 25' - 200 lb. Mushroom or 200 lb. Pyramid - 25' 1/2 chain 12' of 5/8" min rope to chock or (2) 1/2" lines as pendants.

26' to 30' - 300 lb. mushroom or 300 lb. Pyramid - 25' 1/2 chain 15' of 3/4" min rope to chock or (2) 5/8" lines as a pendent.

## (c) Tidal Flats

1. Under 16' - 50 - 75 lb. mushroom or 70 lb. Pyramid - 15' 5/16" chain 7/16" min rope to chock.
2. 17' to 18' - 100 lb. mushroom or 135 lb. Pyramid - 15' 3/8" chain 15' 1/2" min. rope to chock.
3. 19' to 20' - 150 lb. mushroom or 135 lb. Pyramid - 15' 3/8" chain 15' 5/8" min. rope to chock or (2) 1/2" lines as a pendants.
4. 21' to 25' - 200 lb. Mushroom or 200 lb. Pyramid - 15' 1/2" chain 15' 5/8" min. rope to chock or (2) 1/2" lines as a pendants.

## (d) Howland's Landing, Cove Street, West end of the Powder Point Bridge, and the Back River (deepwater)

1. Under 16' - 50 - 100 lb. mushroom or 70 lb. Pyramid - 15' 5/16 chain 15' of 7/16 min rope to cleat.
2. 17' to 25' - 200 lb. mushroom or 200 lb. Pyramid - 25' 1/2 chain 15' of 5/8" min rope to cleat or (2) 1/2" lines as pendants.
3. 26' to 30' - 250 lb. mushroom or 270 lb. Pyramid - 25' 1/2 chain 15' of 3/4" min rope to chock or (2) 5/8" lines as pendants.
4. 31' to 35' - 300 lb. mushroom or 270 lb. Pyramid - 25' 1/2 chain 15' of 3/4" min rope to chock or (2) 5/8" lines as pendants.
5. 36' to 45' - 500 lb. mushroom or 500 lb. Pyramid - 25' 5/8 chain 20' of 1" min rope to chock or (2) 3/4" lines as pendants.

## (e) Two Rock Channel and Clark's Island

1. 20' to 30' - 250 lb. mushroom or 270 lb. Pyramid - 40' 1/2" chain 12' 3/4" min rope to chock or (2) 5/8" lines as pendants.
2. 31' to 35' - 300 lb. mushroom or 270 lb. Pyramid - 40' 1/2" chain 12' 3/4" min rope to chock or (2) 5/8" lines as pendants.
3. 36' to 40' - 500 lb. mushroom or 500 lb. Pyramid - 40' 5/8" chain 12' 1" min rope to chock or (2) 3/4" lines as pendants.

(f) All moorings in designated anchorages shall have an approved non-sinkable chain float with a pendant directly connected to the chain above water.

(g) All mooring chain shall be of the galvanized type; Black iron chain shall not be accepted.

(h) All moorings in designated anchorages shall be in working order, rigged appropriately and placed in assigned locations by 15 June of each calendar year.

(i) All moorings in designated anchorages shall be inspected, serviced and maintained by an Authorized Mooring Service unless otherwise authorized by the Harbormaster.

(j) Cement block of approved type, if used, shall be six times the weight of mushroom required.

(k) All moorings shall be inspected by the Harbormaster or authorized agent before being set.

(l) All moorings shall be hauled at the expense of the owner and inspected by the Harbormaster or his authorized Agent every three years. Any mooring with three-

- eighths inch (3/8") chain shall be hauled at the expense of the owner and inspected by the Harbormaster or his authorized Agent every 2 years.
- (m) Chain size and mooring design are subject to change with or without notice based on the discretion of the Harbormaster for the purpose of public safety.
  - (n) All moorings shall be equipped with properly positioned chafe gear, when applicable, as approved by the Harbormaster.
  - (o) All mooring shackles shall be safe tied with stainless steel wire.
  - (p) Only nylon rope of proper length and diameter shall be used.
  - (q) All moorings above mean low water, utilizing mushroom anchors, shall be buried properly below the surface of the flats within three days of being set.
  - (r) All mooring buoys shall be white with blue stripe as specified by statute.
  - (s) All mooring buoys in any designated anchorage shall legibly designate mooring location and maximum boat size with three (3) to five (5) inch block lettering.
  - (t) All pick up buoys shall legibly designate the owner's last name.
  - (u) All chain flotation buoys shall be plainly and clearly visible above any tide level at all times.
  - (v) Approved winter sticks and winter buoys shall:
    - 1. Be marked with the owner's name.
    - 2. Be marked with the mooring location.
    - 3. Not be installed before November 1<sup>st</sup>.
    - 4. Be removed on or before May 1<sup>st</sup>.
    - 5. If located within a designated anchorage, be removed or marked with unsinkable buoy to be visible at all tides on or before May 1<sup>st</sup>.
    - 6. All fairways be marked with unsinkable buoys to be visible at all tides between November 1<sup>st</sup> and May 15<sup>th</sup>.
    - 7. Inclusive of all Duxbury waterways, except designated anchorages, be marked with an unsinkable buoy to be visible at all tides between November 1<sup>st</sup> and May 15<sup>th</sup>.
    - 8. All Snug Harbor Deep Water (dredged) Basin float moorings, be marked with unsinkable buoys to be visible at all tides between November 1<sup>st</sup> and May 15<sup>th</sup>.

#### 7.6 § 2008     DEFECTIVE MOORINGS

- (a) After appropriate notice, owners of defective moorings shall have seven (7) days to correct said mooring, after which time the mooring will be hauled at the direction of the Harbormaster.
- (b) The expense of such removal or hauling, and any liability incurred therefore, shall be the responsibility of the mooring owner.
- (c) Any defective mooring removed or hauled shall be positioned back in its designated location after completed repairs and inspections as soon as possible, or no later than seven (7) days.

#### 7.6 § 2009     MOORING RECORDS

The Harbormaster will keep a detailed description of all moorings, their location, owner's name, contact numbers, home address, emergency boat yard and description of vessel, to include



length, rig and name for the purposes of identifying and addressing any issues that may arise with that mooring location or the vessel assigned to it.

#### 7.6 § 2010 WAITING LIST POLICIES AND PROCEDURES

(a) Waiting Lists for mooring location assignment.

1. The Harbormaster will identify and determine those mooring fields that have received maximum mooring saturation and establish a waiting list for the issuance of mooring locations and permits for those areas. Each mooring field shall have a separate waiting list.
2. The Harbormaster shall also establish a waiting list for existing mooring owners that wish to be reassigned a different mooring location that would allow them the ability to either increase or decrease the size of their vessel.
3. Applicants shall completely fill out and submit a waiting list application and pay the waiting list application fee.
4. All applications shall be complete, signed, dated, and received by the Harbormaster. Received applications will be signed off by the officer in charge and added to the waiting list in the order it was received.
5. Applications shall not be “passed down” or transferred to another family member. The only exception of this rule is that if an applicant dies, the waiting list application may be transferred to their legal spouse.
6. The Town of Duxbury is not liable or responsible for incomplete or lost applications.
7. All complete applications and attached documents submitted to the Harbormaster Department become the property of the Town of Duxbury.
8. All official mooring waiting lists will be kept at the Duxbury Harbormaster’s office.
9. All mooring waiting lists will be periodically up-dated.
10. A copy of all updated waiting lists will be posted at the following locations: Duxbury Harbormaster’s office, Duxbury Free Library, Duxbury Town Clerk and the Duxbury Town Manager / Board of Selectmen’s office.

(b) Assignment of mooring locations within waiting list areas.

1. When a mooring location within a waiting list area becomes available, the Harbormaster will offer a mooring location to the first applicant on the waiting list.
2. The applicant shall respond to the Harbormaster within forty-eight (48) hours after has been received with an answer. If an answer is not given within that time frame, it will be considered a pass.
3. If the applicant chooses not to accept the location, the Harbormaster will offer a mooring location to the first applicant on the existing mooring list requesting a reassignment of their mooring location.
4. If the existing mooring applicant accepts the reassignment thereby still leaving an open mooring to be assigned, then the Harbormaster shall again offer a location to the first applicant on the waiting list.
5. Whenever a mooring location within a waiting list is available, the Harbormaster shall follow the procedures as outlined in 7.6 § 2010(b2, b3) alternating between the

- first applicant on the waiting list and the first applicant on the existing mooring list requesting a reassignment until all locations have been assigned.
6. Applicants will have three (3) opportunities to accept a mooring location.
  7. If within a calendar year multiple mooring locations, in the same vessel size range become available in the same waiting list area and an applicant who is offered a location refuses that location, it is assumed that the applicant will not accept another location within that area. This shall only be counted as one pass for the applicant.
  8. The Harbormaster may at his discretion reassign mooring locations to existing and new applicants based on the size of their vessel relative to the maximum length of the mooring location.

(c) Waiting list renewal and removal from list.

1. All waiting list applications shall be renewed annually within a time period of, two weeks before and two weeks after the anniversary date, of the applicant's initial active waiting list application date to be placed on a waiting list.
2. Renewals shall be done in person or in writing only by certified (return receipt) mail and will be date stamped and signed by the Harbormaster.
3. Annual mooring applications for any other mooring within the waters of the Town of Duxbury shall not be accepted as renewal on any waiting list.
4. Failure to renew the waiting list application shall result in the original application being voided and the applicant's name being removed from that waiting list.
5. If an applicant chooses not to accept a location on their third offer, the individual's application shall be voided and removed from that waiting list.

7.6 § 2011 GUEST MOORINGS

The following shall apply to mooring permit holders, permitted guests and transient guests:

- (a) Rules and Regulations for the Waterways of Duxbury Chapter 7.6 § 1006: Berthing, shall apply.
- (b) Moorings vacated by the permit holder for twenty-four (24) hours or longer, for any reason, shall be reported to the Harbormaster Department upon vacating the mooring.
- (c) It shall be the responsibility of the owner and / or their agent to report the status of said vessel and / or mooring to the Harbormaster.
- (d) Guests shall register with the Harbormaster prior to picking up a mooring. The guest shall provide the vessel's length, rig, and registration or documentation number if applicable and number of days they wish to use the mooring.
- (e) Guests shall accept any and all responsibility for loss or damage caused by them or their vessel.
- (f) Guests shall secure the vessel properly on a mooring designated by the Harbormaster with chafe gear in positioned correctly when applicable.
- (g) Local boat yards shall be responsible to ensure any vessel being serviced by them and moored in Duxbury as a registered guest has chafe gear positioned correctly. Rules and Regulations for the Waterways of Duxbury Chapter 7.6 § 1006 shall apply to that boat yard where applicable and when not in conflict with existing law.
- (h) Transient guests may secure guest tender if used to the Town Tender Float at Town Pier.

- (i) Persons requesting a guest mooring for a period of less than six (6) hours may be exempt from being charged a guest mooring fee provided that they register with the Harbormaster. All other rules and regulations still apply.
- (j) Transient vessels or permitted vessels who use a mooring other than their own without authorization from the Harbormaster shall still be charged the applicable guest mooring fee and shall be cited.

#### 7.6 § 2012 AUTHORIZED MOORING SERVICE POLICY

The following requirements shall apply to the application and approval of Authorized Mooring Service permits:

- (a) One (1) year mooring service experience. Applicant shall provide:
  - 1. Documented proof of knowledge and experience with Duxbury or similar mooring equipment.
  - 2. Local knowledge of the waterways of Duxbury.
  - 3. References as approved by the Harbormaster.
- (b) Utilize a proper mooring service vessel as approved by the Harbormaster.
- (c) Possess required federal, state and local licenses and permits.
- (d) Supply proof of proper liability insurance and workmen's compensation insurance.
- (e) Agrees to abide by all Federal, State, Local laws, rules and regulations, conditions, terms and the direction of the Harbormaster Department.
- (f) Provide twenty-four (24) hour on call status to respond to emergency or exigent circumstances regarding any mooring serviced by them.
- (g) Accepts any and all responsibility for damage or injury caused by them or their employees. The Town of Duxbury and its employees accept no responsibility or liability.
- (h) Existing services shall reapply for Mooring Service Authorization annually within the month of January on forms supplied by the Town of Duxbury. Approval or denial shall be determined within thirty (30) days.
- (i) Existing services shall inform the Harbormaster in writing that they do not want to reapply for Mooring Service Authorization by July 1<sup>st</sup> of their active permit.
- (j) New applications shall apply for Mooring Service Authorization on forms supplied by the Town of Duxbury. Approval or denial shall be determined within sixty (60) days.
- (k) Applications that are not complete will not be processed. Illegible applications will be considered incomplete. Applications submitted without the requested attachments will be considered incomplete. The Town of Duxbury assumes no responsibility or liability for lost, late or incomplete applications, attachments or documents.
- (l) Failure of the applicant to adhere to the set standards, terms, conditions, rules or regulations, law, or the direction of the Harbormaster may immediately void authorization.
- (m) Minimum standards and requirements may be amended as authorized by law, rule and regulation.
- (n) This authorization is non-transferable.
- (o) The Harbormaster will administer and enforce this policy.

- (p) The Town Manager shall rule on all appeals. Appeals shall be submitted in writing not later than thirty (30) days following a denial or termination of Mooring Service Authorization.

A valid list of authorized mooring services will be posted at the Harbormaster office

7.6 § 3001     NON CRIMINAL DISPOSITION PENALTIES

- (g) The penalty for violations of any of the Town of Duxbury Rules and Regulations for Duxbury Waterways and Town of Duxbury Rules and Regulations for Moorings, Permits and Waiting Lists shall be \$25.00 for the first offense; \$50.00 for the second offense; \$100.00 for the third offense and \$200.00 for the fourth and subsequent offenses.
- (h) Violations of the Town of Duxbury Rules and Regulations for Duxbury Waterways and Town of Duxbury Rules and Regulations for Moorings, Permits and Waiting Lists may be sufficient cause for the Harbormaster to refuse an individual or his vessel the use of town properties or any town waterfront facility for a period of time as determined by the Harbormaster and Board of Selectmen.
- (i) Offenders may be prosecuted by the Harbormaster and all other enforcement agents.
- (j) In the event that any provisions, sections or clauses of the Town of Duxbury Rules and Regulations for Duxbury Waterways and Town of Duxbury Rules and Regulations for Moorings, Permits and Waiting Lists are found to be invalid, such invalidity shall not affect the validity of the remaining portions of those Rules and Regulations.
- (k) The revenue from any said fines, after payment of fees and expenses, shall be paid to the Town of Duxbury, Massachusetts.
- (l) The Town of Duxbury Rules and Regulations for Duxbury Waterways and the Town of Duxbury Rules and Regulations for Moorings, Permits and Waiting Lists may be amended by the Board of Selectmen.

## APPENDIX D: INPUT FROM PUBLIC WORK GROUPS

### AQUACULTURE

#### Implementation Action Plan

Objectives: Develop baseline of existing sites and potential sites. Develop Master Plan. Resolve conflict with boaters (existing complaints successful?). Are there aesthetic conflicts with heavy public use? Depth of Bay and height of cages creates conflicts. Identify non-point source pollution sources. Limit aquaculture grants.

**Table 9-1. Summary of Aquaculture Master Plan**

| What (will be done) | Where (will it be done) | When (will it be done) | Who (will do it)         | How (will it be done)                                               |
|---------------------|-------------------------|------------------------|--------------------------|---------------------------------------------------------------------|
| Develop Master Plan | Duxbury Bay             | ASAP                   | Shellfish Advisory Comm. | Determine existing sites (est. 21 now)                              |
|                     |                         |                        |                          | Identify potential sites                                            |
|                     |                         |                        |                          | Document degree of success                                          |
|                     |                         |                        |                          | Document degree of public use conflicts (based on complaints, tide) |
|                     |                         |                        |                          | Limit to protect navigation safety                                  |
|                     |                         |                        |                          | Limit in other reasonable form                                      |
|                     |                         |                        |                          | Application. Process ensure the grant will be actively used         |
|                     |                         |                        |                          | Study other limits to expansion                                     |

#### **Management Plan and Recommendations Summary**

Summary Implementation Action (Work) Plans (table)

Investigate causes of closure of shellfish beds

Investigate causes of closure of beaches

Investigate causes of loss of eelgrass

Investigate condition of Sensitive receptors – beaches, eelgrass beds, shellfish beds

Investigate effect of storm drains and NPS pollution on sensitive receptors

Investigate effects of boating activity on sensitive receptors

Duxbury Bay Management Goals (Measurable characteristics that would be monitored)

Water Quality

Bacteria

Acre-day shellfish harvest opportunities

Trends in dry/wet-weather bacteria indicators

Trends in beach closings

Nutrients and Eutrophication

Trends in nutrient concentrations (NO<sub>3</sub>, NO<sub>4</sub>, NH<sub>4</sub>, PO<sub>4</sub>)

Trends in particulate concentrations

Trends in biological oxygen demand (BOD)

Trends in dissolved oxygen (DO)

Toxic Contaminants

Trends in shellfish/benthic invertebrate tissue concentrations

Trends in sediment contaminant levels

Plant Species

. Eelgrass distribution

. Other salt marsh distribution

Animal Species

. Anadromous fish returns

. Abundance of wintering waterfowl

Abundance of shellfish harvest

Habitat Protection

. Conservation lands

. Shoreline development

Habitat Restoration

- . Anadromous fish returns (e.g., Island Creek herring fish ladder)
- . Shellfish grants
- . Restored eelgrass beds ?

## BOATING, SAFETY, AND NAVIGATION

### Implementation Action Plan

Objective: Insure that the Bay is safe for all users

**Table 2. Summary of Boating Master Plan**

| <b>What (will be done)</b>                 | <b>Where (will it be done)</b>    | <b>When (will it be done)</b> | <b>Who (will do it)</b>          | <b>How (will it be done)</b> |
|--------------------------------------------|-----------------------------------|-------------------------------|----------------------------------|------------------------------|
| Speed posters where people swim            | Town landings/ area before bridge | By May 31, 04                 | Town and harbormaster to enforce | Town Depts.                  |
| Mark swim areas warning?                   | Shipyard, Howlands                |                               |                                  |                              |
| Mark launch areas with signage             | Landing Rd.                       |                               |                                  |                              |
| Have smaller aquaculture cages and markers |                                   |                               |                                  |                              |
| Better educate boating public              |                                   |                               |                                  |                              |
| Establish emergency storm procedure        |                                   |                               |                                  |                              |

## PUBLIC ACCESS

Summary Implementation Action (Work) Plans (table)

Topic: Parking and Access to the Bay

Objective: Reduce traffic congestion for Snug harbor to Powder Point and thereby increase safety.

**Table 9-2. Summary of Bay Access**

| What (will be done)                          | Where (will it be done)                | When (will it be done) | Who (will do it)                                                  | How (will it be done)                      |
|----------------------------------------------|----------------------------------------|------------------------|-------------------------------------------------------------------|--------------------------------------------|
| Investigate Shuttle System/Satellite Parking | High School parking lot to Snug Harbor | Summer 2004            | Maritime School, Yacht Club, Bayside, DR and HS, Bay Rider Launch | Joint venture of Town and private entities |
| Install bike racks at Town Landings          | Various landings                       | ASAP                   | Town                                                              | Town set up/Town Meeting                   |
| Survey town landings                         | All town landings and public ways      | Start Summer 04        | Town                                                              | Volunteers/Consultant                      |



APPENDIX E: CHARACTERISTICS OF TOWN LANDINGS AND WAYS TO THE WATER, DUXBURY, MASSACHUSETTS

Characteristics of Town Landings and Ways to the Water, Duxbury, Massachusetts

| Types of Public Access Properties                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Date of Transfer or Acceptance by Town | 1899/1900, Report on Town Landings | Contained in, 2002 Open Space and Recreation Plan | Contained in Conservation Land and Other Points of Interest 2000 (ConCom Map) | 1893 Surveyed Plans | Acres | Registry of Deeds |      | Town of Duxbury |                | Existing Uses    |             |               |           |      |                   |             |              |               |                | Boat Ramp | Swimming | Picnic | Viewing | Recreational Fishing, Fowling and Navigation Access | Existing Uses |            |                     | Ecologically Sensitive Receptors in Area |     |     |                                                                                                                                                                                                        | Comments                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------|---------------------|-------|-------------------|------|-----------------|----------------|------------------|-------------|---------------|-----------|------|-------------------|-------------|--------------|---------------|----------------|-----------|----------|--------|---------|-----------------------------------------------------|---------------|------------|---------------------|------------------------------------------|-----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                        |                                    |                                                   |                                                                               |                     |       | Parking           |      |                 |                |                  |             | Resident Only | Over Sand | Cars | Time Restrictions | Handicapped | Boat Trailer | Seasonal Uses | Shellfish Beds |           |          |        |         |                                                     | Eelgrass Beds | Salt Marsh | Anadromous Fish Run |                                          |     |     |                                                                                                                                                                                                        |                                                                                                                                                             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                        |                                    |                                                   |                                                                               |                     |       | Plan Book         | Page | Plan Book       | Page           |                  |             |               |           |      |                   |             |              |               |                |           |          |        |         |                                                     |               |            |                     |                                          |     |     |                                                                                                                                                                                                        |                                                                                                                                                             |
| Town Landings <sup>a</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                        |                                    |                                                   |                                                                               |                     |       |                   |      |                 |                |                  |             |               |           |      |                   |             |              |               |                |           |          |        |         |                                                     |               |            |                     |                                          |     |     |                                                                                                                                                                                                        |                                                                                                                                                             |
| 1. Blue Fish River, by Old Mill Dam                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1883 and 1909                          | Yes                                | Yes                                               | Yes, Blue Fish River Landing                                                  | Yes                 |       | 2                 | 6    | 1               | 5              | No               |             |               | No        | No   | No                | No          | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           |            |                     |                                          |     |     | 1975 Report says, Town Report 1904 Article to rescind ban on buildings at Town Landings allowing Engine House to be placed there.                                                                      |                                                                                                                                                             |
| 2. Cove Street, By Old Cove                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1834                                   | Designated #5                      | Yes                                               | Yes, Old Cove Landing                                                         | Yes                 |       | 2                 | 6    |                 |                | No               | recommended |               | No        | No   | Yes               | Yes         | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      |     |     | 1975 Report says, "1924 Town Report shows land exchange with Clapp and Evans"                                                                                                                          |                                                                                                                                                             |
| 3. Drew Salt Works                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1834                                   | Designated #4                      | Yes                                               | Yes                                                                           | Yes                 |       | 2                 | 6    |                 |                | No               | recommended |               | No        | No   | No                | No          | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      | Yes |     | End of Bay Pond Rd.                                                                                                                                                                                    |                                                                                                                                                             |
| 4. Ford's Stand                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1897                                   |                                    | Yes                                               | Yes                                                                           |                     |       |                   |      |                 |                | No               | recommended |               | No        | No   | no                | No          | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | No                                       | No  |     | Off Gurnet Road on Cape Cod Bay, 1897,98 and 1931 TMA to extend this as highway                                                                                                                        |                                                                                                                                                             |
| 5. Gurnet Bridge, At Powder Point End                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1898,County 1870                       | Designated #1                      | yes (2)                                           | Yes                                                                           | Yes                 |       |                   |      |                 |                | No               | recommended | 50            | No        | 2    | Yes               | Yes         | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      |     | No  |                                                                                                                                                                                                        | 1975 Report says, Voted be recorded as a "Highway to the Sea" to conform to the layout of the County Commissioners (1901). Several other improvement votes. |
| 6. Hardin Hill Road                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                        |                                    | Yes                                               | Yes                                                                           |                     |       |                   |      | 1               | 17, 75 revised | No               | recommended |               | No        | No   | Yes               | Yes         | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      |     | Yes |                                                                                                                                                                                                        | At end of Hardin Hill Rd. named 1941                                                                                                                        |
| 7. Hicks Point Road                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1909                                   |                                    |                                                   | Yes                                                                           |                     |       |                   |      |                 |                | recommended      |             |               | No        | No   | Yes               | Yes         | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      | Yes |     | Various TMA                                                                                                                                                                                            |                                                                                                                                                             |
| 8. Howlands Lane                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1895                                   |                                    | Yes                                               | Yes                                                                           |                     |       |                   |      |                 |                | No               | recommended | 18            | No        | No   | 3                 | Yes         | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     | n/a                                                 | Yes           |            | Yes                 |                                          | Yes |     | 1896 TM accepts the road and landing near Allen's Wharf.                                                                                                                                               |                                                                                                                                                             |
| 9. Josselyn Avenue                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1907                                   |                                    | Yes                                               | Yes                                                                           |                     |       |                   |      |                 |                | No               |             |               | No        | No   | No                | No          | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      | Yes |     | 1921 hard surfaced as far as possible                                                                                                                                                                  |                                                                                                                                                             |
| 10. Landing Road                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1909                                   |                                    | Yes                                               | Yes                                                                           |                     | 0.21  |                   |      |                 |                | No               | recommended |               | No        | No   | yes               | Yes         | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      | Yes | No  |                                                                                                                                                                                                        | Mrs. Foote property?                                                                                                                                        |
| 11. Mattakeeset Court                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1902, 1916                             | 1900                               | Yes                                               | Yes                                                                           |                     | 1.05  |                   |      | 1               | 4              | No               | recommended | 60            | Yes       | 4    | 12                | Yes         | No           | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | No                                       | No  |     | 1902 Town Builds Road, 1916 buy 100 feet of frontage Flagg land, 1921 landing bathing beach sea                                                                                                        |                                                                                                                                                             |
| 12. Peterson's Landing                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1898                                   | Designated #3                      | Yes                                               | Yes                                                                           |                     |       | 2                 | 6    |                 |                | No               | recommended |               | No        | No   | Yes               | Yes         | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      | Yes |     |                                                                                                                                                                                                        |                                                                                                                                                             |
| 13. Soule's Landing                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1898                                   | Designated #3                      | Yes                                               | Yes                                                                           |                     |       |                   |      |                 |                | No               | recommended |               | No        | No   | Yes               | Yes         | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      | Yes |     |                                                                                                                                                                                                        |                                                                                                                                                             |
| 14. St. George St., by Old Mill Pond                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1834                                   | Designated #6                      | Yes                                               | Yes, "Anchorage Lane"                                                         | Yes                 | 0.21  | 2                 | 6    |                 |                | No               |             |               | No        | No   | No                | No          | Yes          | Yes           | Yes            | Yes       | No       | No     | No      |                                                     | No            | No         | No                  | No                                       | Yes |     |                                                                                                                                                                                                        | 1715 Grant, 1902 enlarged settle question(?)                                                                                                                |
| 15. Water Street                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1899                                   |                                    | Yes                                               | Yes                                                                           |                     |       |                   |      |                 |                | No               |             |               | No        | No   | No                | No          | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      | Yes |     |                                                                                                                                                                                                        | 1900 Accepted as Highway as an extension Water St. to the shore                                                                                             |
| 16. Winsor Street                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1919                                   |                                    | Yes                                               | Yes                                                                           |                     |       |                   |      |                 |                | No               |             |               | No        | No   | No                | No          | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      | Yes | Yes |                                                                                                                                                                                                        | 1922 Voted to widen eastern end for turning                                                                                                                 |
| Ways to the Water <sup>b</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                        |                                    |                                                   |                                                                               |                     |       |                   |      |                 |                |                  |             |               |           |      |                   |             |              |               |                |           |          |        |         |                                                     |               |            |                     |                                          |     |     |                                                                                                                                                                                                        |                                                                                                                                                             |
| 1. Shipyard Lane                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1944                                   |                                    | Yes                                               | Yes, Shipyard Lane Beach                                                      |                     | 1.22  | 1859              | 374  |                 |                | √ (not enforced) |             |               | 27        | No   | 2                 |             | No           | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           |            |                     |                                          |     |     | Gift of land at end of Shipyard Lane by Mr. Eben Ellison including strip of land 16 feet wide from Washington St. Restricted to Duxbury residents and guests for recreation, no pier shall be erected. |                                                                                                                                                             |
| 2. Bay Farm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1972                                   |                                    | Yes                                               |                                                                               |                     |       |                   |      |                 |                | No               |             |               |           |      | No                | No          | Yes          | Yes           | Yes            | Yes       | no       | no     | yes     | Yes                                                 |               | Yes        |                     | Yes                                      |     |     | TMA, recreation purpose                                                                                                                                                                                |                                                                                                                                                             |
| 3. Myles Standish Cellar Hole, Shore Way                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1930                                   |                                    | Yes                                               | Yes, Myles Standish Homestead                                                 |                     |       |                   |      | 2               | 144            | No               | No          |               | Yes       | No   | No                | No          | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      |     | Yes |                                                                                                                                                                                                        | Gift of Trustees of the Standish Monument Association.                                                                                                      |
| 4. Samoset Rd.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                        |                                    |                                                   | No                                                                            |                     |       |                   |      | 1               | 70-71          | No               | No          |               | No        | No   | No                | No          | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      |     |     |                                                                                                                                                                                                        | As public ways. Various years.                                                                                                                              |
| 5. Sagamore Rd.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                        |                                    |                                                   | No                                                                            |                     |       |                   |      | 1               | 70-71          | No               | No          |               | No        | No   | No                | No          | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      |     |     |                                                                                                                                                                                                        | As public ways. Various years.                                                                                                                              |
| 6. Elder Brewster Rd.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                        |                                    |                                                   | No                                                                            |                     |       |                   |      | 1               | 70-71          | No               | No          |               | No        | No   | No                | No          | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      |     | Yes |                                                                                                                                                                                                        | As public ways. Various years.                                                                                                                              |
| 7. Massasoit Rd.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                        |                                    |                                                   | No                                                                            |                     |       |                   |      | 1               | 70-71          | No               | No          |               | No        | No   | No                | No          | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           | Yes        | Yes                 | Yes                                      |     |     |                                                                                                                                                                                                        | As public ways. Various years.                                                                                                                              |
| 8. Island Creek Pond                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                        |                                    | Yes                                               | Yes                                                                           |                     |       |                   |      |                 |                | No               | recommended |               | Yes       | No   | Yes               | Yes         | Yes          | Yes           | Yes            | Yes       | Yes      | Yes    | Yes     |                                                     | Yes           |            | Yes                 | no                                       |     |     |                                                                                                                                                                                                        | As a public way to shore                                                                                                                                    |
| 9. Elderberry Ln.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                        |                                    |                                                   |                                                                               |                     |       |                   |      |                 |                |                  |             |               |           |      |                   |             |              |               |                |           |          |        |         |                                                     |               |            |                     |                                          |     |     |                                                                                                                                                                                                        |                                                                                                                                                             |
| <sup>a</sup> Town Landings are, "designated areas to which the Town has an undoubted right, which have been surveyed and recorded with the Plymouth County Registry of Deeds to the low water mark (mean).                                                                                                                                                                                                                                                                                    |                                        |                                    |                                                   |                                                                               |                     |       |                   |      |                 |                |                  |             |               |           |      |                   |             |              |               |                |           |          |        |         |                                                     |               |            |                     |                                          |     |     |                                                                                                                                                                                                        |                                                                                                                                                             |
| <sup>b</sup> Ways to the Water are, "designated areas to which the Town has rights as public ways to the water, by gift, or otherwise which may or not be restricted as to their usage and which have not been specifically laid out and surveyed to the low water mark and recorded as Town Landings. Included in this right are roads properly laid out, improved and extended to the water by funds raised and appropriated by the Town and designated as public highways (by Town Meeting |                                        |                                    |                                                   |                                                                               |                     |       |                   |      |                 |                |                  |             |               |           |      |                   |             |              |               |                |           |          |        |         |                                                     |               |            |                     |                                          |     |     |                                                                                                                                                                                                        |                                                                                                                                                             |
| <sup>c</sup> As reported in, <i>Report of the Town Landing Study Committee</i> , 1975, Duxbury.                                                                                                                                                                                                                                                                                                                                                                                               |                                        |                                    |                                                   |                                                                               |                     |       |                   |      |                 |                |                  |             |               |           |      |                   |             |              |               |                |           |          |        |         |                                                     |               |            |                     |                                          |     |     |                                                                                                                                                                                                        |                                                                                                                                                             |